

How many faces does a cube have?

Faces: _____

How many faces does a rectangular prism have?

- A.** 4
- B.** 6
- C.** 8
- D.** 12

True or false?

A square pyramid has 5 faces.

A. True

B. False

How many faces does a triangular prism have?

Faces: _____

How many triangular faces does a square pyramid have?

A. 1

B. 3

C. 4

D. 5

SET 65 • 6.G.A.4

BACKWARD 2

Shapes of faces • Grade 3 • Geometry • 3.G.A.1

What shape are the faces of a cube?

- A.** Triangles
- B.** Pentagons
- C.** Circles
- D.** Squares

True or false?

Every square is also a rectangle.

A. True

B. False

What shape is the base of a square pyramid?

- A.** Square
- B.** Circle
- C.** Triangle
- D.** Rectangle that is not a square

True or false?

The two ends of a triangular prism are triangles.

A. True

B. False

What shape are the side faces of a triangular prism?

Shape: _____

SET 65 • 6.G.A.4

FORWARD 1

Cross sections of three-dimensional figures • Grade 7 • Geometry • 7.G.A.3

A rectangular prism is sliced parallel to its base.
What shape is the cross section?

- A.** Rectangle
- B.** Circle
- C.** Pentagon
- D.** Triangle

A square pyramid is sliced parallel to its base.
What shape is the cross section?

- A.** Square
- B.** Rectangle that is not a square
- C.** Circle
- D.** Triangle

A cube is sliced through the middle, parallel to one face.
What shape is the cross section?

Shape: _____

A square pyramid is sliced vertically through its top vertex.

What shape is the cross section?

A. Pentagon

B. Square

C. Rectangle

D. Triangle

True or false?

A triangular prism sliced parallel to its triangular bases makes a triangle cross section.

- A.** True
- B.** False

SET 65 • 6.G.A.4

FORWARD 2

Volume of a cylinder described by its net • Grade 8 • Geometry • 8.G.C.9

A net of a cylinder has two circles with a radius of 3 cm and a rectangle that is 8 cm tall.

What is the volume of the cylinder in terms of π ?

Volume: _____

A net of a cylinder has two circles with a radius of 5 in. and a rectangle that is 4 in. tall.

What is the volume in terms of π ?

Volume: _____

A net of a cylinder has circles with a diameter of 6 m.
The rectangle is 10 m tall.

What is the volume?

- A.** 30π cubic meters
- B.** 90π cubic meters
- C.** 360π cubic meters
- D.** 60π cubic meters

A net of a cylinder has two circles with a radius of 2 ft and a rectangle that is 7 ft tall.

What is the volume? Use 3.14 for π .

Volume: _____

True or false?

In the net of a cylinder, the height of the rectangle is the height of the cylinder.

- A. True
- B. False

Use nets to find surface area

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area of triangles	6.G.A.1
BACKWARD 3	Adding the areas of parts	3.MD.C.7.d
FORWARD 1	Surface area of prisms in real-world problems	7.G.B.6
FORWARD 2	Pythagorean Theorem in three dimensions	8.G.B.7

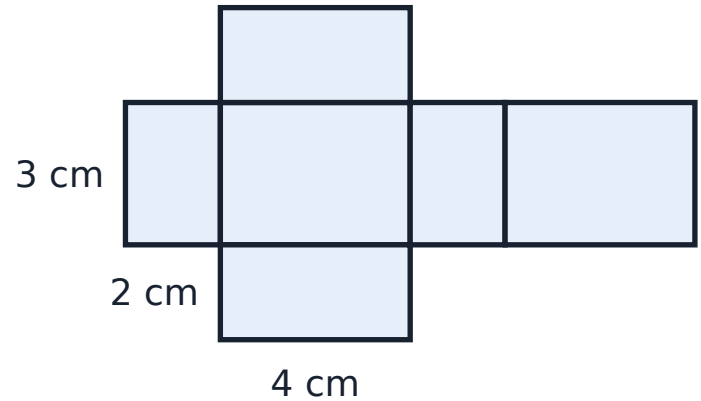
SET 66 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

The net folds into a rectangular prism.

What is the surface area of the prism?

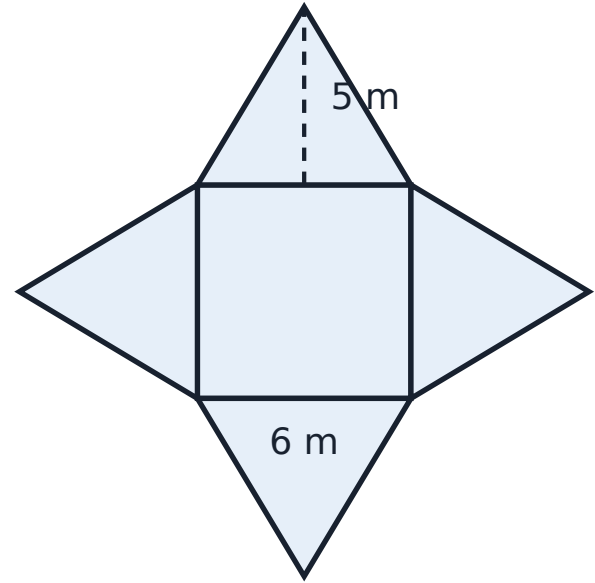


Surface area: _____

The net folds into a square pyramid.

What is the surface area of the pyramid?

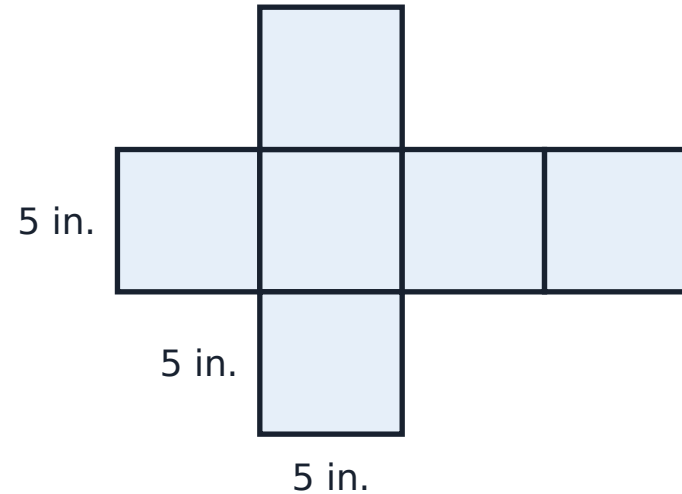
Surface area: _____



The net folds into a cube.

What is the surface area of the cube?

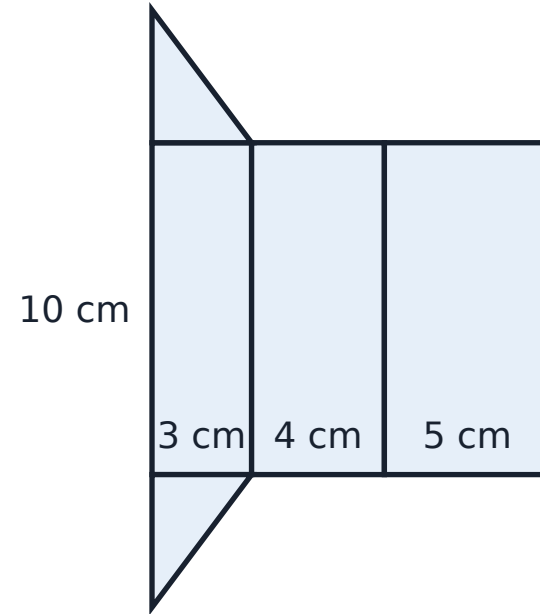
Surface area: _____



The net folds into a triangular prism.

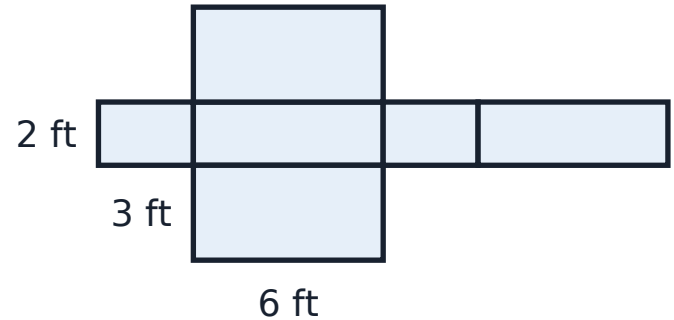
What is the surface area of the prism?

Surface area: _____



The net folds into a rectangular prism.

What is the surface area of the prism?



Surface area: _____

SET 66 • 6.G.A.4

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



A rectangle is 5 m by 7 m.

What is its area?

Area: _____

True or false?

A rectangle that is 3 units by 6 units has an area of 18 square units.

- A. True
- B. False

A rectangle is 6 m by 2 m.

What is its area?

- A.** 8 square meters
- B.** 12 square meters
- C.** 16 square meters
- D.** 24 square meters

A square face has 5-inch edges.

What is its area?

Area: _____

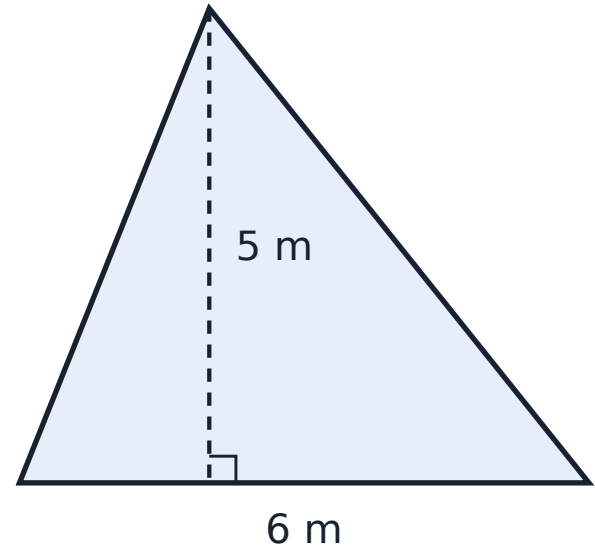
SET 66 • 6.G.A.4

BACKWARD 2

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



A triangle has a base of 8 cm and a height of 3 cm.

What is its area?

Area: _____

True or false?

A triangle with a base of 10 and a height of 4 has an area of 40 square units.

- A. True
- B. False

A right triangle has legs of 3 cm and 4 cm.

What is its area?

- A.** 6 square cm
- B.** 7 square cm
- C.** 10 square cm
- D.** 12 square cm

Four triangles each have a base of 6 m and a height of 5 m.

What is their total area?

Total area: _____

SET 66 • 6.G.A.4

BACKWARD 3

Adding the areas of parts • Grade 3 • Measurement & Data • 3.MD.C.7.d

A figure is made of a 4-by-3 rectangle and a 4-by-2 rectangle that do not overlap.

What is its total area?

Area: _____

True or false?

A figure made of two 5-by-5 squares has an area of 50 square units.

- A. True
- B. False

A figure is made of rectangles with areas of 12, 8, and 6 square units.

What is its total area?

A. 14 square units

B. 20 square units

C. 26 square units

D. 576 square units

Six squares each have an area of 9 square inches.
What is their total area?

Area: _____

A figure is made of rectangles that are 2 by 6, 2 by 6,
and 3 by 6.

What is its total area?

Area: _____

SET 66 • 6.G.A.4

FORWARD 1

Surface area of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A cube has edges of 2.5 cm.

What is its surface area?

Surface area: _____

A rectangular prism is 5 in. by 4 in. by 2.5 in.

What is its surface area?

- A.** 42.5 square inches
- B.** 50 square inches
- C.** 65 square inches
- D.** 85 square inches

A gift box is 8 in. by 6 in. by 4 in.

How much wrapping paper covers the box exactly?

Paper: _____

A box with no lid is 10 in. long, 4 in. wide, and 3 in. tall.
How much cardboard is needed to make its 5 faces?

Cardboard: _____

True or false?

The surface area of a prism is the sum of the areas of all of its faces.

A. True

B. False

SET 66 • 6.G.A.4

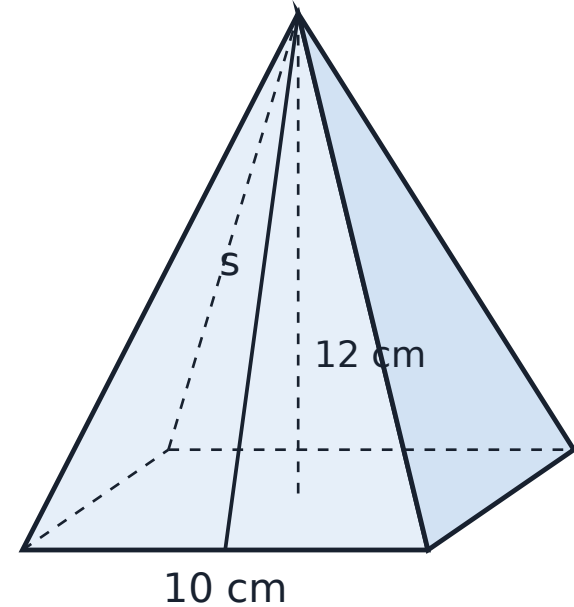
FORWARD 2

Pythagorean Theorem in three dimensions • Grade 8 • Geometry • 8.G.B.7

The square pyramid has a base edge of 10 cm and a height of 12 cm.

What is its slant height s ?

$s =$ _____



A box is 3 in. by 4 in. by 12 in.

How long is the longest diagonal from one corner to the opposite corner?

Length: _____

A square pyramid has a base edge of 16 cm and a slant height of 10 cm.

What is its height?

A. 6 cm

B. 8 cm

C. 12.8 cm

D. 18.9 cm

A cone has a radius of 5 in. and a slant height of 13 in.
What is its height?

Height: _____

True or false?

The bottom of a box is 6 in. by 8 in. The diagonal across the bottom is 10 in.

A. True

B. False

Recognize statistical questions

MAIN Grade 6 main question

BACKWARD 1 Reading a bar graph

BACKWARD 2 Reading a line plot

FORWARD 1 Random samples and representative samples

FORWARD 2 Statistical questions about two variables and scatter plots

6.SP.A.1

3.MD.B.3

4.MD.B.4

7.SP.A.1

8.SP.A.1

SET 67 • 6.SP.A.1

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.1

Which question is a statistical question?

- A.** How many hours did I sleep last night?
- B.** How many days are in a week?
- C.** What time does our school start?
- D.** How many hours of sleep do students in my class get each night?

True or false?

"How tall is the flagpole at our school?" is a statistical question.

A. True

B. False

Which question is NOT a statistical question?

- A.** How much do the dogs at the shelter weigh?
- B.** What is the population of our city right now?
- C.** How long do students spend on homework each day?
- D.** How many pets do the students in grade 6 have?

True or false?

"What are the ages of the teachers at our school?" is a statistical question.

A. True

B. False

Which question is a statistical question?

- A.** How many minutes do the players on our team practice each week?
- B.** How many minutes are in an hour?
- C.** How many minutes did Coach Lee practice today?
- D.** How many players are on a soccer field at once?

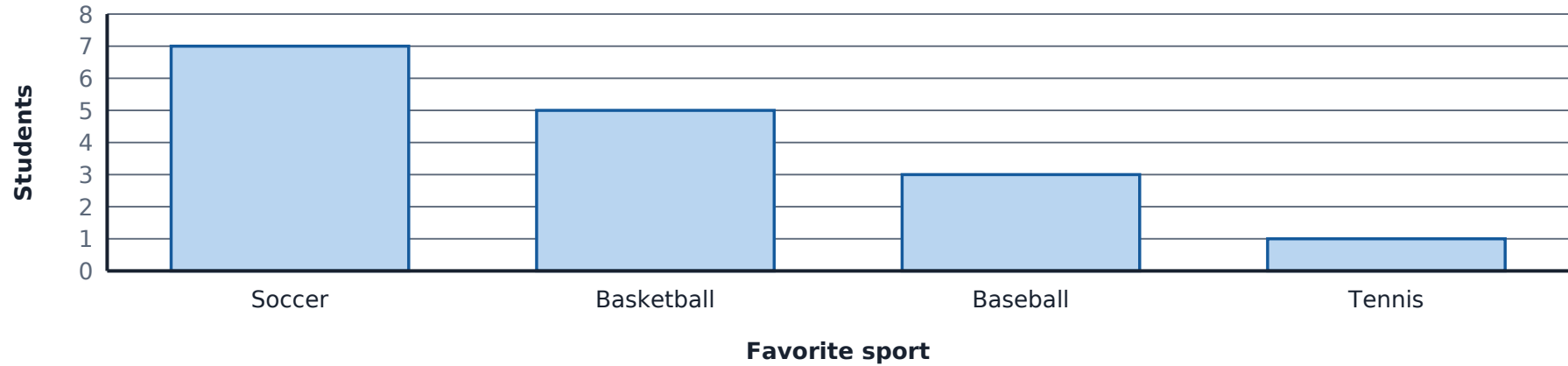
SET 67 • 6.SP.A.1

BACKWARD 1

Reading a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

The graph shows the favorite sports of a class.

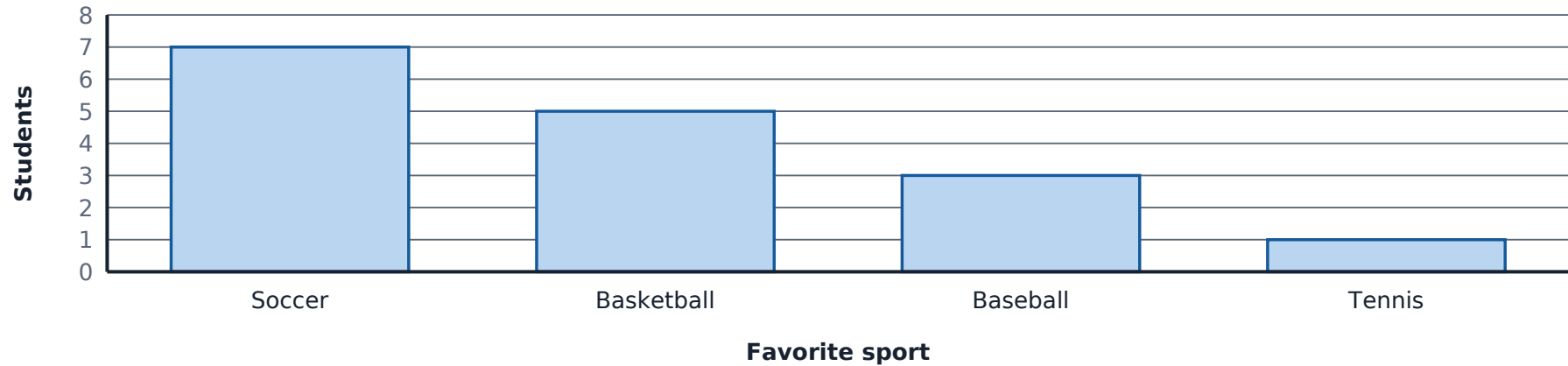
How many students chose basketball?



Students: _____

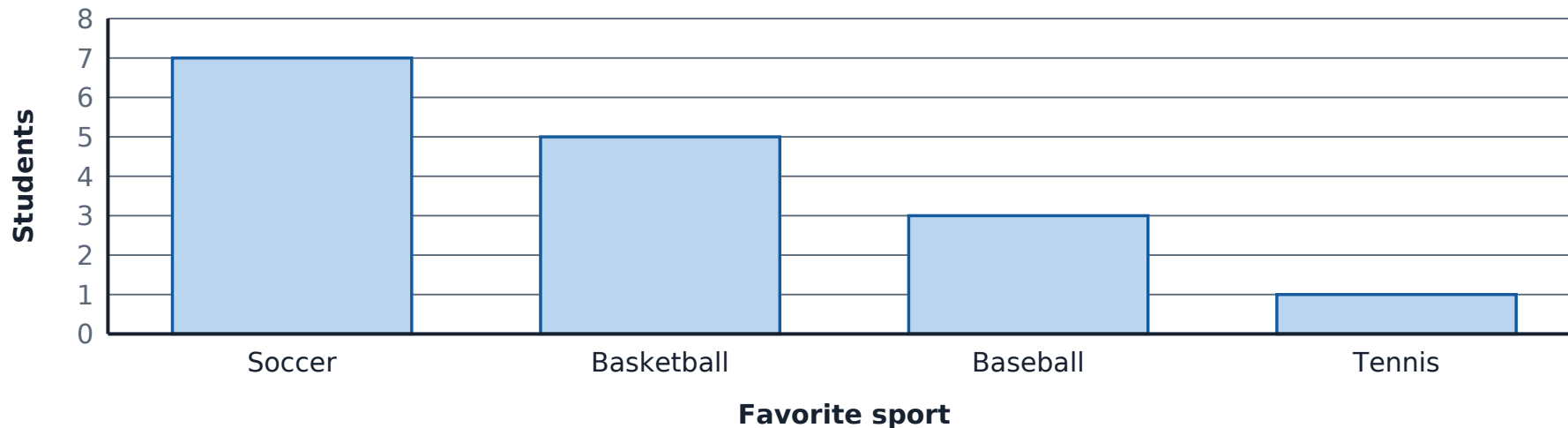
The graph shows the favorite sports of a class.

How many more students chose soccer than baseball?



Students: _____

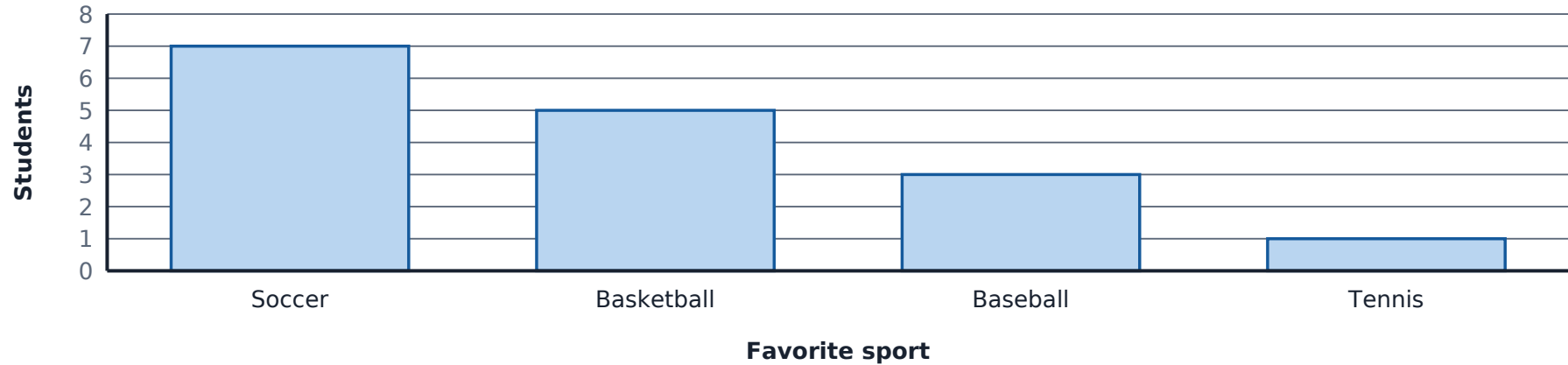
Which sport did 3 students choose?



- A.** Baseball **B.** Tennis **C.** Soccer **D.** Basketball

The graph shows the favorite sports of a class.

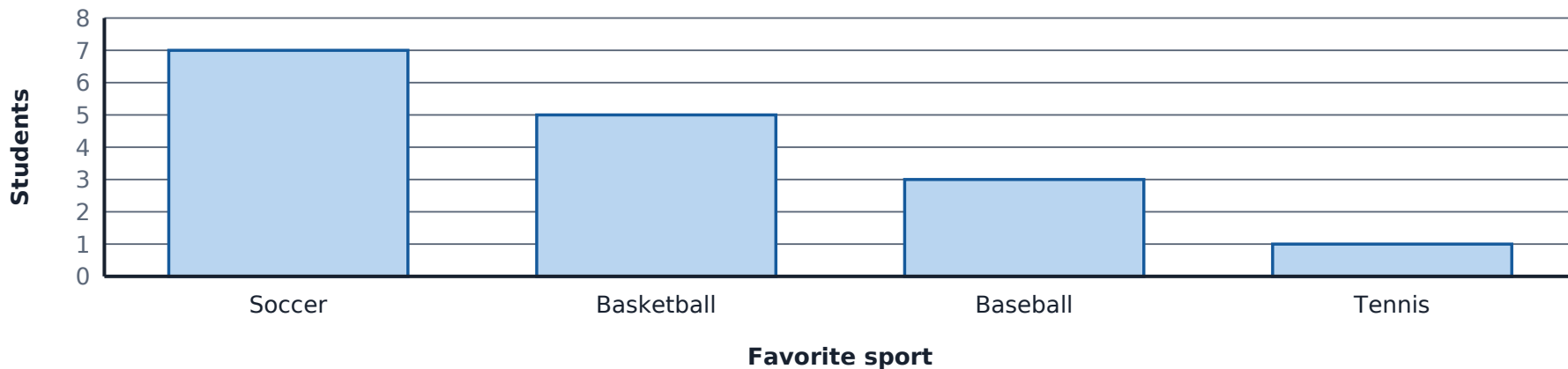
How many students answered in all?



Students: _____

True or false?

More students chose basketball than soccer.



A. True

B. False

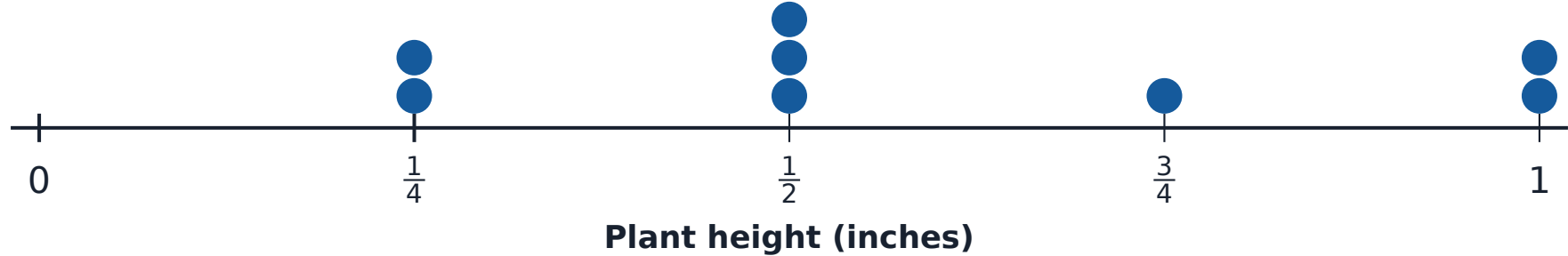
SET 67 • 6.SP.A.1

BACKWARD 2

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the heights of plants.

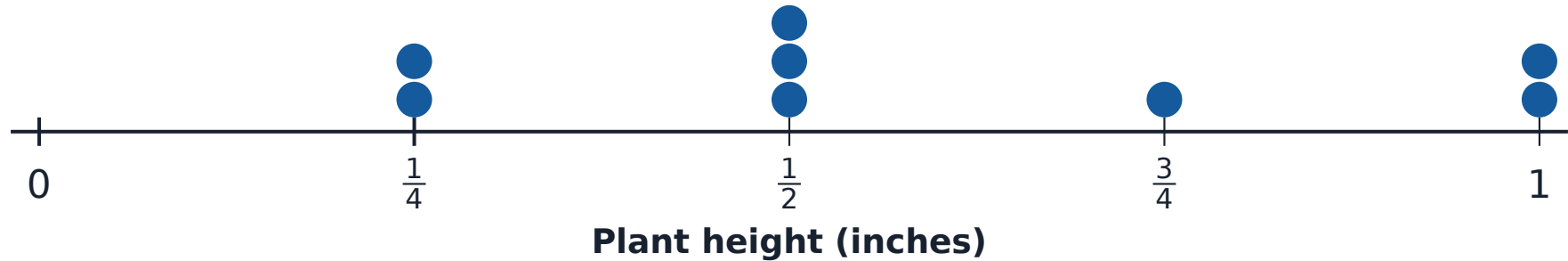
How many plants are $\frac{1}{2}$ inch tall?



Plants: _____

The line plot shows the heights of plants.

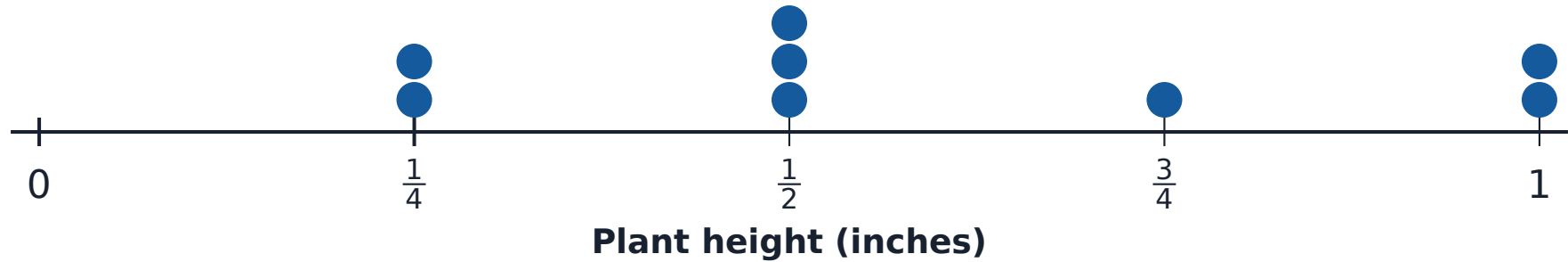
How many plants were measured?



Plants: _____

True or false?

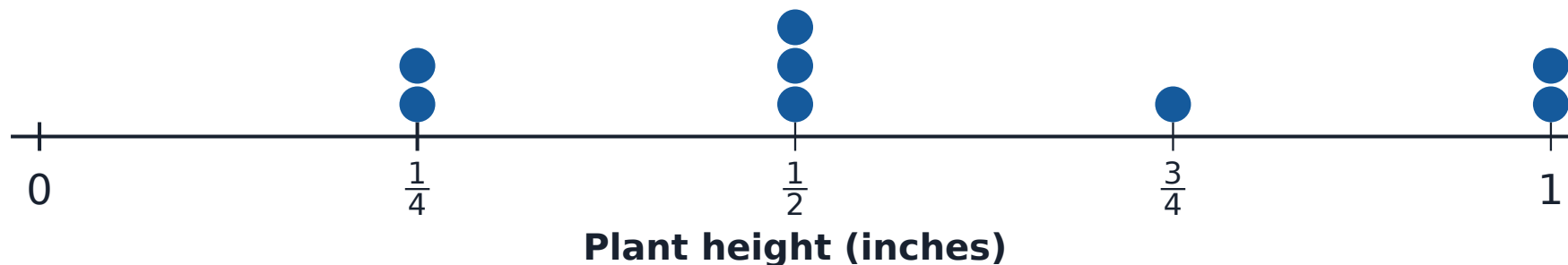
The most common plant height is $\frac{3}{4}$ inch.



A. True

B. False

Which height appears least often?



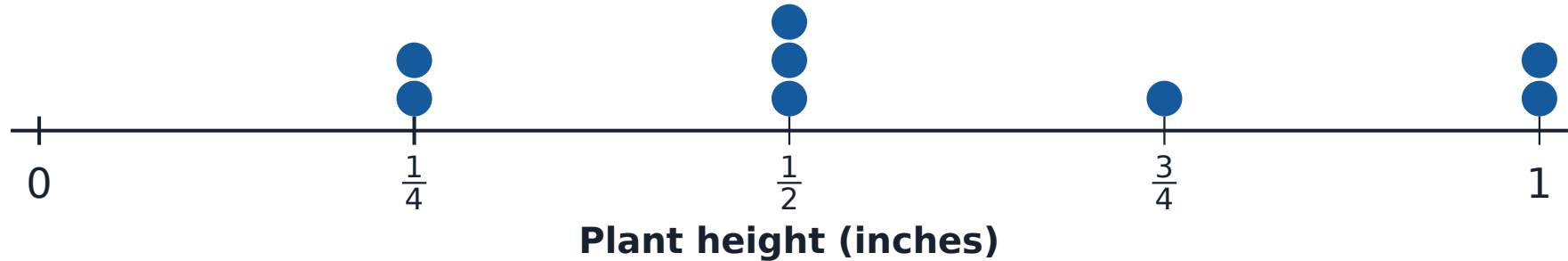
A. $\frac{1}{4}$ inch

B. $\frac{1}{2}$ inch

C. $\frac{3}{4}$ inch

D. 1 inch

What is the difference between the tallest and the shortest plant heights?



Difference: _____

SET 67 • 6.SP.A.1

FORWARD 1

Random samples and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know the favorite lunch of all 600 students.

Which sample is most likely to represent all students?

- A.** The students on the soccer team
- B.** 50 students chosen at random from the whole school
- C.** The 30 students in one class
- D.** The first 50 students in line on pizza day

True or false?

A random sample is more likely to represent a population than a sample of your friends.

A. True

B. False

A reporter asks only basketball players, "What is your favorite sport?"

Why is this sample biased?

Reason: _____

A random sample of 40 students out of 800 shows that 10 walk to school.

About how many of the 800 students walk to school?

A. 10

B. 40

C. 80

D. 200

True or false?

Conclusions about a population from a sample are valid only if the sample represents the population.

- A. True
- B. False

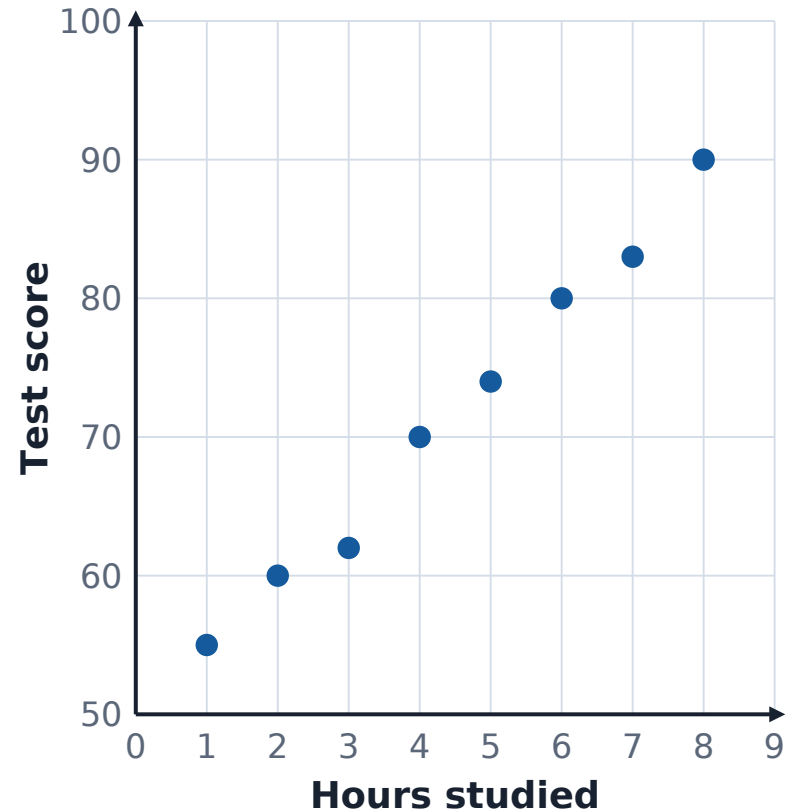
SET 67 • 6.SP.A.1

FORWARD 2

Statistical questions about two variables and scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

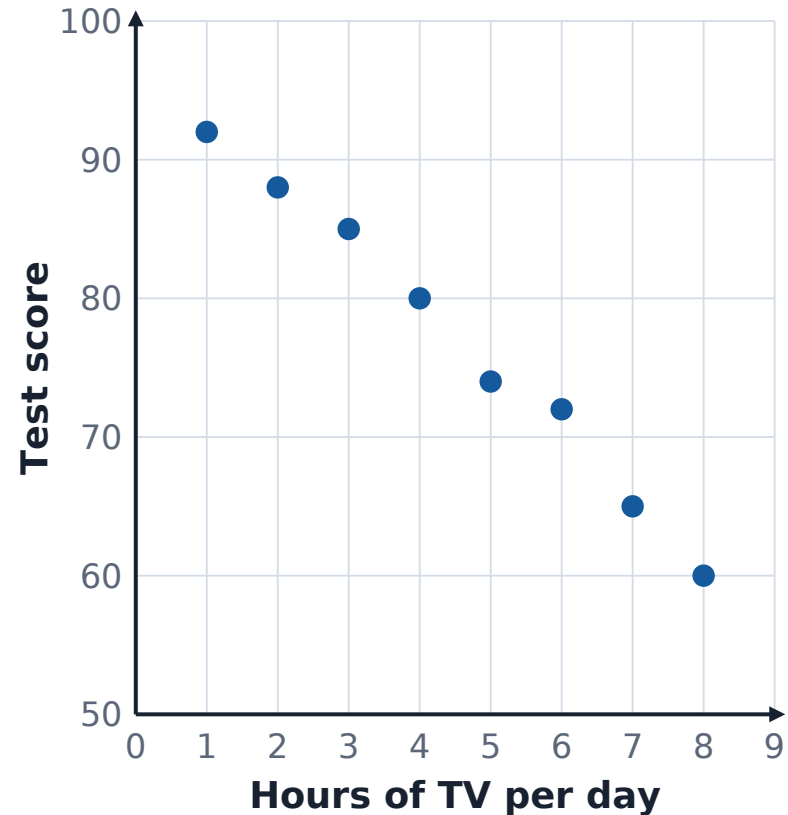
What type of association does the scatter plot show?

- A.** No association
- B.** Nonlinear
- C.** Positive linear
- D.** Negative linear



What type of association does the scatter plot show?

- A.** Negative linear
- B.** Positive linear
- C.** No association
- D.** Nonlinear



True or false?

A scatter plot shows data about two variables for each person or object.

- A.** True
- B.** False

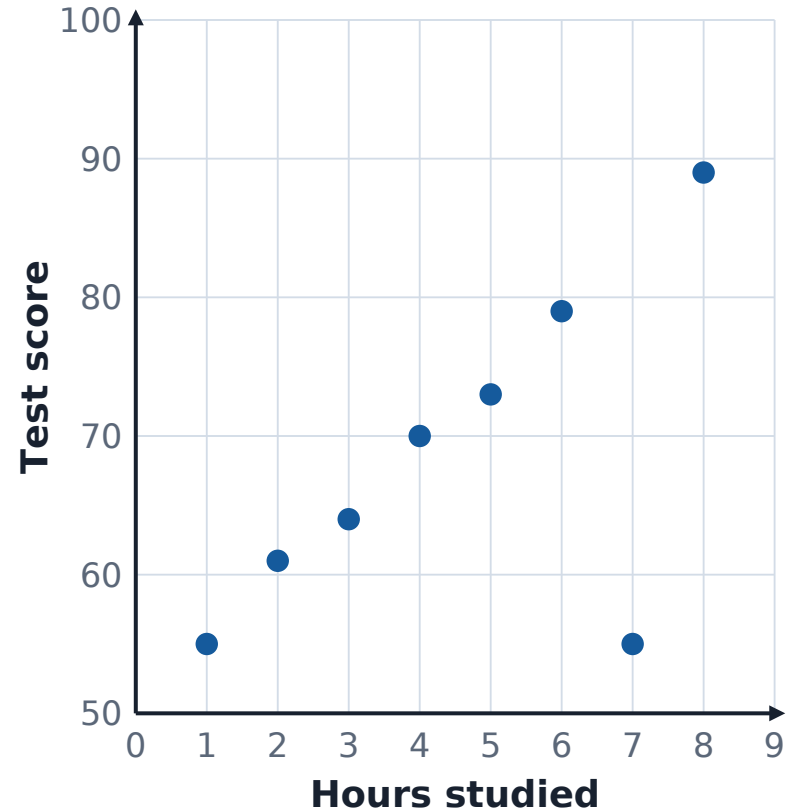
Which question would be best answered with a scatter plot?

- A.** How many students ride the bus?
- B.** What is the most common eye color?
- C.** Is there a relationship between height and arm span?
- D.** What is the median age in the class?

Which point is an outlier?

Write its coordinates.

Outlier: _____



Describe the overall shape of a distribution

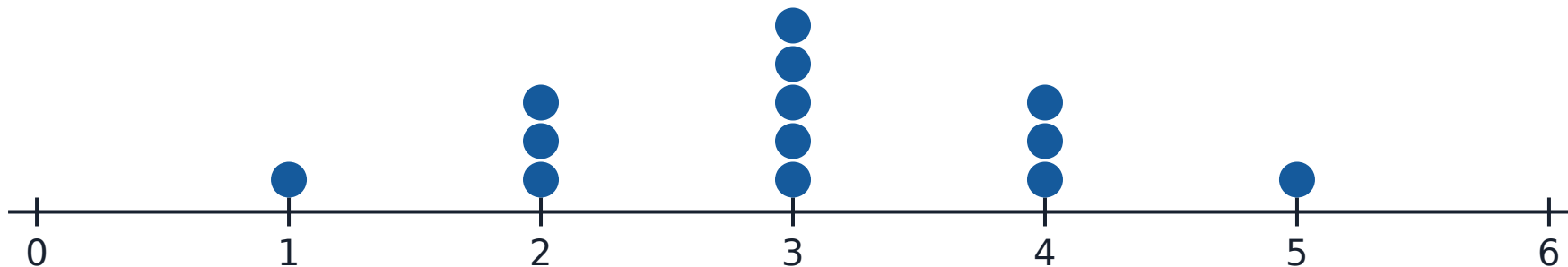
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Most and least in a bar graph	3.MD.B.3
FORWARD 1	Compare two distributions visually	7.SP.B.3
FORWARD 2	Linear and nonlinear patterns in scatter plots	8.SP.A.1

SET 68 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word best describes the shape of the data?



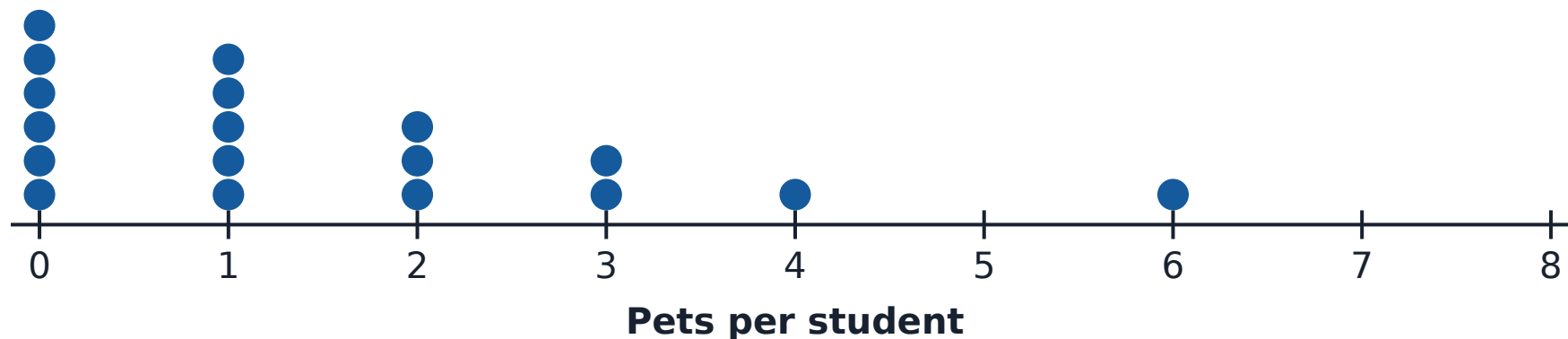
A. Uniform

B. Symmetric

C. Skewed left

D. Skewed right

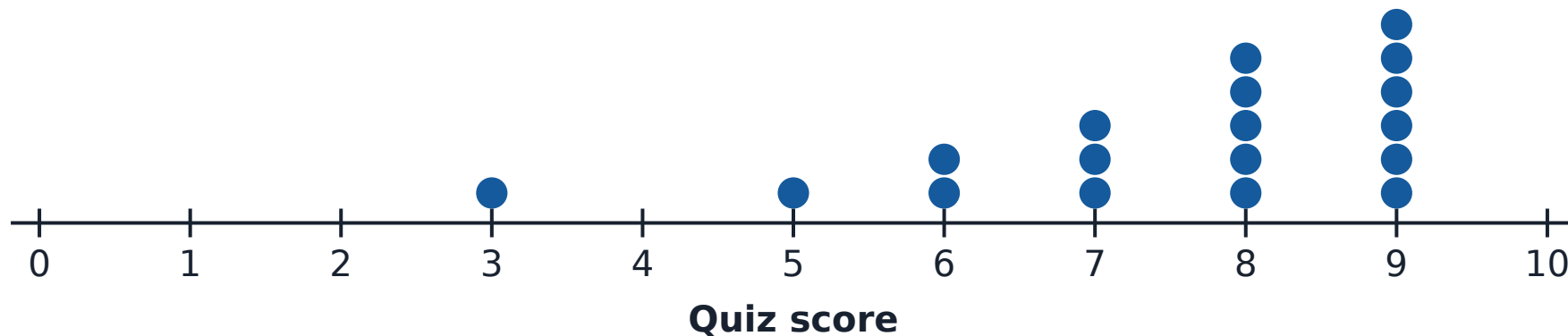
Which word best describes the shape of the data?



- A.** Skewed left
- C.** Skewed right

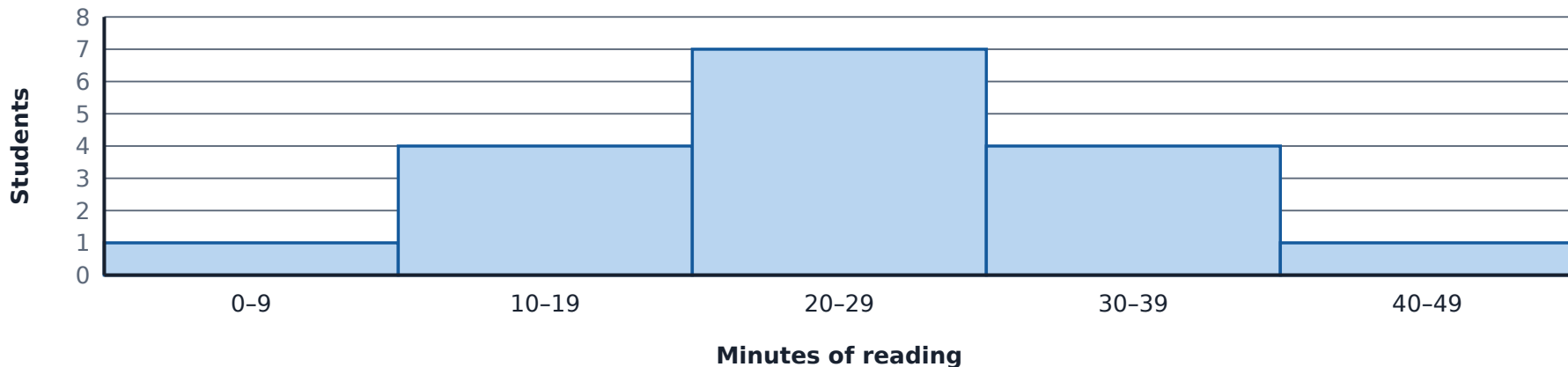
- B.** Symmetric
- D.** Uniform

Which word best describes the shape of the data?



- A.** Symmetric
- B.** Skewed left
- C.** Skewed right
- D.** Uniform

Which word best describes the shape of the data in the histogram?



A. Skewed right

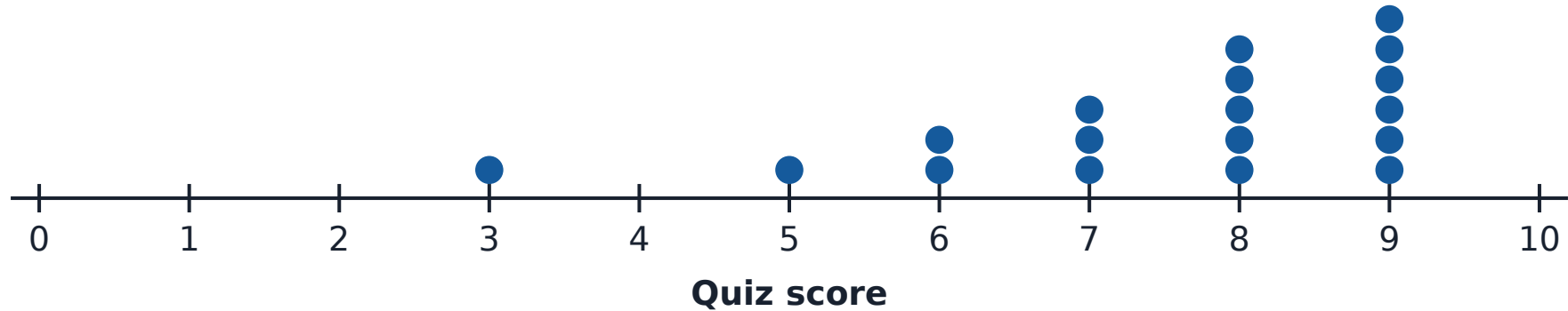
B. Skewed left

C. Uniform

D. Symmetric

True or false?

The data are skewed right.



A. True

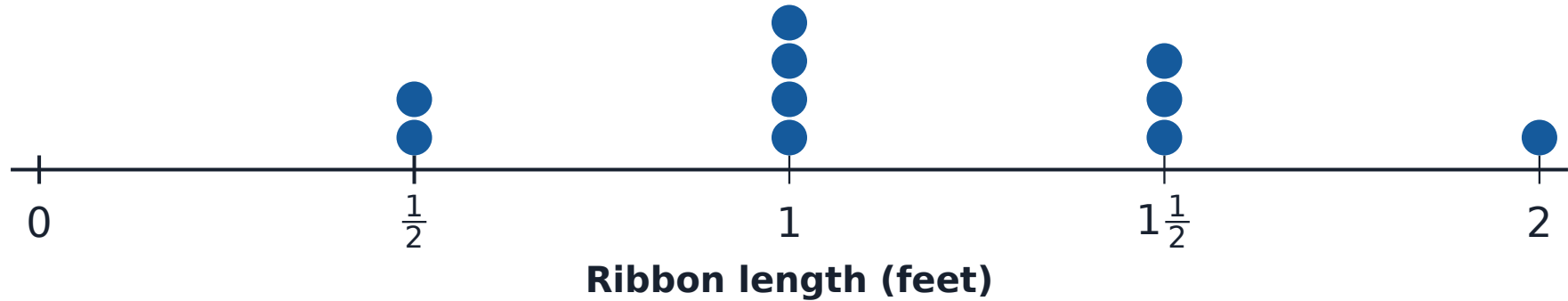
B. False

SET 68 • 6.SP.A.2

BACKWARD 1

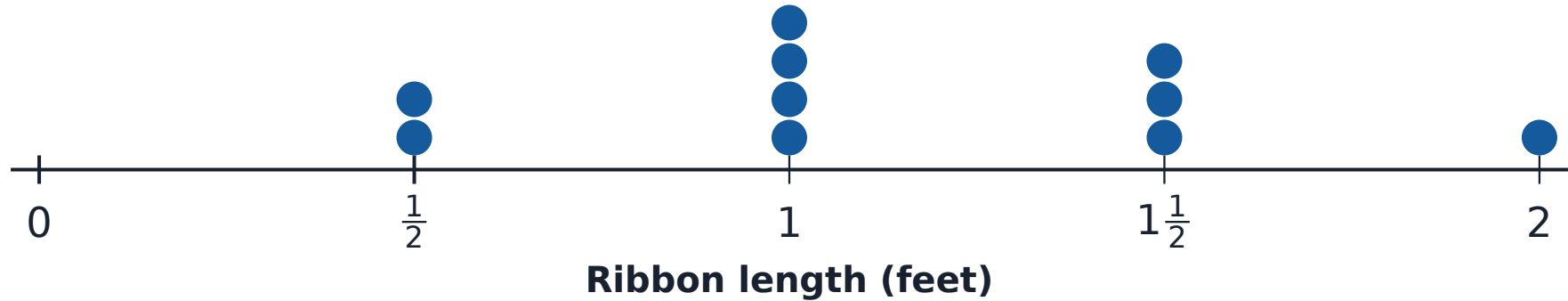
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many ribbons are 1 foot long?



Ribbons: _____

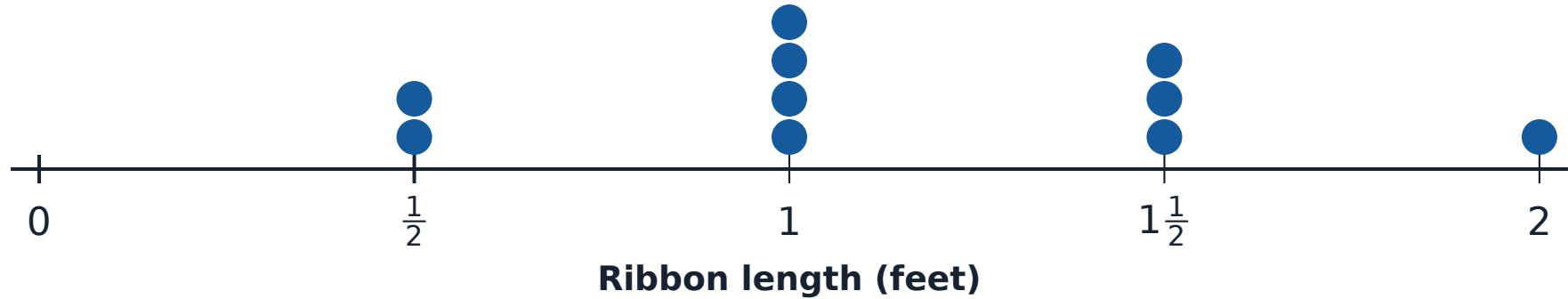
How many ribbons are there in all?



Ribbons: _____

True or false?

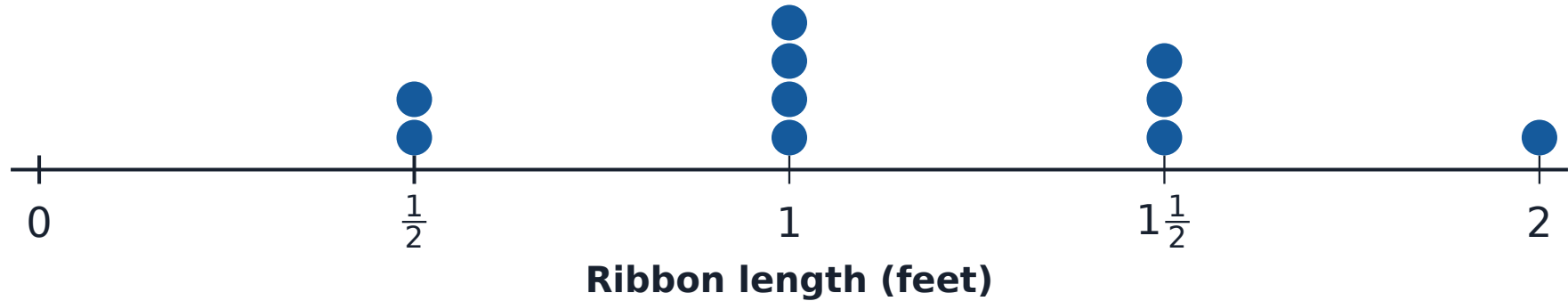
Two ribbons are $\frac{1}{2}$ foot long.



A. True

B. False

Which ribbon length is most common?



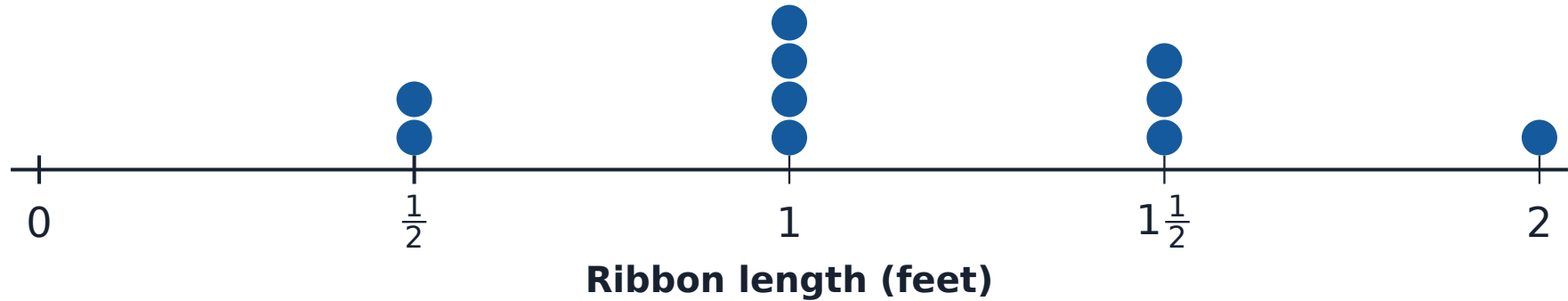
A. $\frac{1}{2}$ foot

B. 1 foot

C. $1\frac{1}{2}$ feet

D. 2 feet

How many ribbons are longer than 1 foot?



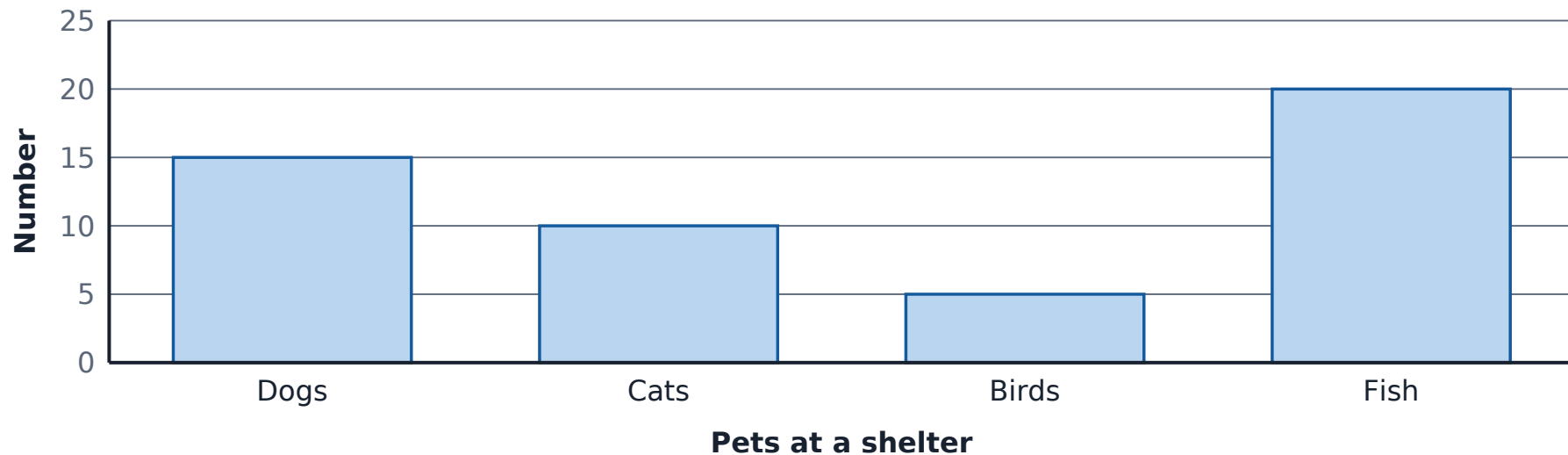
Ribbons: _____

SET 68 • 6.SP.A.2

BACKWARD 2

Most and least in a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

Which pet does the shelter have the most of?



A. Birds

B. Fish

C. Dogs

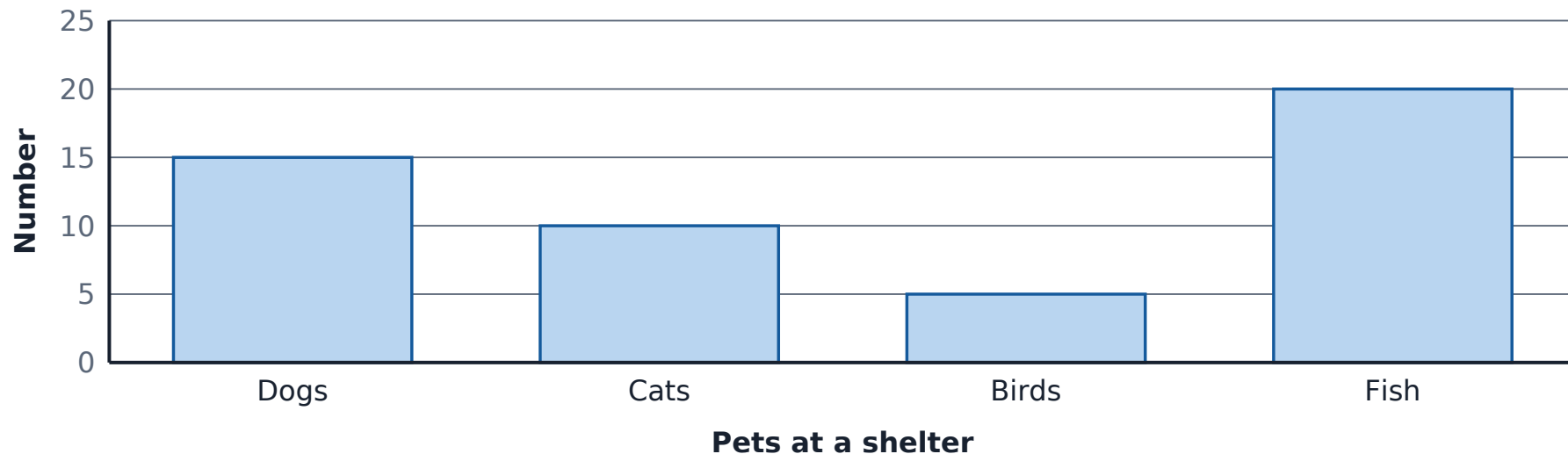
D. Cats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q1/5

Which pet does the shelter have the fewest of?



A. Dogs

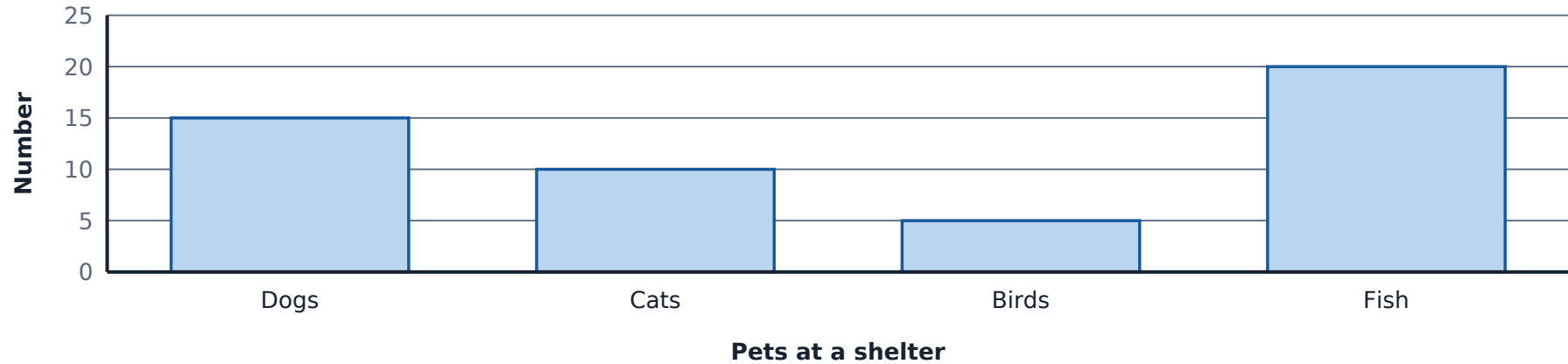
B. Birds

C. Cats

D. Fish

True or false?

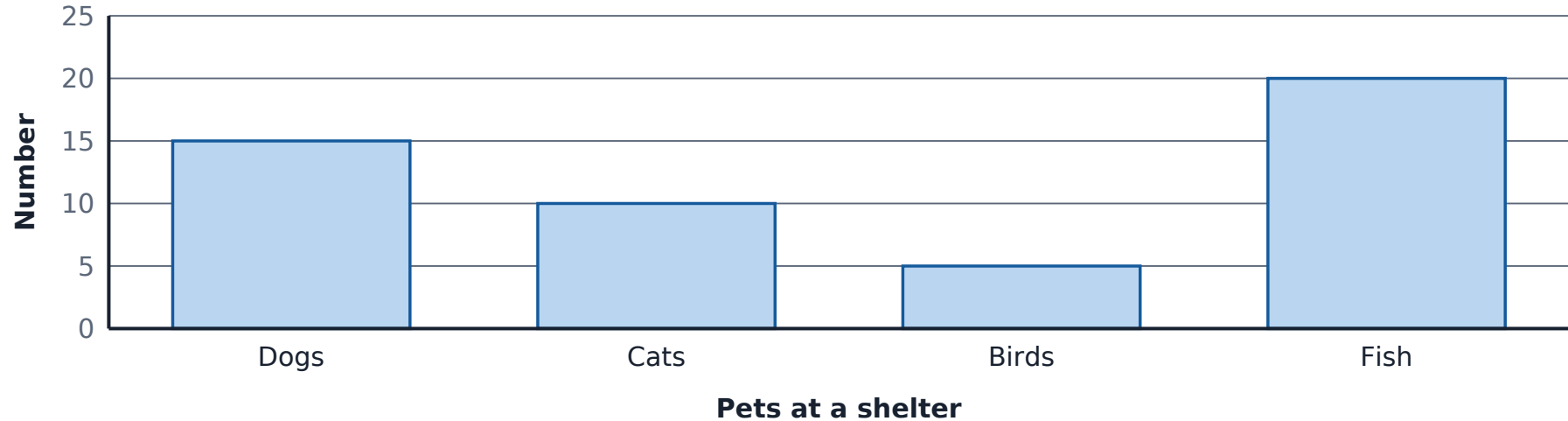
The shelter has more dogs than cats.



A. True

B. False

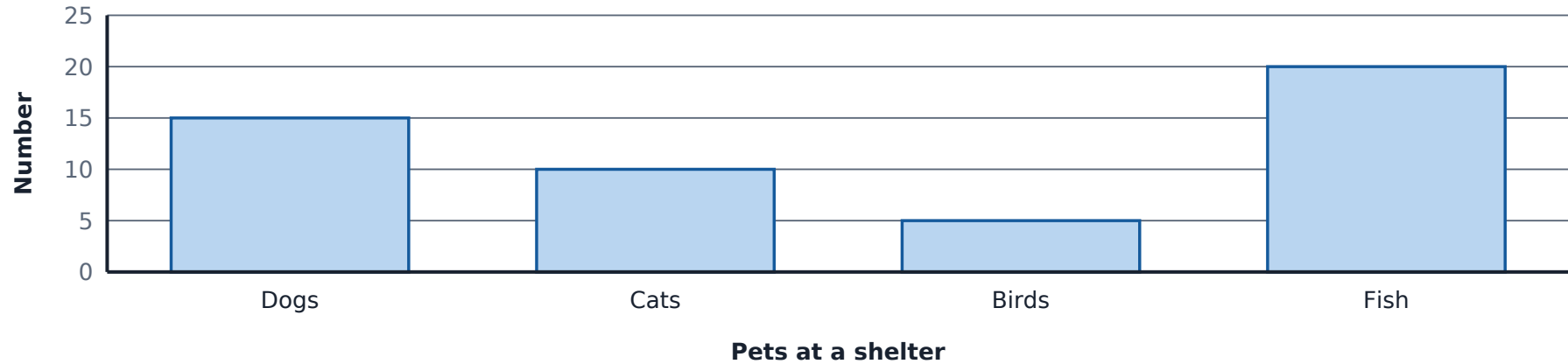
How many more fish than birds are at the shelter?



Number: _____

True or false?

The tallest bar shows the smallest number.



A. True

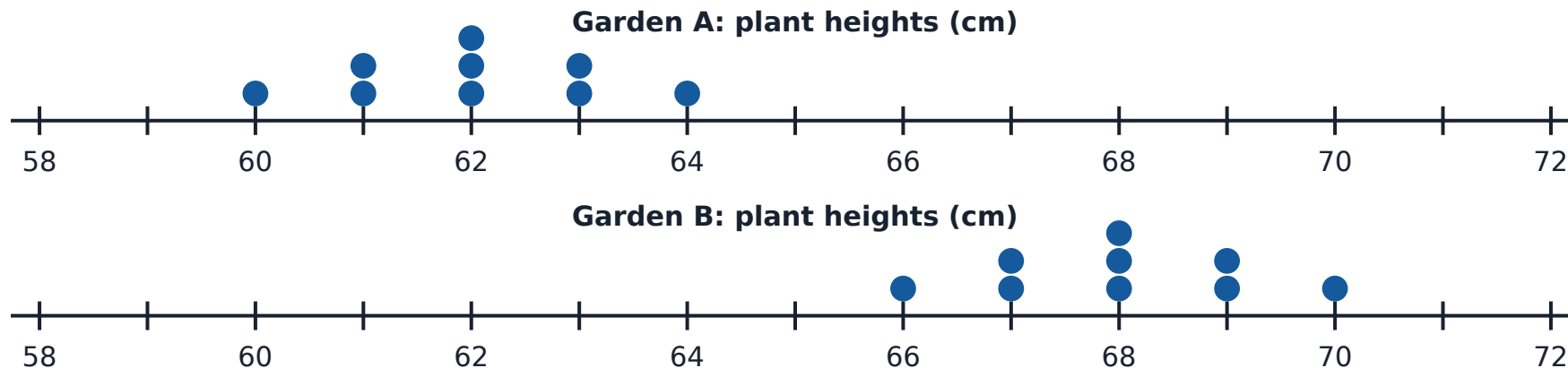
B. False

SET 68 • 6.SP.A.2

FORWARD 1

Compare two distributions visually • Grade 7 • Statistics & Probability •
7.SP.B.3

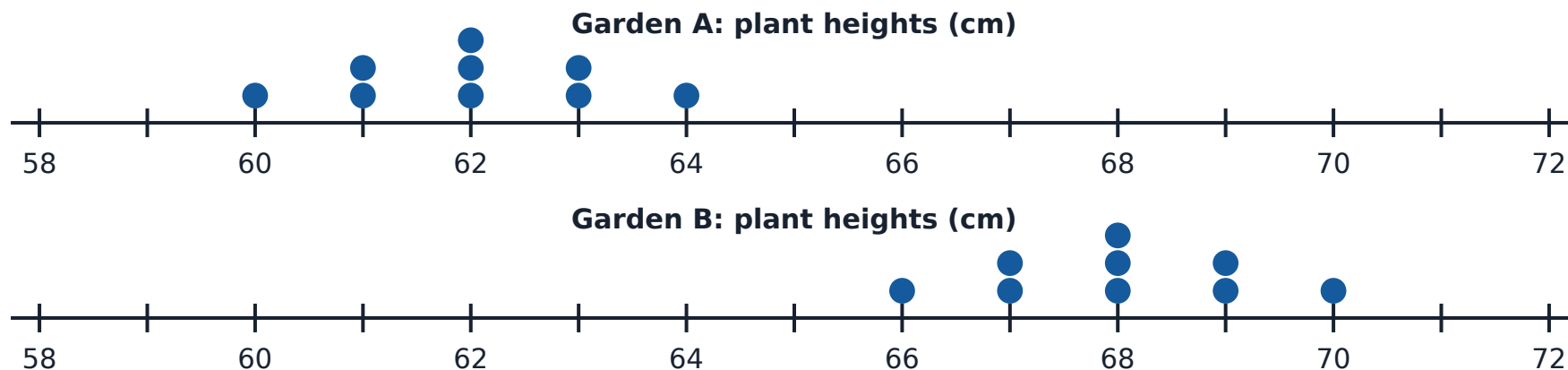
About how many centimeters greater is the center of Garden B's plant heights?



Centimeters: _____

True or false?

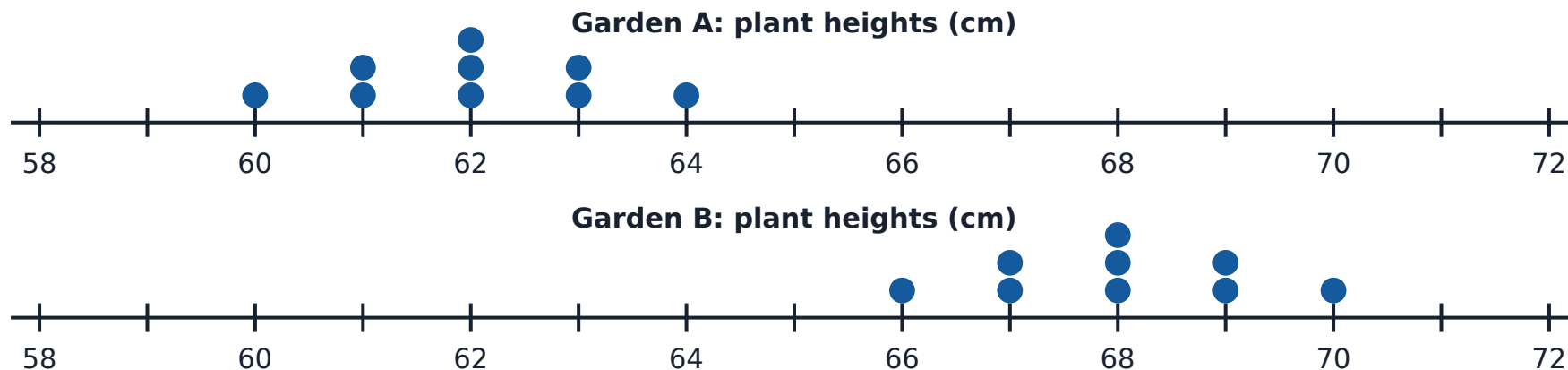
The two distributions have about the same shape and spread.



A. True

B. False

Which statement is true?



- A.** Garden B has more variability. **B.** Garden A's plants are typically taller.
- C.** The distributions do not overlap. **D.** The centers are equal.

Garden A's mean height is 62 cm and Garden B's is 68 cm. Each has a MAD of about 1 cm.

The difference in means is about how many times the MAD?

Times: _____

Two classes have mean scores of 78 and 84. Each has a MAD of 6.

The difference in means is how many times the MAD?

A. 0.5

B. 1

C. 2

D. 6

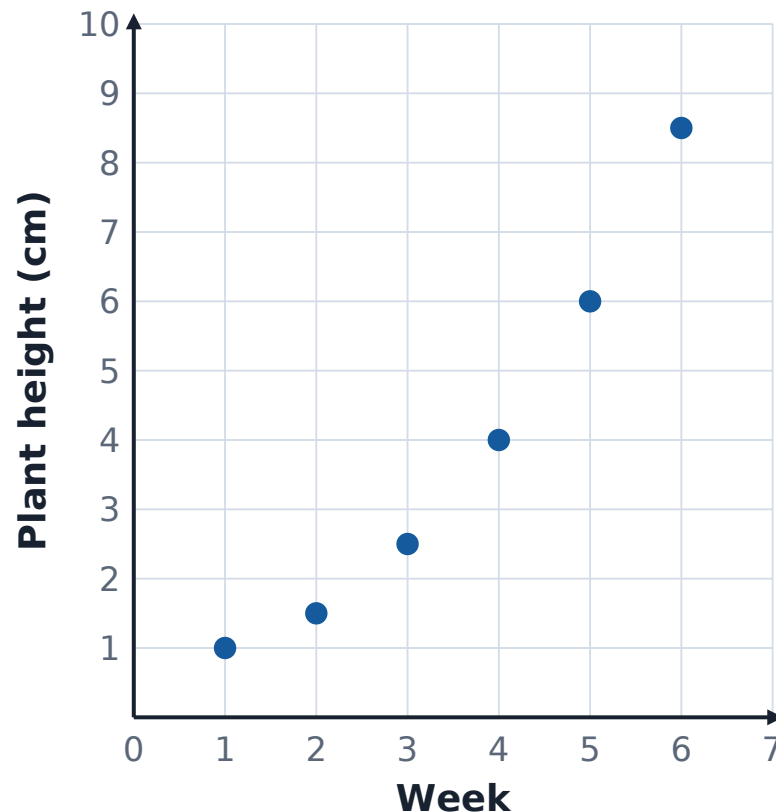
SET 68 • 6.SP.A.2

FORWARD 2

Linear and nonlinear patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

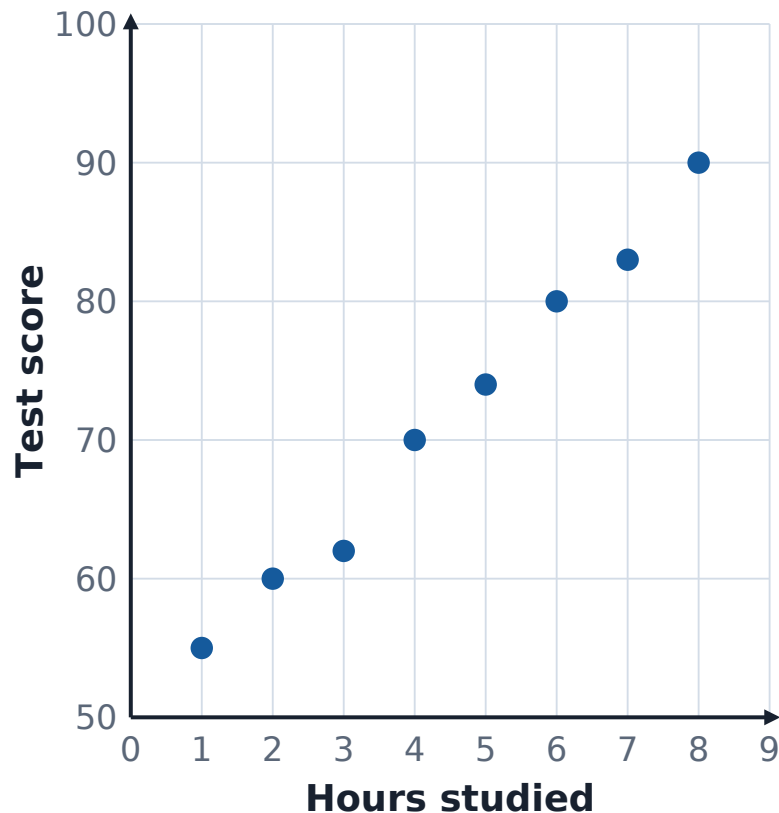
Which describes the pattern in the scatter plot?

- A.** Nonlinear
- B.** No association
- C.** Negative linear
- D.** Constant



Is the pattern in the scatter plot linear or nonlinear?

- A.** Linear
- B.** Nonlinear

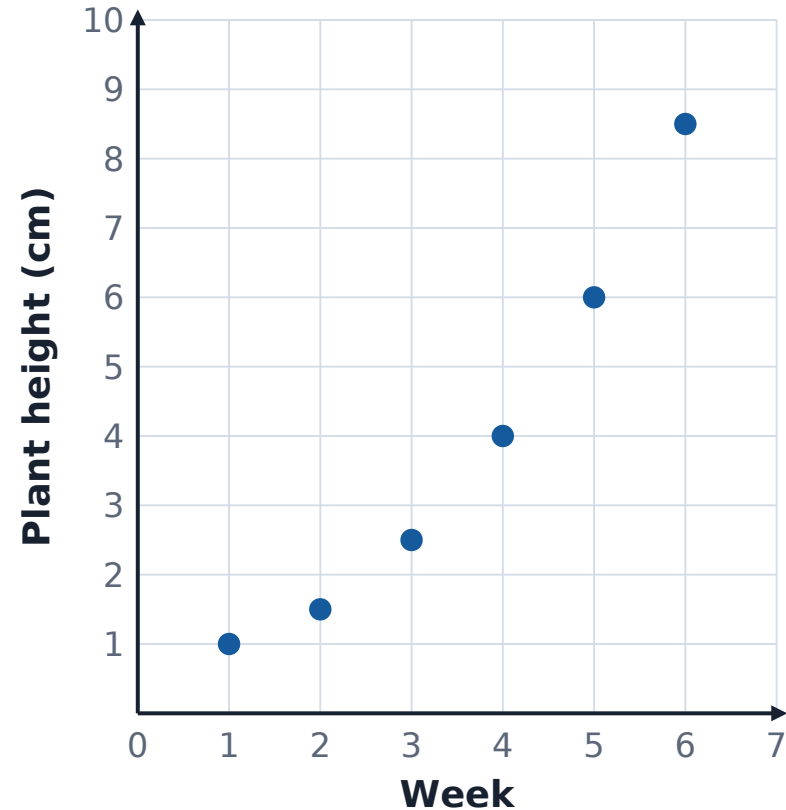


True or false?

The pattern in the scatter plot is linear.

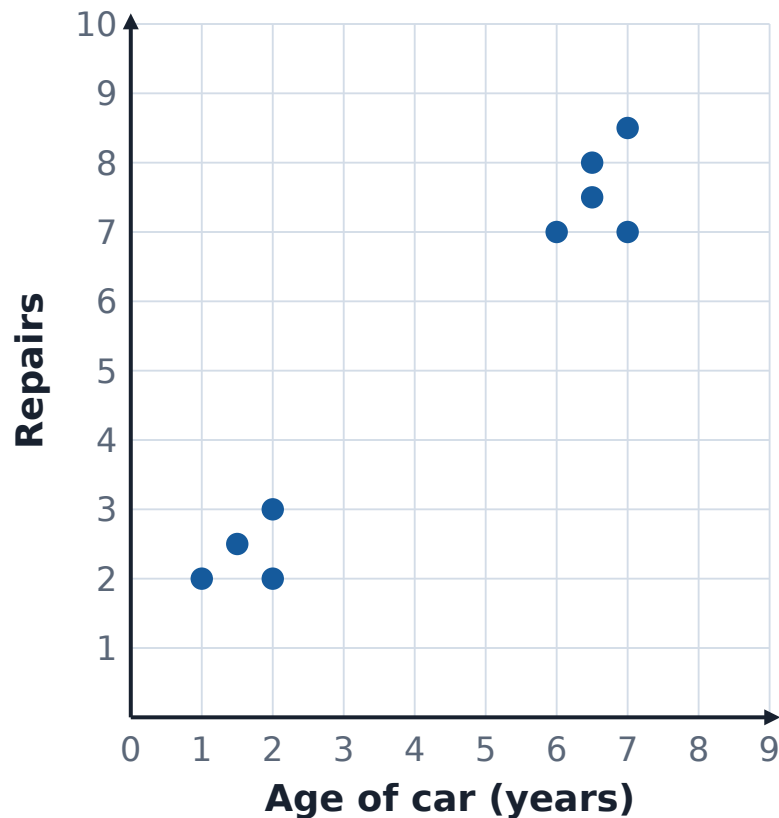
A. True

B. False



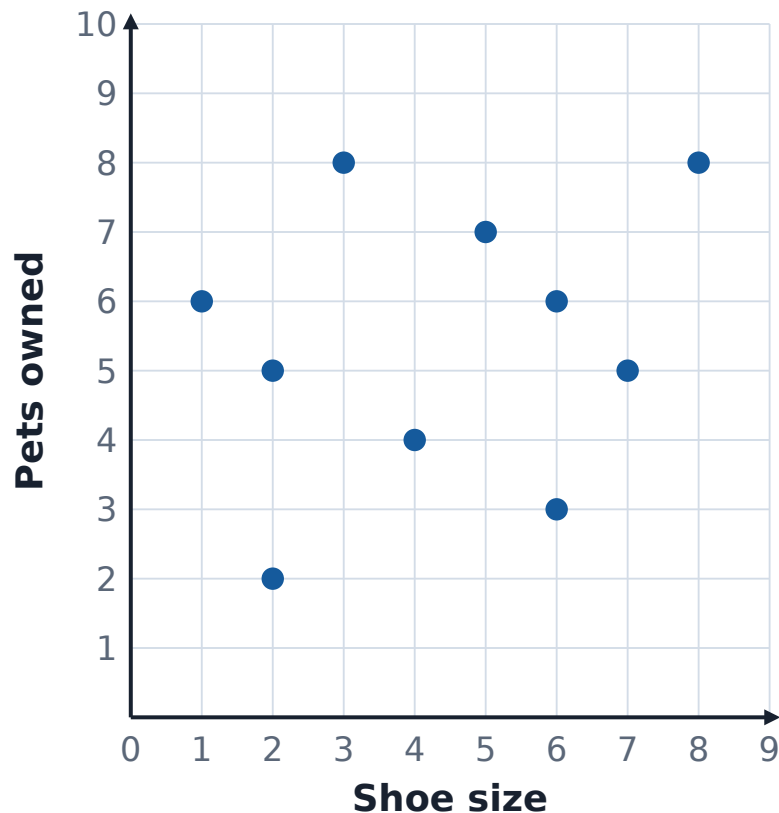
How many clusters does the scatter plot show?

Clusters: _____



Which describes the scatter plot?

- A.** Positive linear
- B.** Nonlinear
- C.** Negative linear
- D.** No association



Describe the center of a distribution

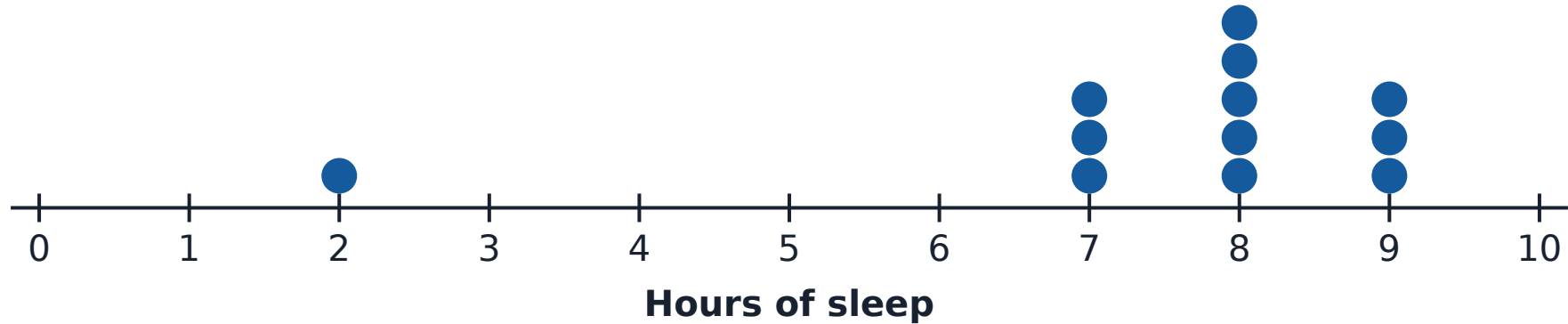
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Ordering numbers	2.NBT.A.4
BACKWARD 3	Halfway between two numbers	2.MD.B.6
FORWARD 1	Compare the centers of two populations	7.SP.B.4
FORWARD 2	Use a line of best fit to make predictions — nearest related (no di...	8.SP.A.2

SET 69 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

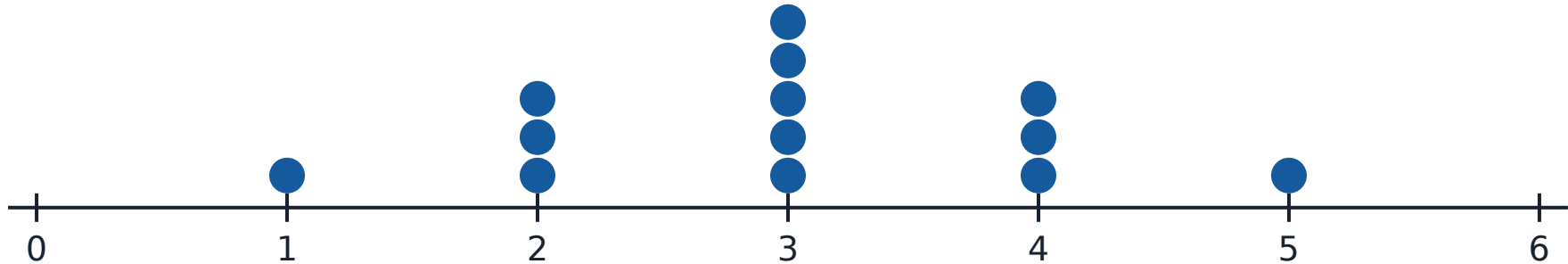
Around what value do most of the data cluster?



Value: _____

The data are symmetric.

What value is at the center of the distribution?



Value: _____

In which interval is the center of the data?



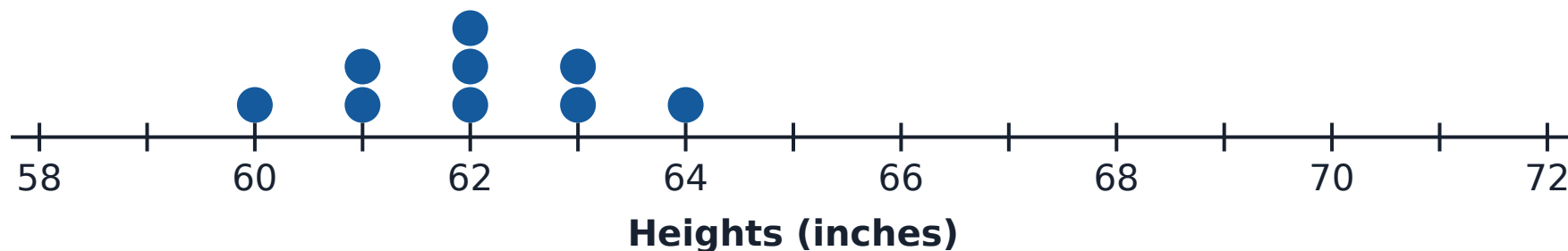
- A.** 0-9 **B.** 10-19 **C.** 20-29 **D.** 30-39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q3/5

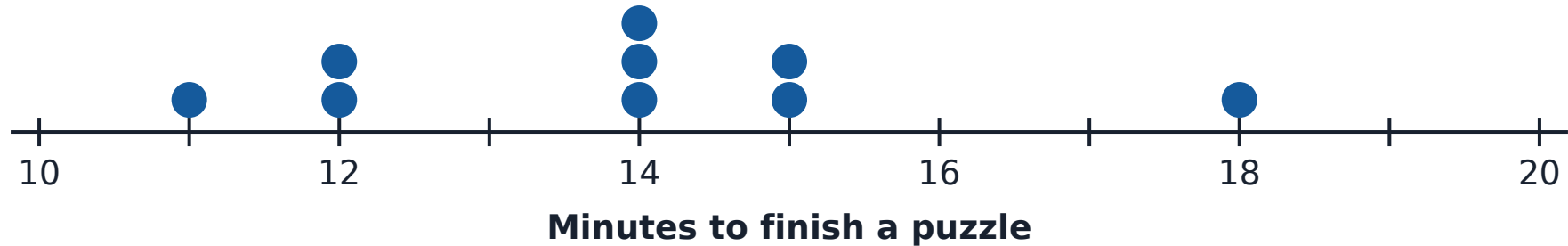
What value is at the center of the heights?



Value: _____

True or false?

The center of the distribution is about 12.



A. True

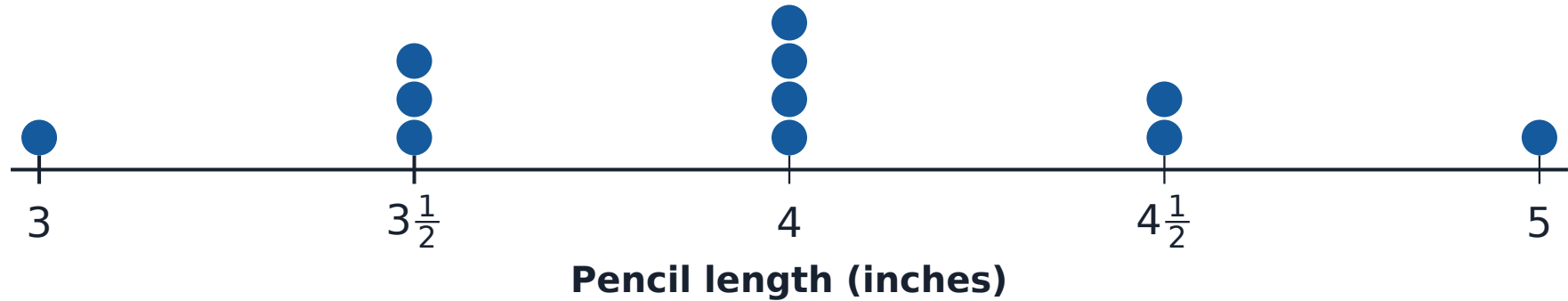
B. False

SET 69 • 6.SP.A.2

BACKWARD 1

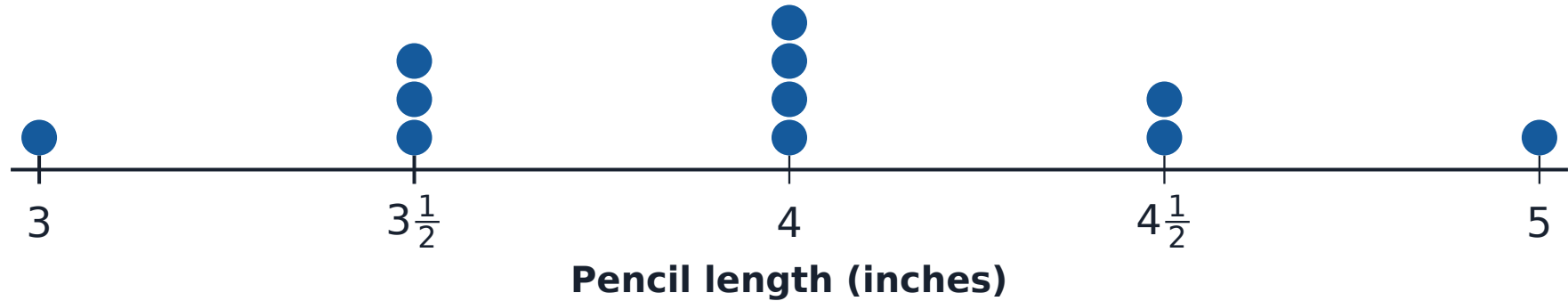
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many pencils are 4 inches long?



Pencils: _____

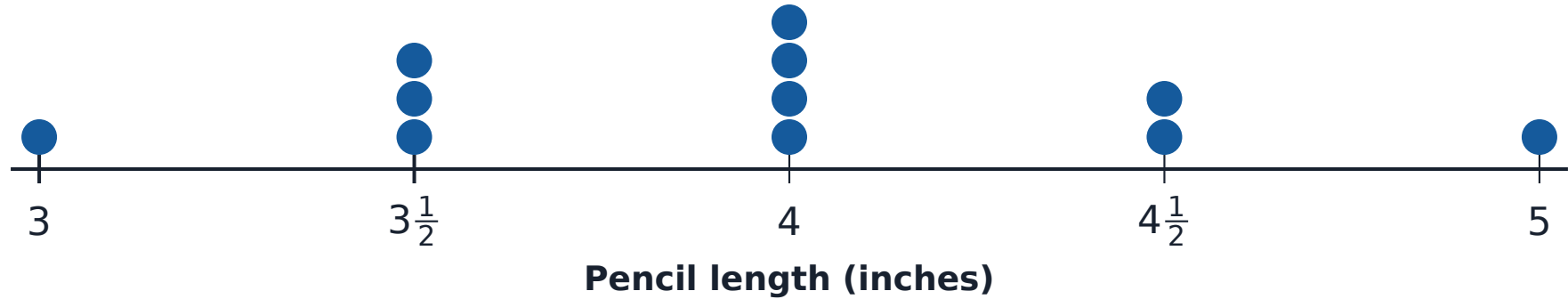
How many pencils were measured?



Pencils: _____

True or false?

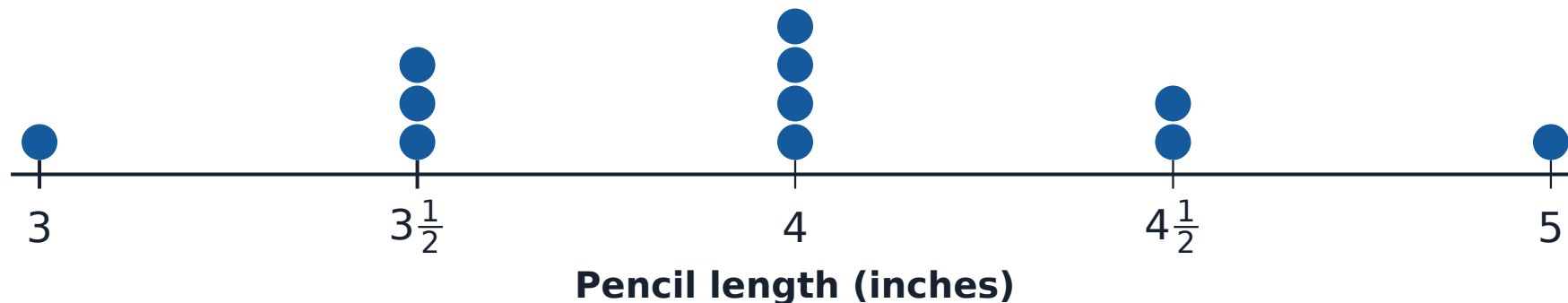
Three pencils are $3\frac{1}{2}$ inches long.



A. True

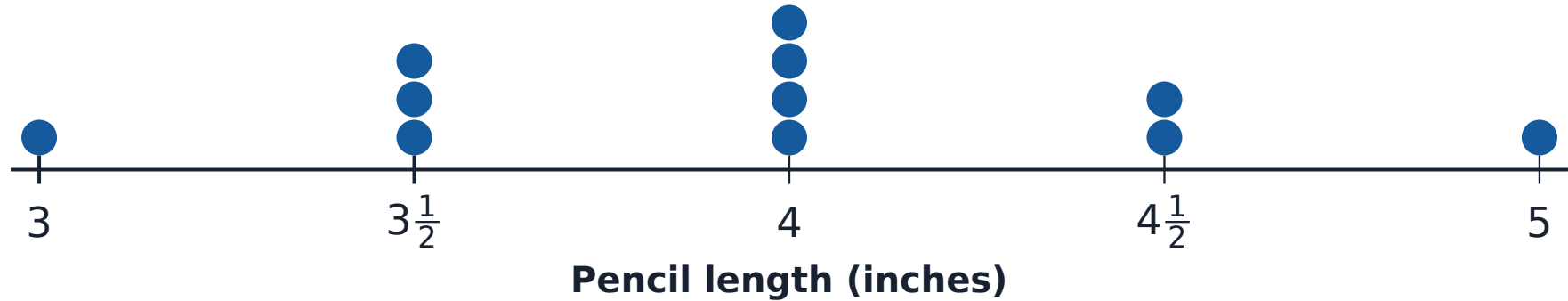
B. False

Which pencil length is most common?



- A.** 3 inches **B.** $3\frac{1}{2}$ inches **C.** 4 inches **D.** 5 inches

How many pencils are shorter than 4 inches?



Pencils: _____

SET 69 • 6.SP.A.2

BACKWARD 2

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

42, 24, 36, 18

Order: _____

Which number is greatest?

- A.** 53
- B.** 299
- C.** 305
- D.** 350

True or false?

These numbers are in order from least to greatest: 12, 21, 19, 30

- A.** True
- B.** False

Order from least to greatest.

105, 150, 115

Order: _____

Which number is least?

- A.** 89
- B.** 91
- C.** 98
- D.** 100

SET 69 • 6.SP.A.2

BACKWARD 3

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 10 and 20?

Number: _____

What number is halfway between points A and B?



Number: _____

True or false?

The number halfway between 60 and 64 is 62.

- A. True
- B. False

What number is halfway between 4 and 12?

- A.** 6
- B.** 8
- C.** 10
- D.** 16

What number is halfway between 0 and 18?

Number: _____

SET 69 • 6.SP.A.2

FORWARD 1

Compare the centers of two populations • Grade 7 • Statistics & Probability •
7.SP.B.4

Class A has a median score of 82. Class B has a median score of 75. The variability is similar.

Which statement is true?

- A.** The classes scored the same.
- B.** Class A typically scored higher.
- C.** Class B typically scored higher.
- D.** Class A's scores vary more.

Random samples: Town A has a mean age of 34 years.
Town B has a mean age of 42 years.

What is the difference in mean ages?

Difference: _____

In random samples, 7th graders sleep a median of 8 hours, and 6th graders sleep a median of 9 hours.

Which group typically sleeps more?

Group: _____

Sample A has a mean of 12 and Sample B has a mean of 15. The samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Values in population A tend to be greater.
- C.** Values in population B tend to be greater.
- D.** No conclusion is possible.

True or false?

To compare the centers of two populations, you can compare the medians of random samples.

- A.** True
- B.** False

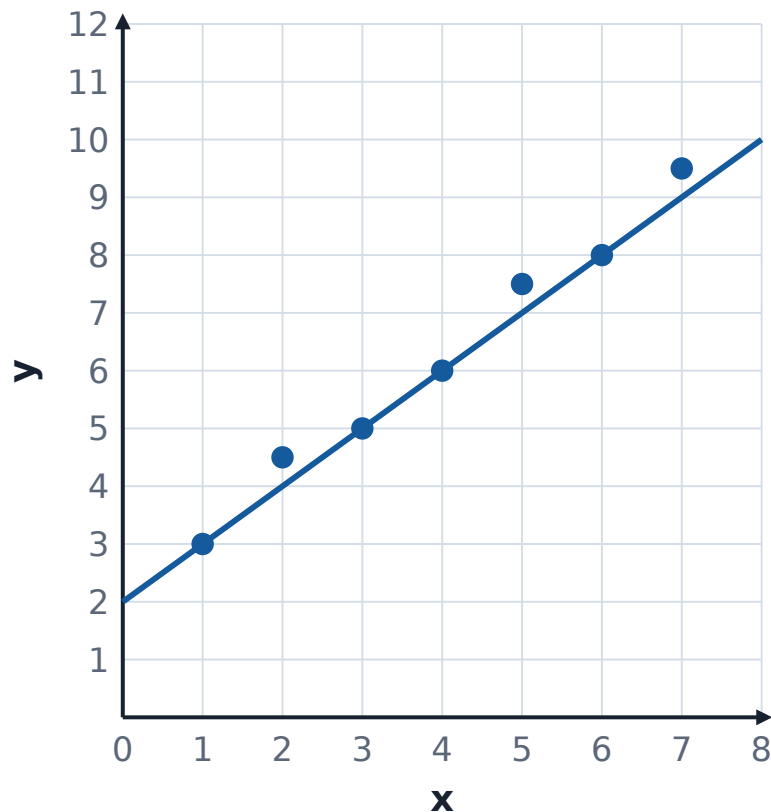
FORWARD 2

Use a line of best fit to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

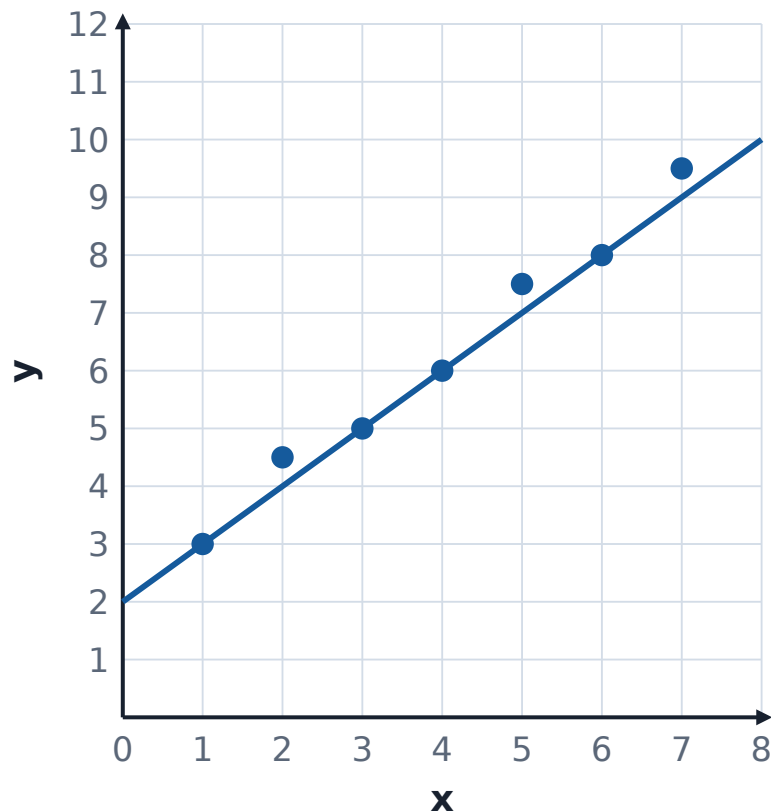
Use the line of best fit to
predict y when $x = 6$.

$y =$ _____



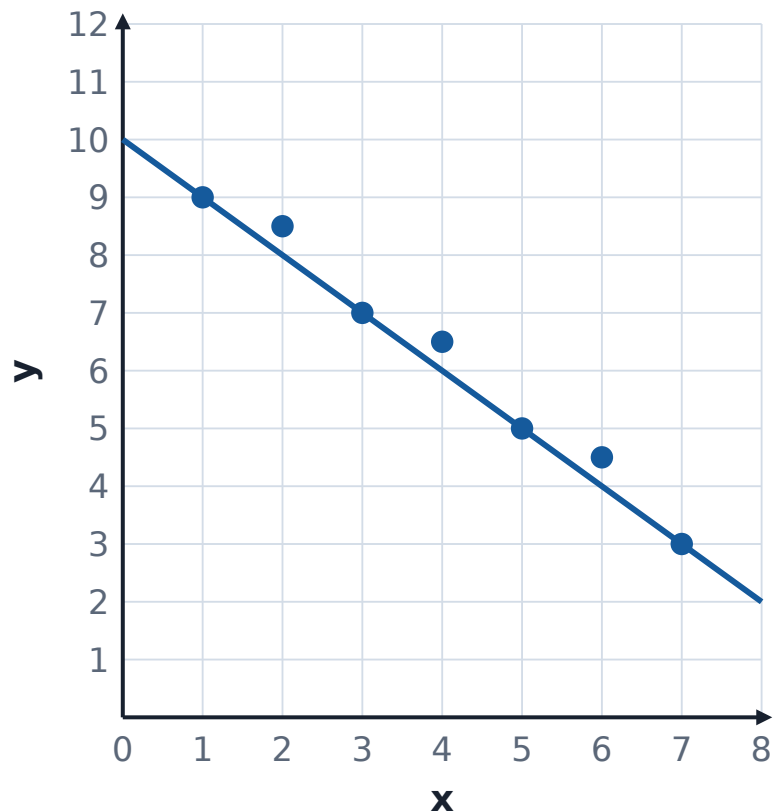
Use the line of best fit to
predict y when $x = 2$.

$y =$ _____



Use the line of best fit to
predict y when $x = 4$.

$y =$ _____



Which statement about a line of best fit is true?

- A.** It must pass through every point.
- B.** It shows the overall trend of the data.
- C.** It must pass through the origin.
- D.** It is always horizontal.

True or false?

A line of best fit usually has about the same number of points above it as below it.

- A. True
- B. False

Describe the spread of a distribution

MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Finding a difference	2.NBT.B.5
BACKWARD 2	Reading a line plot	4.MD.B.4
BACKWARD 3	Greatest and least numbers	2.NBT.A.4
FORWARD 1	Compare variability of two distributions	7.SP.B.3
FORWARD 2	Judge how closely data fit a line	8.SP.A.2

SET 70 • 6.SP.A.2

MAIN

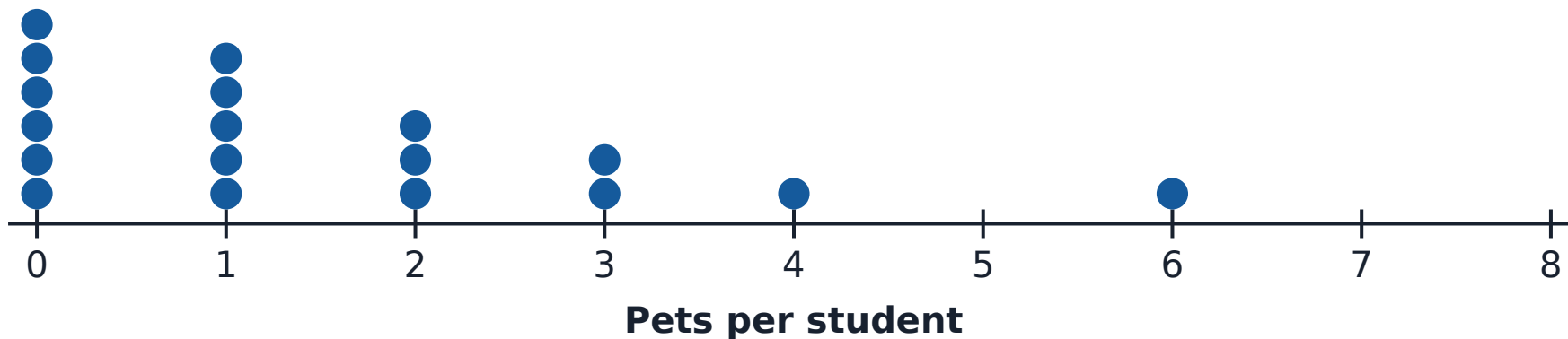
Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

What is the range of the data?



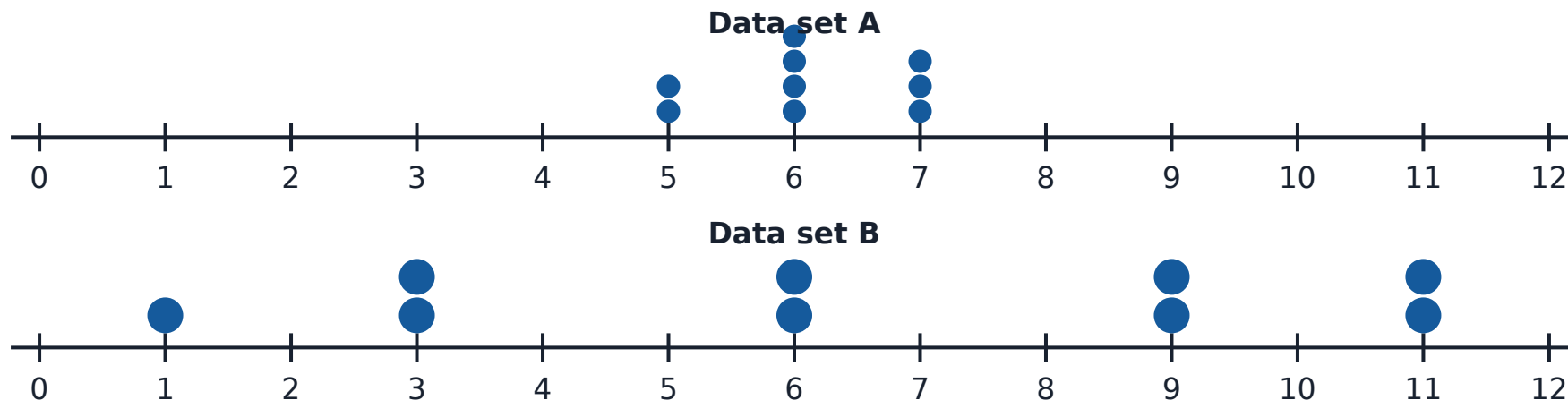
Range: _____

What is the range of the data?



Range: _____

Which data set has the greater spread?

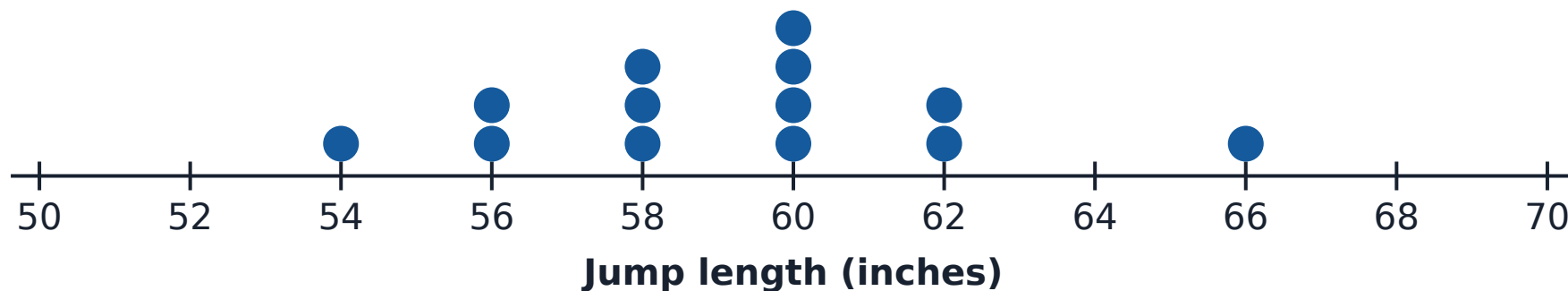


A. Data set A

B. Data set B

C. They have the same spread.

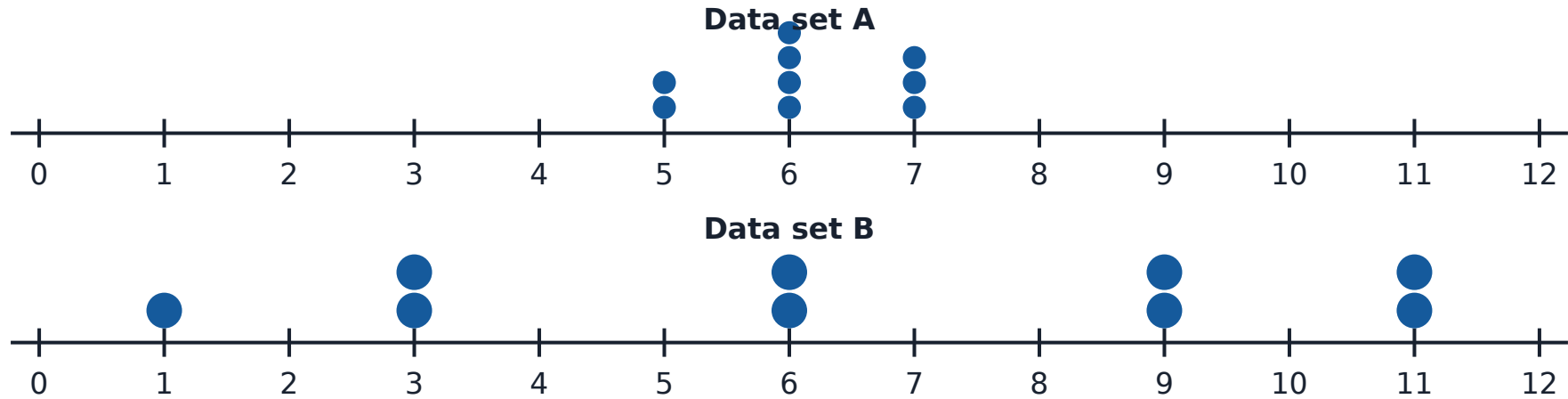
What is the range of the jump lengths?



Range: _____

True or false?

Data set A has a range of 2.



A. True

B. False

SET 70 • 6.SP.A.2

BACKWARD 1

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

The greatest value is 47. The least value is 18.

What is the difference?

Difference: _____

Subtract.

$$83 - 29$$

Difference: _____

True or false?

$$62 - 35 = 27$$

A. True

B. False

$$91 - 46 = ?$$

A. 45

B. 47

C. 55

D. 137

What is the difference between 66 and 54?

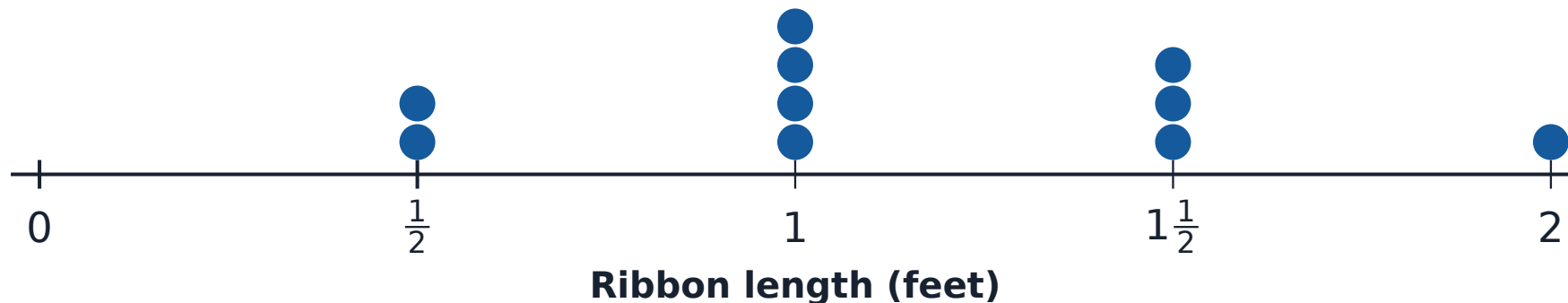
Difference: _____

SET 70 • 6.SP.A.2

BACKWARD 2

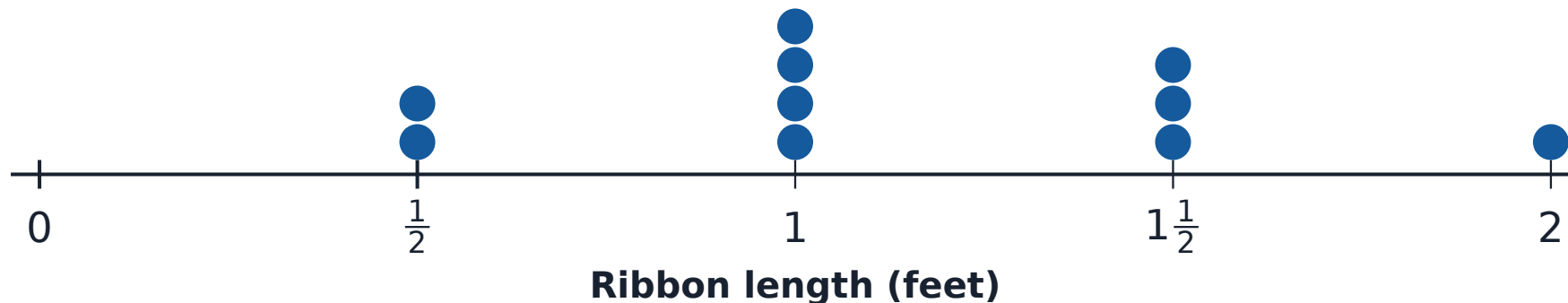
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

What is the longest ribbon length shown?



Length: _____

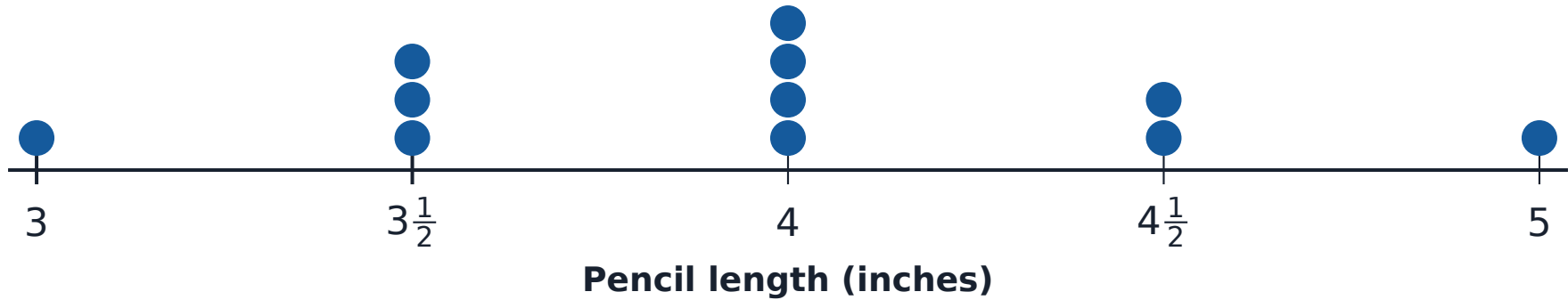
What is the shortest ribbon length shown?



Length: _____

True or false?

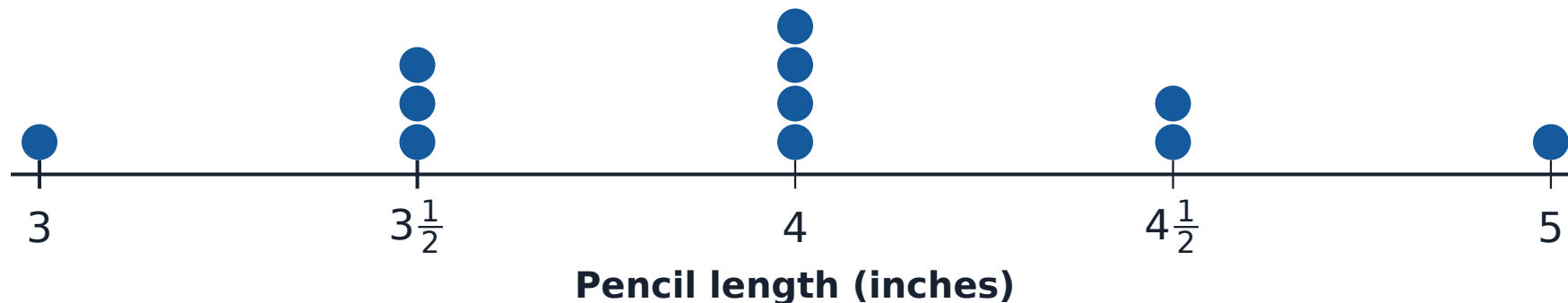
The longest pencil is 5 inches.



A. True

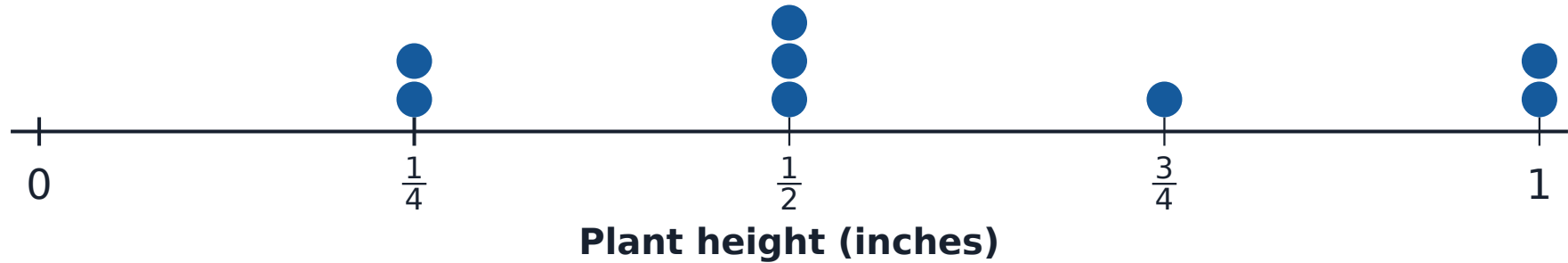
B. False

What is the shortest pencil length?



- A.** 3 inches **B.** $3\frac{1}{2}$ inches **C.** 4 inches **D.** 5 inches

How many plants are 1 inch tall?



Plants: _____

SET 70 • 6.SP.A.2

BACKWARD 3

Greatest and least numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

What is the greatest number?

18, 81, 58, 85

Greatest: _____

What is the least number?

204, 240, 402, 42

Least: _____

True or false?

The least number in the list 33, 30, 13, 31 is 13.

A. True

B. False

Which number is greatest?

A. 207

B. 270

C. 702

D. 720

What is the least number?

66, 56, 65, 60

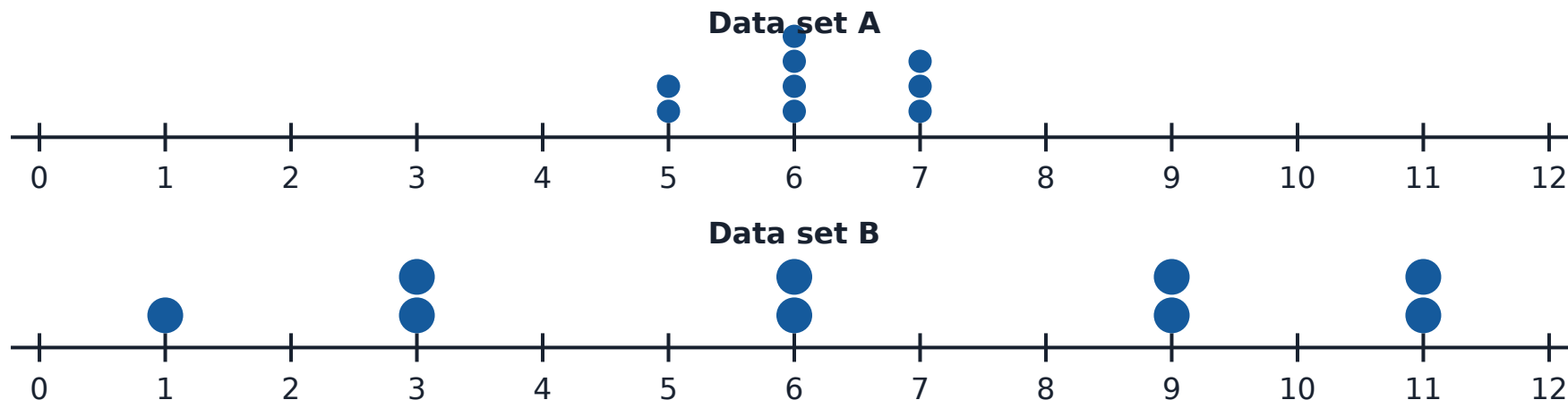
Least: _____

SET 70 • 6.SP.A.2

FORWARD 1

Compare variability of two distributions • Grade 7 • Statistics & Probability •
7.SP.B.3

Which data set has more variability?



A. Data set A

B. Data set B

C. They are the same.

Class A's mean is 60 and Class B's mean is 66. Each class has a MAD of 3.

The difference in means is how many times the MAD?

Times: _____

True or false?

Two data sets with means of 50 and 56 and MADs of 3 have means that differ by 2 MADs.

- A. True
- B. False

Data set A has a mean of 20 and a MAD of 2. Data set B has a mean of 21 and a MAD of 6.

Which statement is true?

- A.** The means differ by 3 MADs.
- B.** Data set A's values are more spread out.
- C.** Data set B's values are more spread out.
- D.** The data sets do not overlap.

Two data sets have means of 8 and 14. Each has a MAD of 1.5.

The difference in means is how many times the MAD?

Times: _____

SET 70 • 6.SP.A.2

FORWARD 2

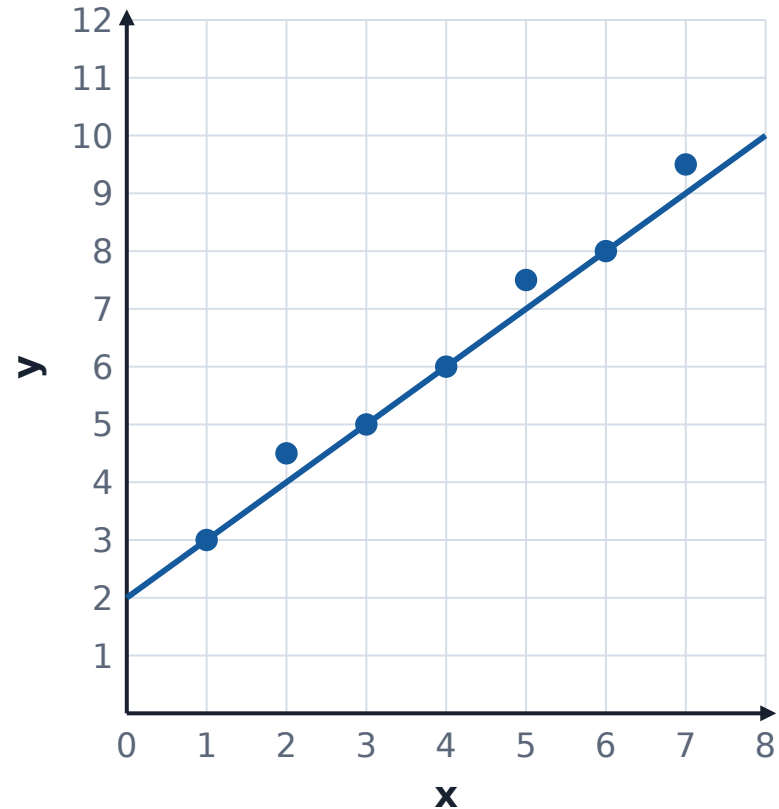
Judge how closely data fit a line • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

The line is a good fit for the data.

A. True

B. False

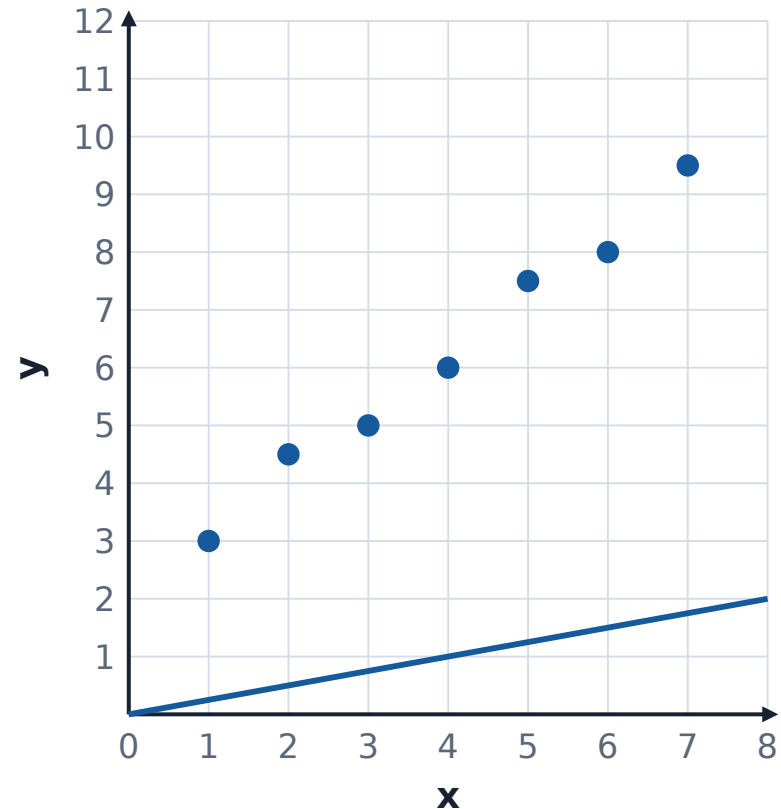


True or false?

The line is a good fit for the data.

A. True

B. False



Which is a sign of a good line of best fit?

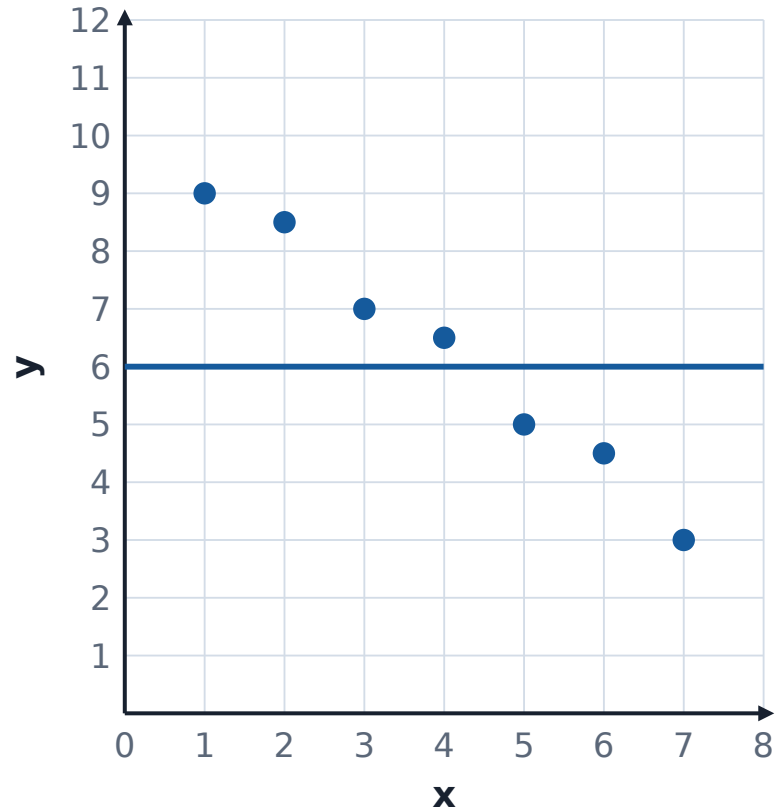
- A.** The line goes through only one point.
- B.** All points are below the line.
- C.** Points are close to the line on both sides.
- D.** All points are above the line.

True or false?

The line is a good fit for the data.

A. True

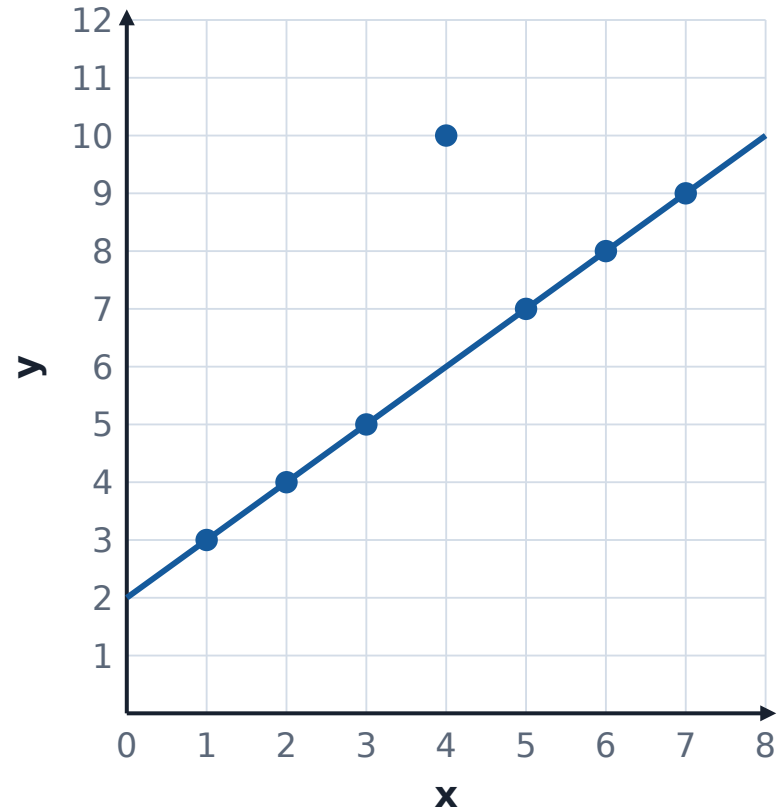
B. False



Which point is farthest from the line?

Write its coordinates.

Point: _____



A measure of center vs. a measure of variation

MAIN	Grade 6 main question	6.SP.A.3
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Sharing a total equally	5.MD.B.2
BACKWARD 3	Finding a difference	2.NBT.B.5
FORWARD 1	Compare populations using center and variability	7.SP.B.4
FORWARD 2	Interpret the slope and intercept of a linear model — nearest rela...	8.SP.A.3

SET 71 • 6.SP.A.3

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.3

Which measure describes the center of a data set?

- A.** Interquartile range
- B.** Mean absolute deviation
- C.** Median
- D.** Range

Which measure describes how much the values in a data set vary?

- A.** Mean
- B.** The middle value
- C.** Median
- D.** Interquartile range

True or false?

A measure of center summarizes all of the values in a data set with a single number.

- A.** True
- B.** False

True or false?

A measure of variation describes how the values in a data set vary with a single number.

- A.** True
- B.** False

Which measure is a measure of variation?

- A.** Median
- B.** Mean
- C.** Mean absolute deviation
- D.** The middle value

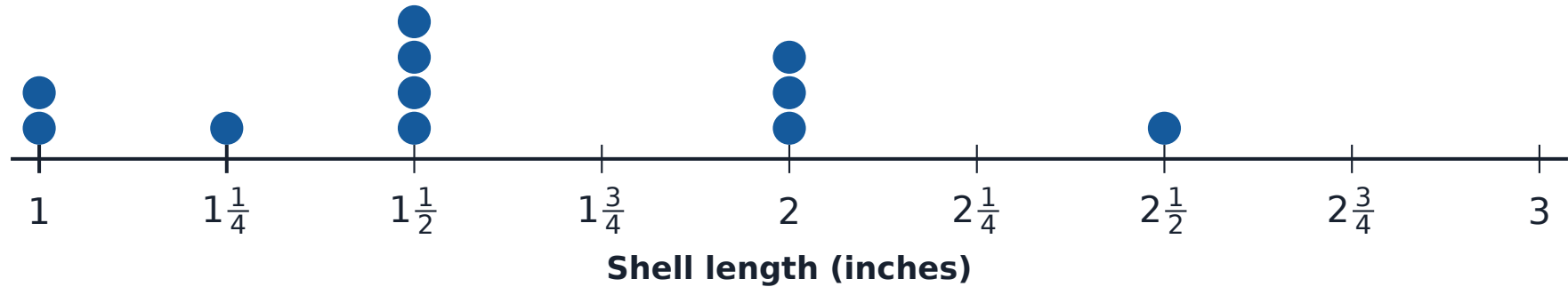
SET 71 • 6.SP.A.3

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows shell lengths.

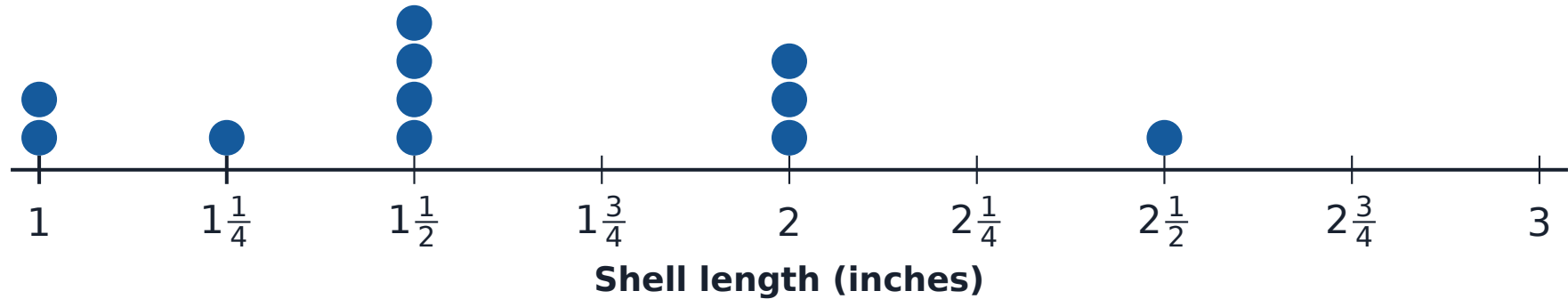
How many shells are $1\frac{1}{2}$ inches long?



Shells: _____

The line plot shows shell lengths.

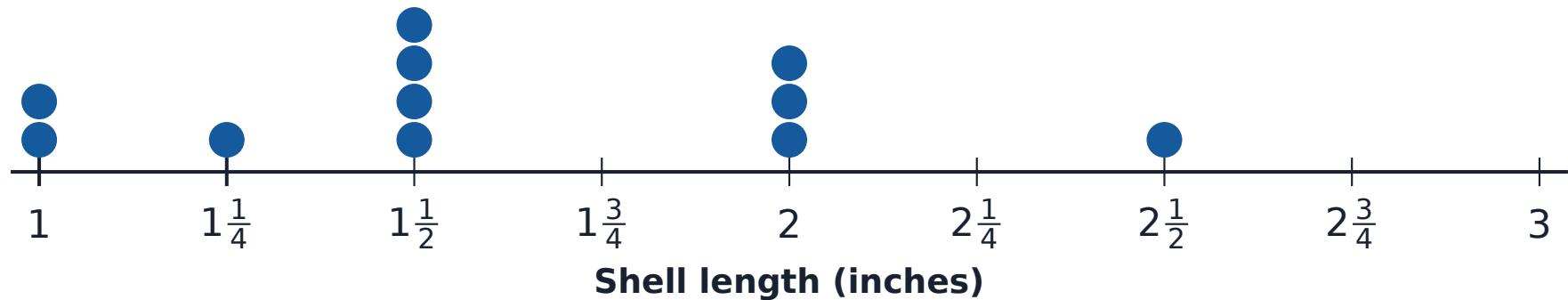
How many shells are there in all?



Shells: _____

True or false?

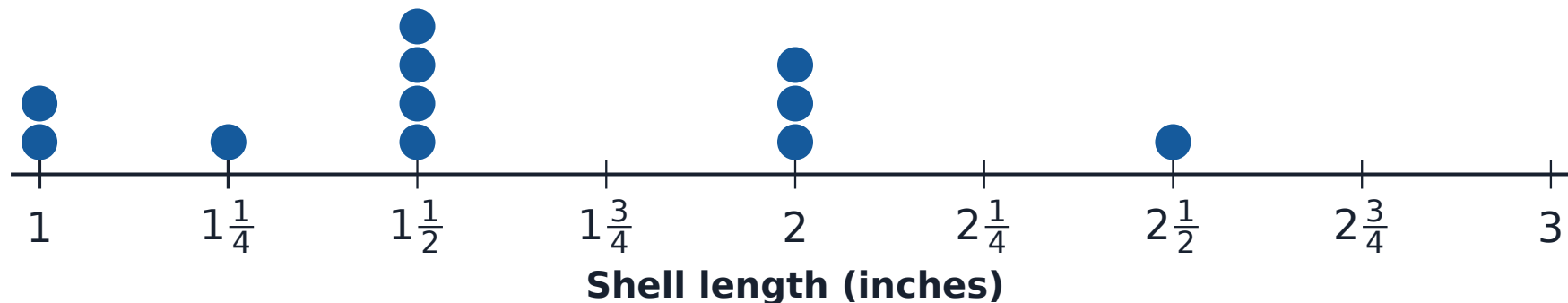
Three shells are 2 inches long.



A. True

B. False

Which shell length is most common?



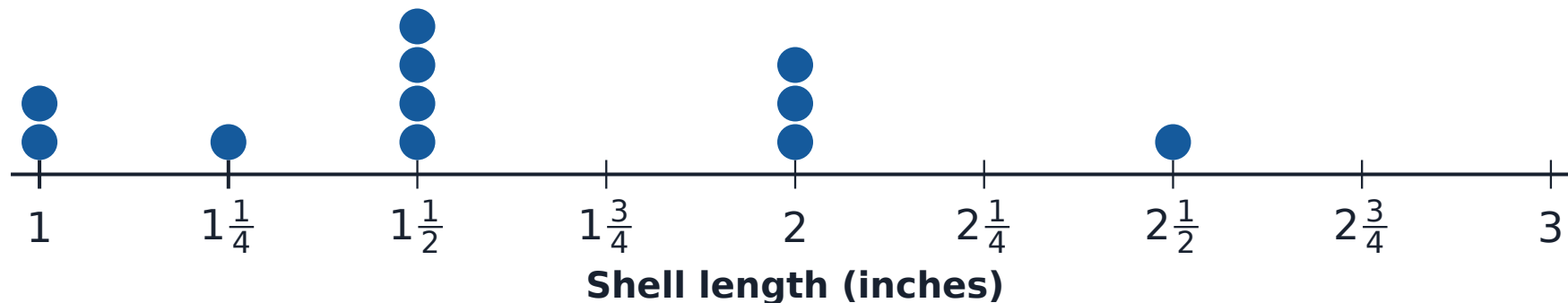
A. 1 inch

B. $1\frac{1}{2}$
inches

C. 2 inches

D. $2\frac{1}{2}$
inches

What is the total length of the 2-inch shells?



Total length: _____

SET 71 • 6.SP.A.3

BACKWARD 2

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Three friends have 4, 6, and 8 stickers. They share all of the stickers equally.

How many stickers does each friend get?

Stickers: _____

Four jars hold 2, 3, 5, and 6 liters of water. The water is shared equally among the jars.

How much water is in each jar?

Liters: _____

True or false?

Cups holding 1, 2, and 3 liters are shared equally.
Each cup then holds 2 liters.

- A. True
- B. False

Five friends collected 3, 5, 6, 7, and 9 cans. They share the cans equally.

How many cans does each friend get?

A. 5

B. 6

C. 7

D. 30

Two bottles hold $\frac{1}{4}$ liter and $\frac{3}{4}$ liter. The juice is shared equally between the bottles.

How much juice is in each bottle?

Liters: _____

SET 71 • 6.SP.A.3

BACKWARD 3

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$76 - 38$$

Difference: _____

The highest score is 95 and the lowest is 58.

What is the difference?

Difference: _____

True or false?

$$100 - 47 = 53$$

A. True

B. False

$$84 - 29 = ?$$

- A.** 45
- B.** 55
- C.** 65
- D.** 113

Subtract.

$$62 - 17$$

Difference: _____

SET 71 • 6.SP.A.3

FORWARD 1

Compare populations using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.4

Class A has a median score of 82 and an IQR of 10.

Class B has a median score of 75 and an IQR of 10.

Which statement is true?

A. Class A typically scored higher.

B. Class A's scores vary more.

C. Class B typically scored higher.

D. The classes scored the same.

True or false?

Two samples have the same median. The sample with the greater IQR has more variability.

- A.** True
- B.** False

Which measure is best for comparing the variability of two samples?

- A.** IQR
- B.** Sample size
- C.** Median
- D.** Mean

Sample A: mean 40, MAD 2. Sample B: mean 40, MAD 7.
Which sample has values that vary more?

Sample: _____

Store A's daily sales have a median of \$520. Store B's have a median of \$610. Their IQRs are similar.

Which store typically has greater daily sales?

Store: _____

FORWARD 2

Interpret the slope and intercept of a linear model • Grade 8 • Statistics & Probability • 8.SP.A.3

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The model $y = 2.5x + 40$ predicts a test score y after x hours of study.

What does the slope 2.5 mean?

Meaning: _____

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What does 12 represent?

- A.** The number of weeks
- B.** The height at week 0
- C.** The height after 12 weeks
- D.** The growth per week

True or false?

In the model $y = -0.5x + 30$, y decreases by 0.5 for each increase of 1 in x .

- A. True
- B. False

The model $y = 1.2x + 5$ gives the length y (cm) of a fish at x months old.

What does 5 represent?

Meaning: _____

In the model $y = 45x + 200$, what does 45 represent if y is dollars saved after x weeks?

- A.** Total savings
- B.** Starting savings
- C.** Number of weeks
- D.** Dollars saved per week

Display numerical data in a dot plot

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Line plots of whole-number measurements	2.MD.D.9
BACKWARD 2	Line plots with fractional measurements	4.MD.B.4
BACKWARD 3	Ordering numbers	2.NBT.A.4
FORWARD 1	Compare two dot plots	7.SP.B.3
FORWARD 2	Construct a scatter plot	8.SP.A.1

SET 72 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Data: 2, 4, 4, 5, 7, 7, 7, 9

Make a dot plot of the data on the number line.



Number of pets: 0, 1, 1, 2, 2, 2, 3, 5

Make a dot plot of the data on the number line.



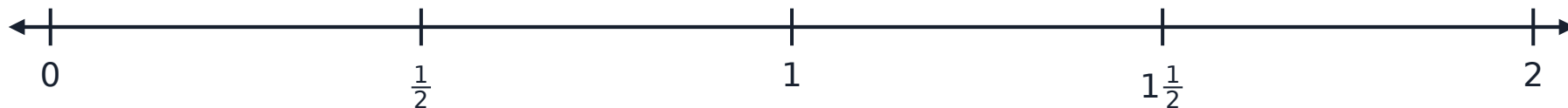
Quiz scores: 6, 7, 7, 8, 8, 8, 9, 10, 10

Make a dot plot of the data on the number line.



Ribbon lengths (feet): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 2, 1

Make a dot plot of the data on the number line.



Hours of sleep: 7, 8, 9, 8, 10, 7, 8

Make a dot plot of the data on the number line.



SET 72 • 6.SP.B.4

BACKWARD 1

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 3, 4, 4, 5, 4

In a line plot, how many marks are drawn at 4?

Marks: _____

Lengths (cm): 2, 3, 3, 5

Make a line plot of the lengths.



True or false?

In a line plot of 6, 6, 7, 9, there are two marks above 6.

A. True

B. False

Which number line is best for a line plot of 12, 14, 15, 15?

- A.** A number line from 0 to 100 by tens
- B.** A number line from 20 to 30
- C.** A number line from 10 to 16
- D.** A number line from 0 to 5

Lengths (feet): 8, 9, 9, 9, 11

Which value gets the most marks?

Value: _____

SET 72 • 6.SP.B.4

BACKWARD 2

Line plots with fractional measurements • Grade 4 • Measurement & Data •
4.MD.B.4

Lengths (inches): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 1, $\frac{1}{2}$

In a line plot, how many X's are drawn at 1?

X's: _____

True or false?

In a line plot of 2, $2\frac{1}{2}$, $2\frac{1}{2}$, 3, there are two X's above $2\frac{1}{2}$.

A. True

B. False

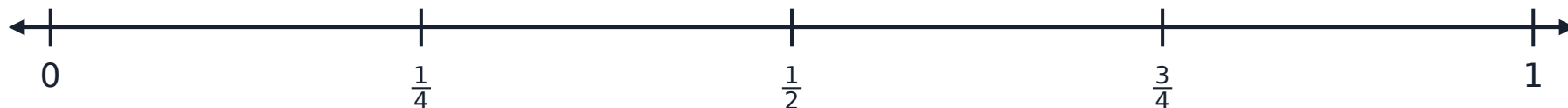
Which number line would you use for these lengths?

$$\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, \frac{1}{2}$$

- A.** A number line from 0 to 10 marked in ones
- B.** A number line from 0 to 100 marked in tens
- C.** A number line from 1 to 2 marked in halves
- D.** A number line from 0 to 1 marked in fourths

Lengths (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

Make a line plot of the lengths.



True or false?

A line plot shows each data value as a mark above a number line.

- A.** True
- B.** False

SET 72 • 6.SP.B.4

BACKWARD 3

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

7, 2, 9, 4, 4

Order: _____

Which list is in order from least to greatest?

A. 6, 7, 8, 10

B. 10, 8, 7, 6

C. 7, 6, 8, 10

D. 6, 8, 7, 10

True or false?

These numbers are in order from least to greatest: 1, 2, 5, 3

- A. True
- B. False

What are the least and greatest values?

8, 10, 7, 9, 7

Least: _____

Greatest: _____

Which number is least?

A. 0

B. 1

C. 2

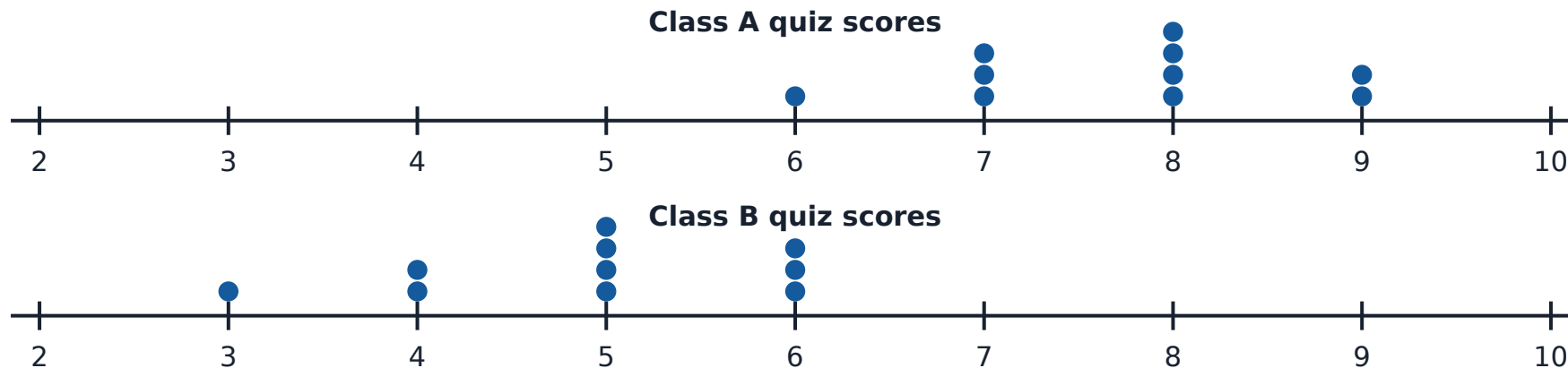
D. 5

SET 72 • 6.SP.B.4

FORWARD 1

Compare two dot plots • Grade 7 • Statistics & Probability • 7.SP.B.3

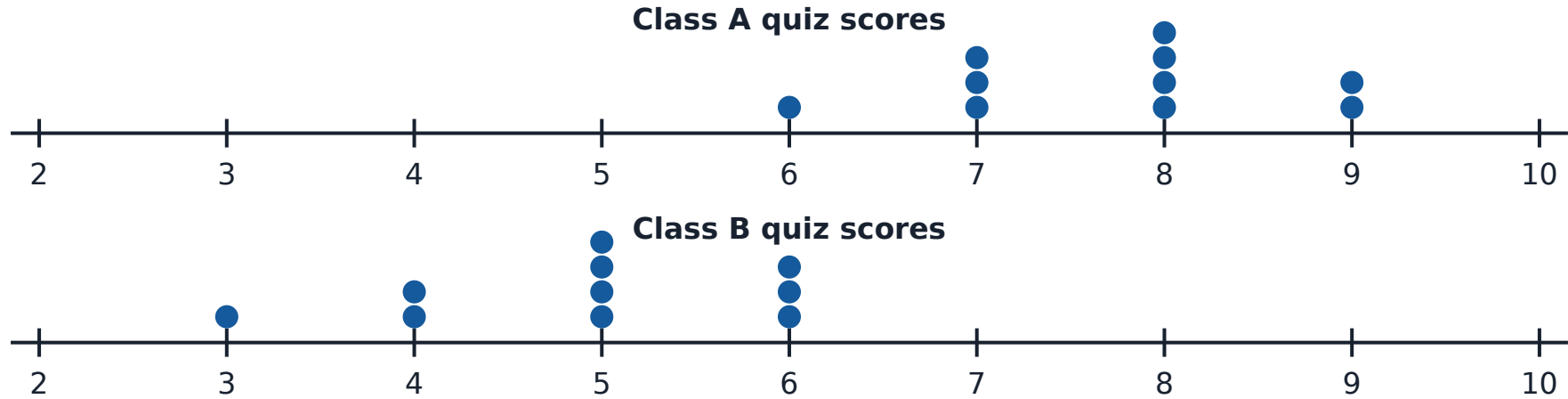
About how many points greater is the center of Class A's scores than Class B's?



Points: _____

True or false?

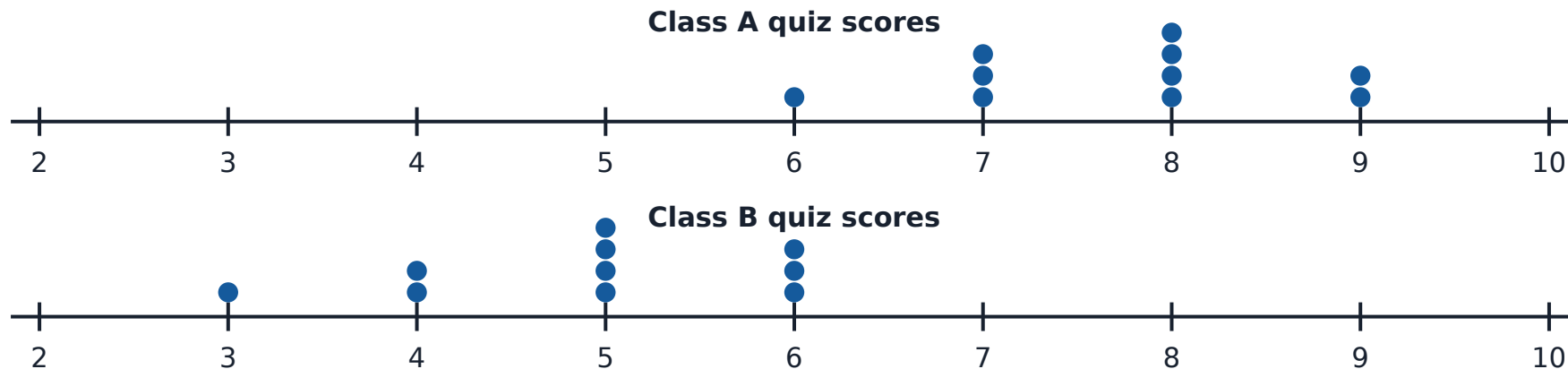
The two dot plots overlap.



A. True

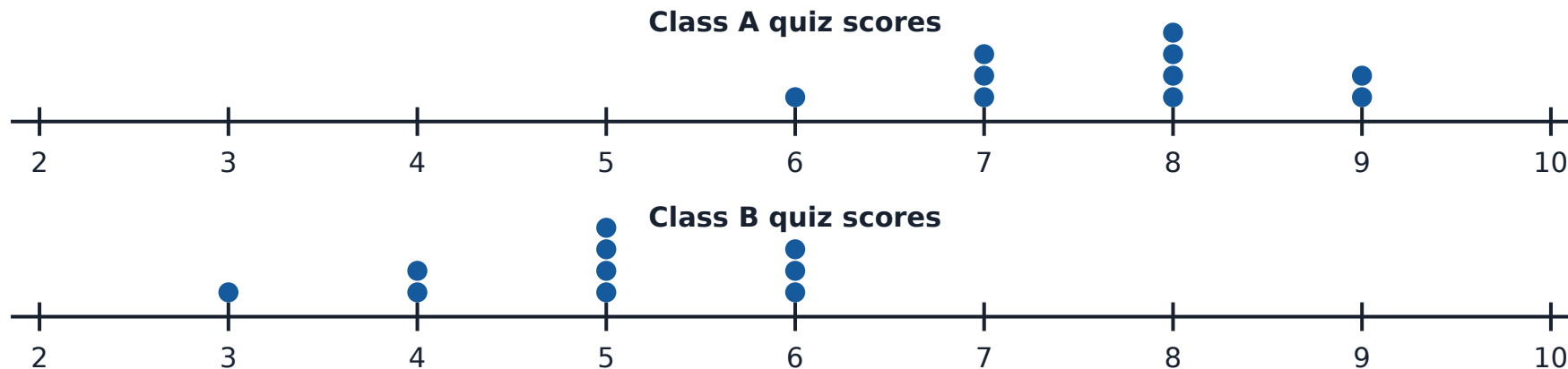
B. False

Which statement is true?



- A.** Class B has more variability.
- B.** The medians are equal.
- C.** Class B's scores are typically higher.
- D.** Class A's scores are typically higher.

What is the range of each class's scores?

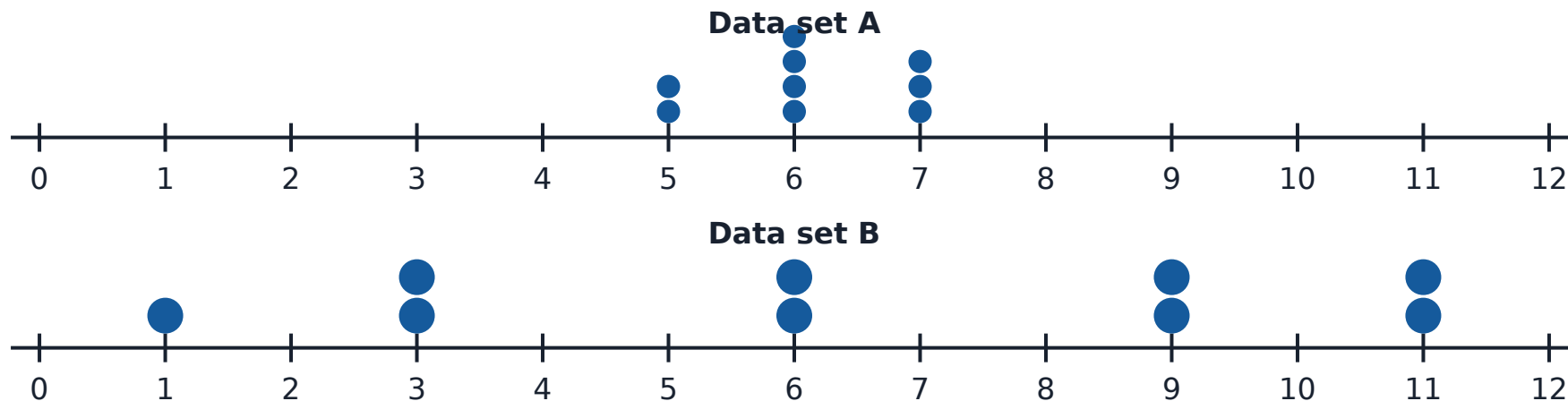


Class A: _____

Class B: _____

True or false?

Data set A and data set B have the same median.



A. True

B. False

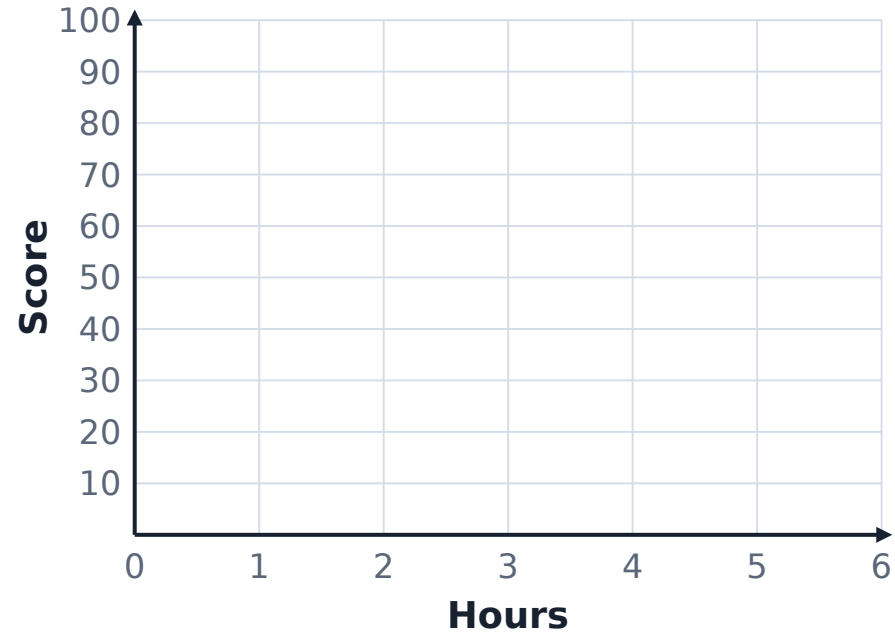
SET 72 • 6.SP.B.4

FORWARD 2

Construct a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

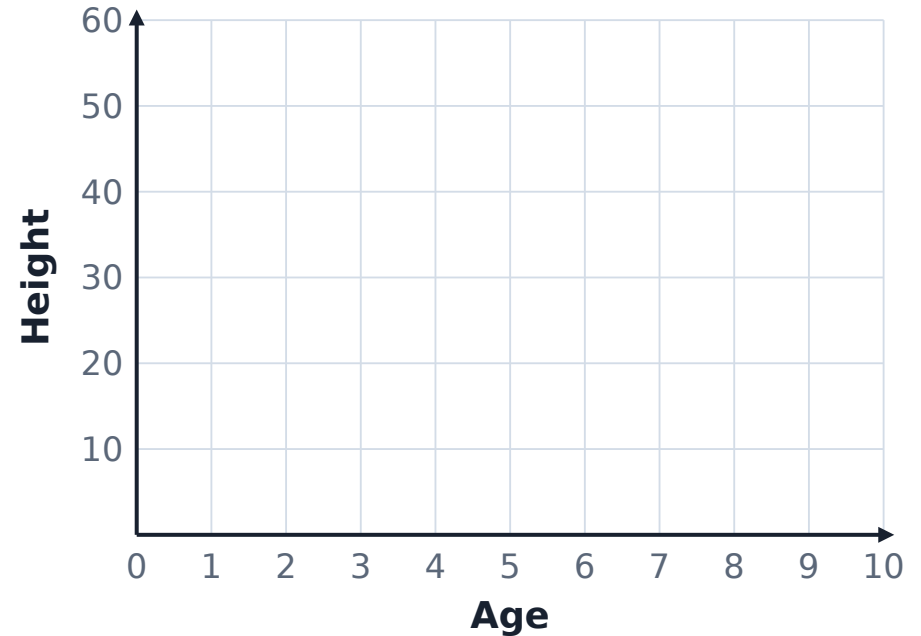
Make a scatter plot of the data.

Hours	1	2	3	4	5
Score	60	65	75	80	90



Make a scatter plot of the data.

Age	2	4	6	8
Height (in.)	34	40	46	50



A scatter plot shows age on the x-axis and height on the y-axis.

A 10-year-old is 54 inches tall. Which point shows this person?

A. (54, 10)

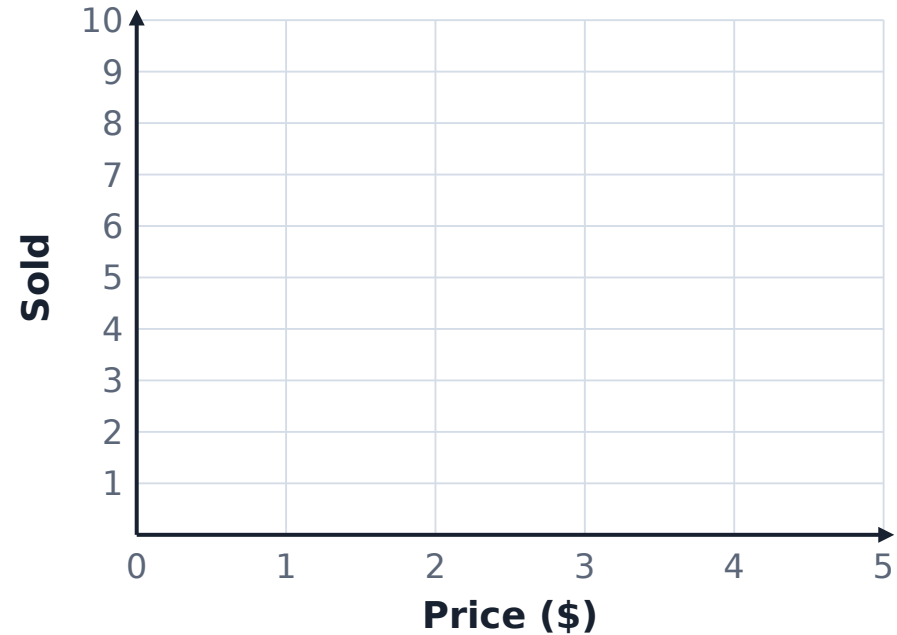
B. (54, 54)

C. (10, 10)

D. (10, 54)

Make a scatter plot of the data.

Price (\$)	1	2	3	4
Sold	9	7	4	2



True or false?

To make a scatter plot of height and arm span, each person is shown as one point.

- A.** True
- B.** False

Display numerical data in a histogram

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Drawing a scaled bar graph	3.MD.B.3
BACKWARD 2	Placing numbers in intervals	2.NBT.A.4
BACKWARD 3	Organizing data into categories	1.MD.C.4
FORWARD 1	Compare two populations shown in histograms	7.SP.B.4
FORWARD 2	Display numerical data for two variables in a scatter plot — neares...	8.SP.A.1

SET 73 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

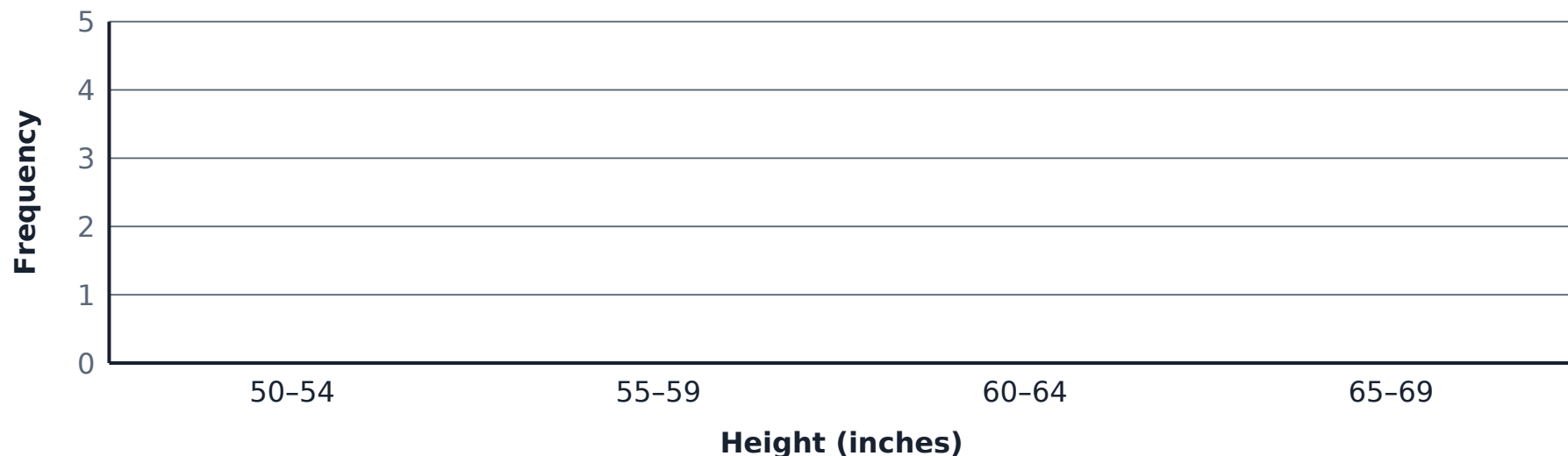
Books read: 3, 12, 15, 8, 22, 17, 25, 11, 31, 19

Make a histogram of the data.



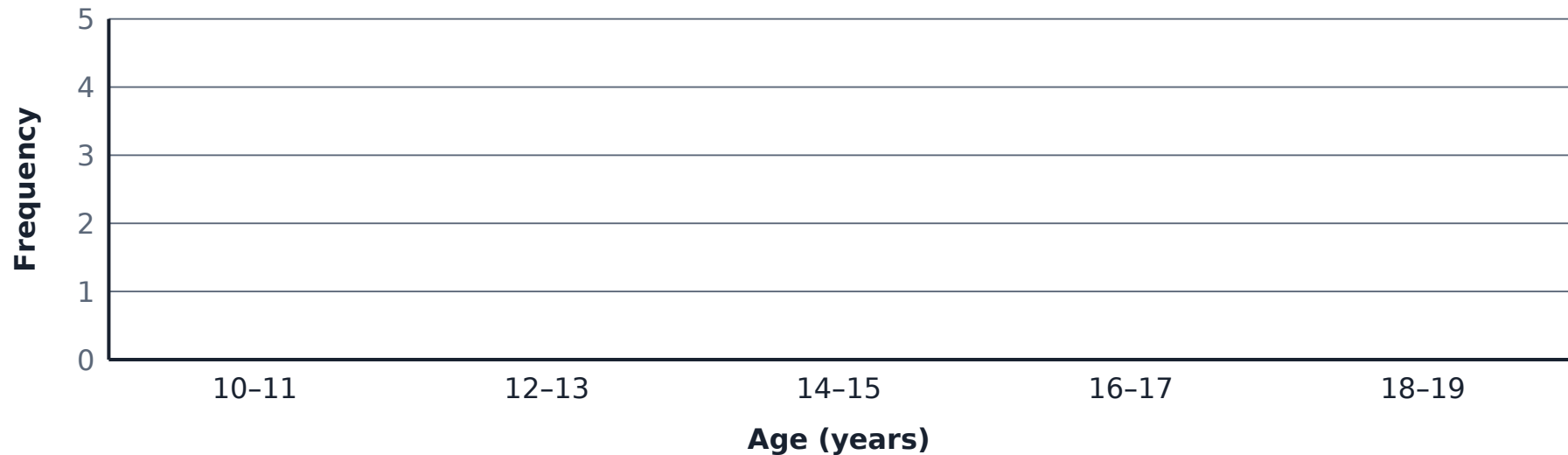
Heights (in.): 52, 58, 61, 55, 63, 57, 66, 60, 54, 59

Make a histogram of the data.



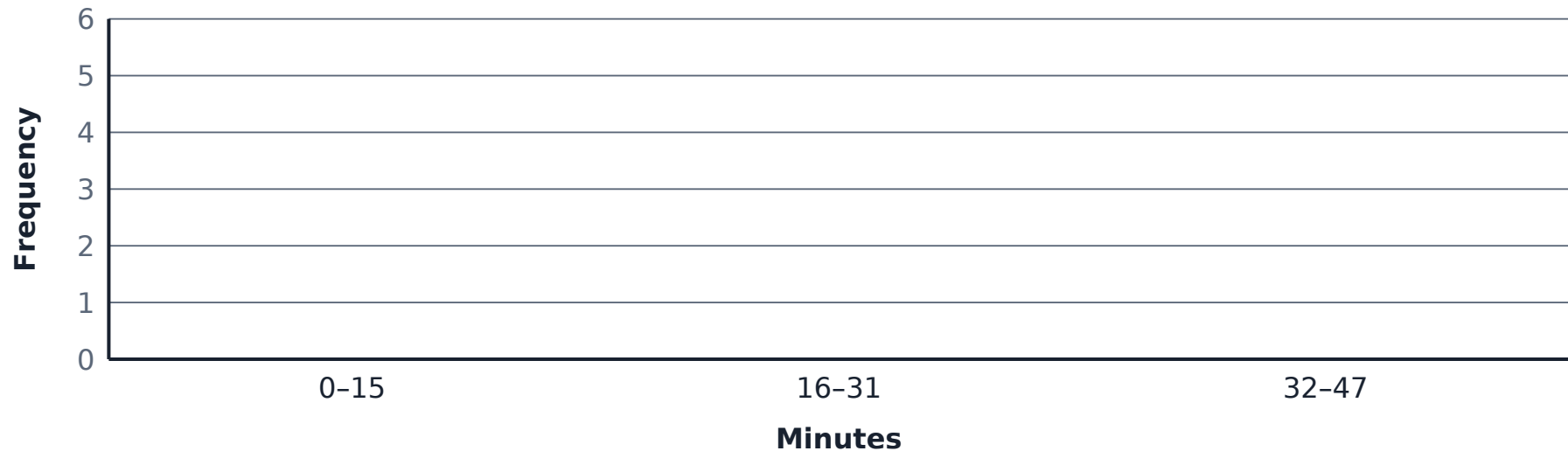
Ages: 11, 12, 12, 13, 14, 14, 14, 15, 16, 17, 19

Make a histogram of the data.



Minutes of exercise: 5, 18, 22, 25, 30, 34, 41, 45, 12, 28

Make a histogram of the data.



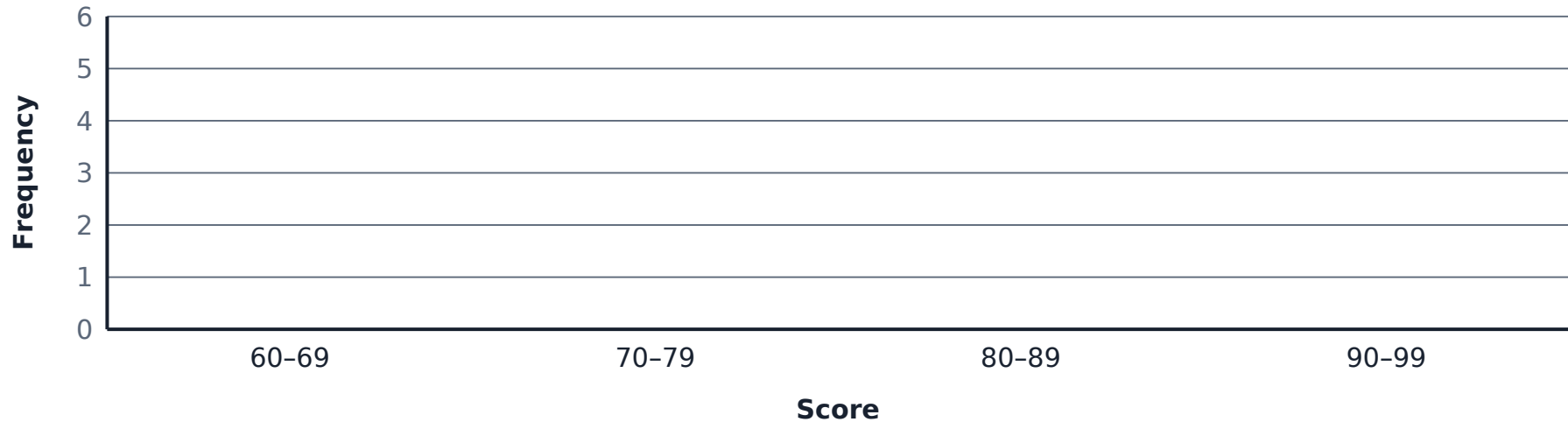
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q4/5

Test scores: 62, 75, 78, 81, 84, 88, 90, 93, 95, 71, 86, 79

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

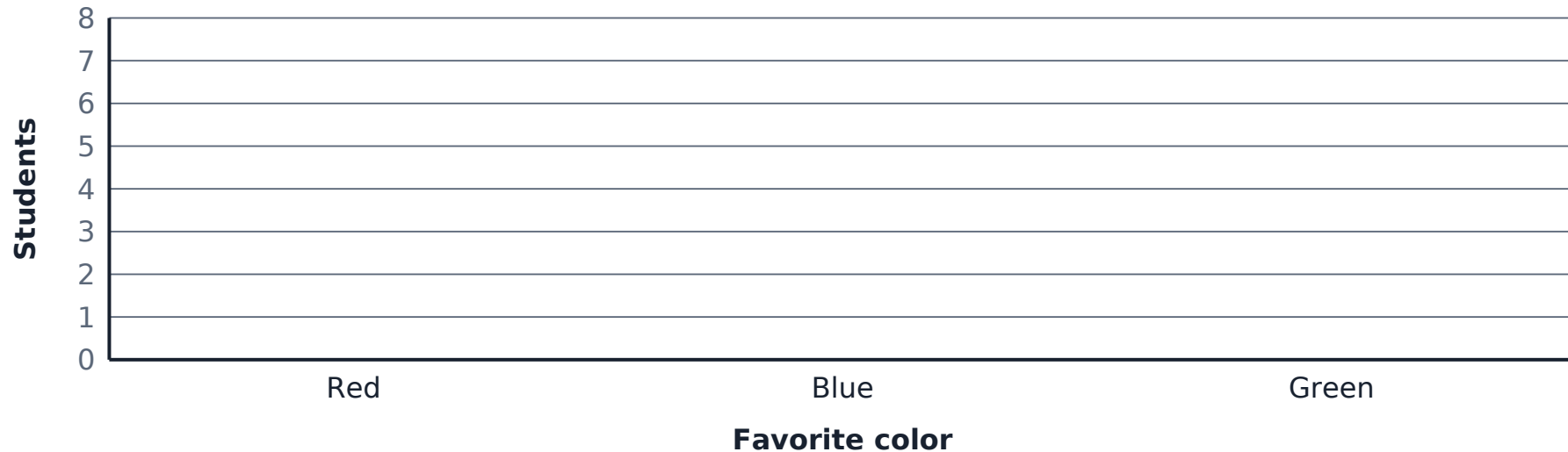
Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q5/5

SET 73 • 6.SP.B.4

BACKWARD 1

Drawing a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

Draw a bar graph: Red 4, Blue 6, Green 2.



On a bar graph, the bar for "Cats" ends at 5 on a scale that counts by 1s.

How many cats does the bar show?

Cats: _____

True or false?

On a bar graph with a scale of 2, a bar that stops halfway between 6 and 8 shows 7.

- A. True
- B. False

A bar graph has a scale that counts by 5s. A bar shows 20.

How many grid lines tall is the bar?

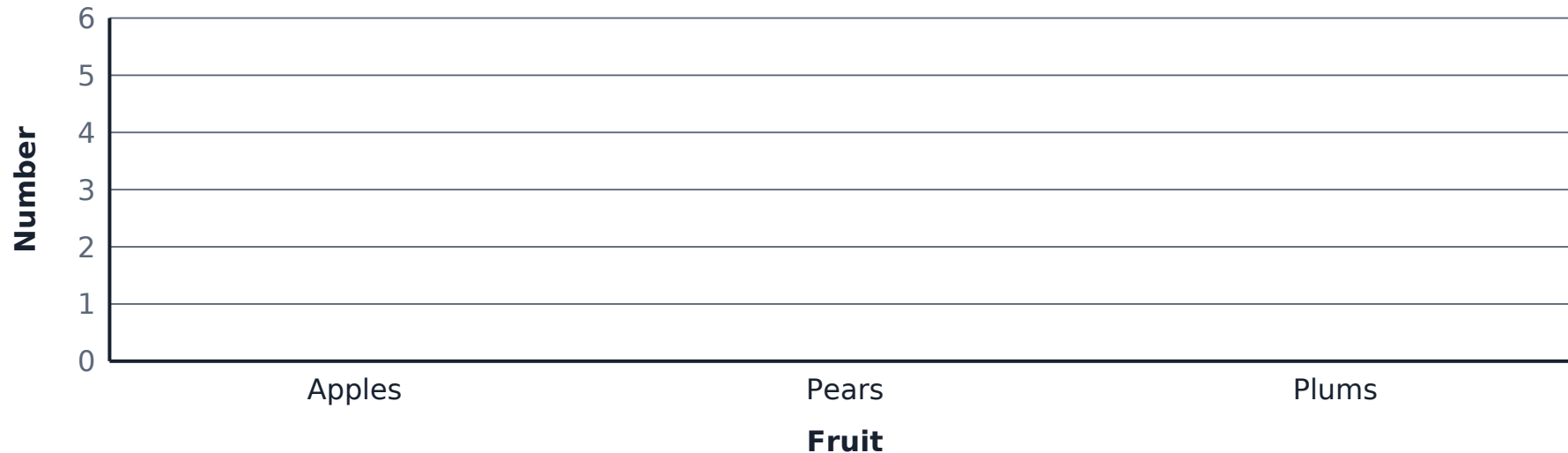
A. 4

B. 5

C. 15

D. 20

Draw a bar graph: Apples 3, Pears 1, Plums 5.



SET 73 • 6.SP.B.4

BACKWARD 2

Placing numbers in intervals • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Which interval contains 17?

- A.** 0–9
- B.** 10–19
- C.** 20–29
- D.** 30–39

True or false?

25 is in the interval 20–29.

A. True

B. False

Data: 3, 12, 15, 18, 22

How many of the numbers are in the interval 10–19?

Numbers: _____

Which number is in the interval 30–39?

- A.** 3
- B.** 29
- C.** 34
- D.** 40

True or false?

19 is in the interval 20–29.

A. True

B. False

SET 73 • 6.SP.B.4

BACKWARD 3

Organizing data into categories • Grade 1 • Measurement & Data • 1.MD.C.4

Data: cat, dog, cat, fish, dog, cat

How many answers are "cat"?

Cat: _____

True or false?

Data: red, blue, red, red. There are 3 answers of "red."

A. True

B. False

Data: A, B, A, C, A, B

Which category has the most?

A. A

B. B

C. C

Data: yes, no, yes, yes, no

How many more "yes" answers than "no" answers are there?

More: _____

Data: 3, 7, 12, 14, 18

How many numbers are less than 10?

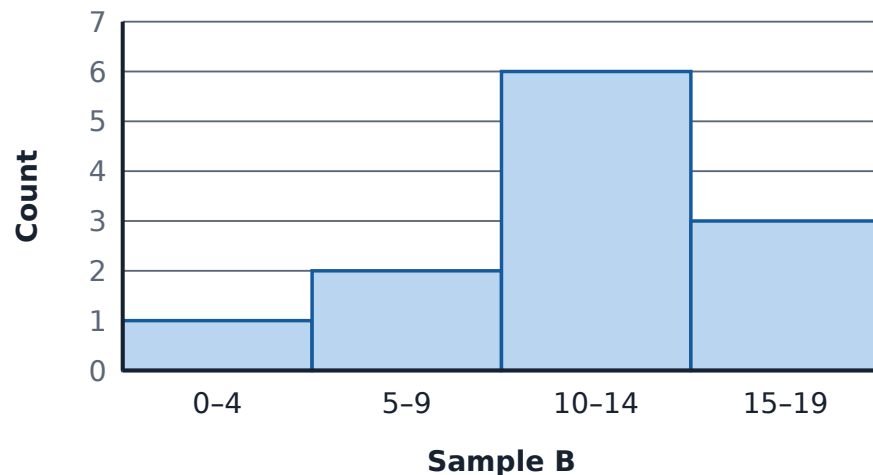
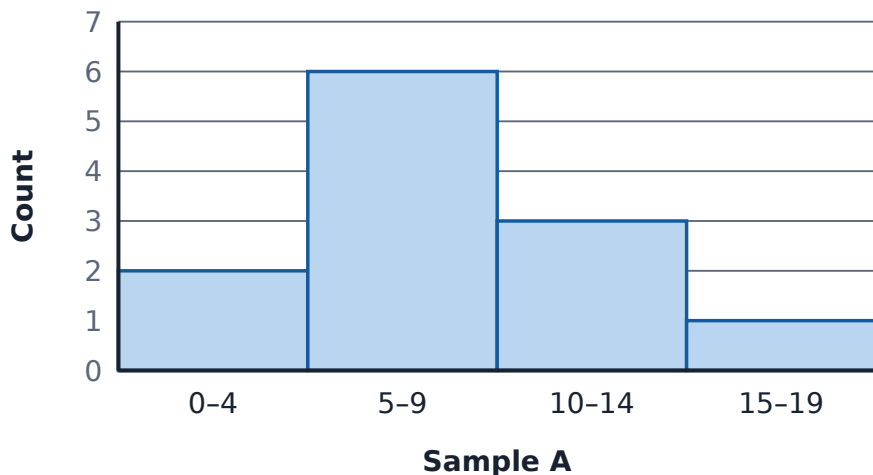
Numbers: _____

SET 73 • 6.SP.B.4

FORWARD 1

Compare two populations shown in histograms • Grade 7 • Statistics & Probability • 7.SP.B.4

Which sample tends to have greater values?

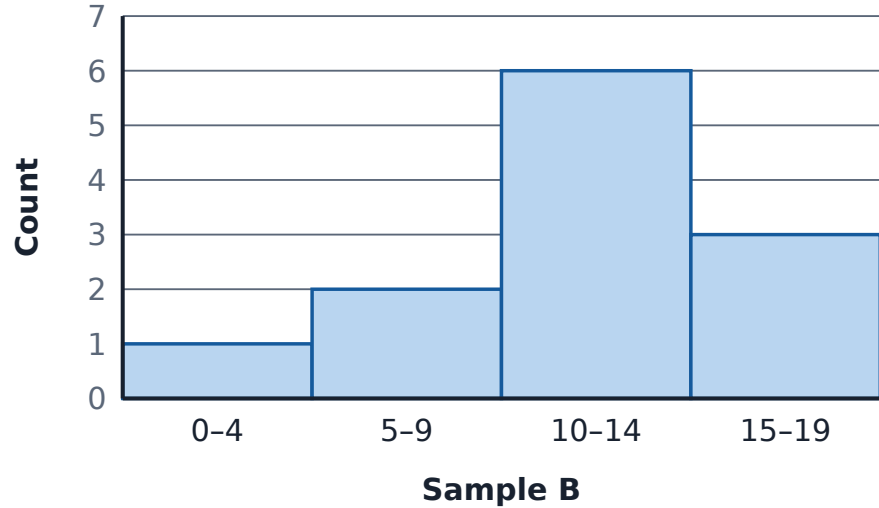
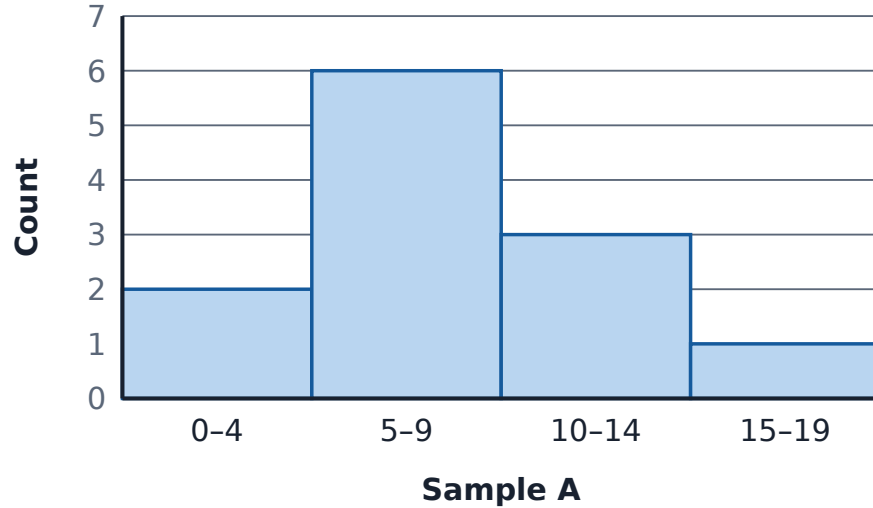


A. Sample A

B. Sample B

C. They are the same.

In which interval is the median of Sample A?



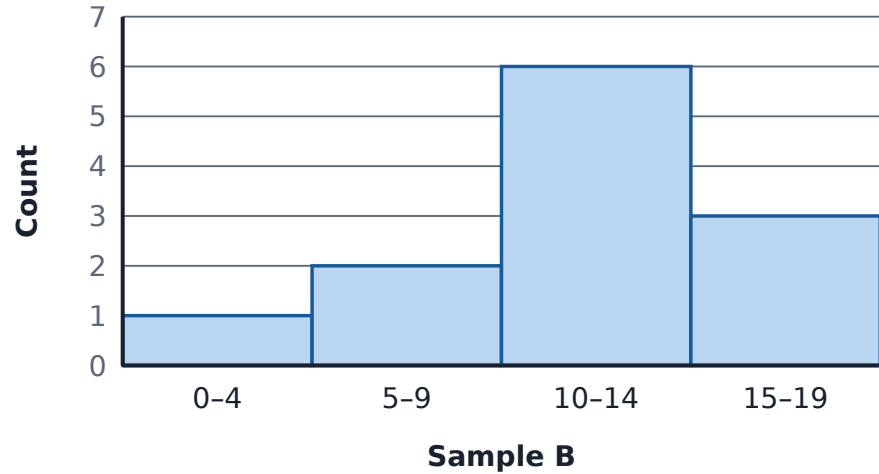
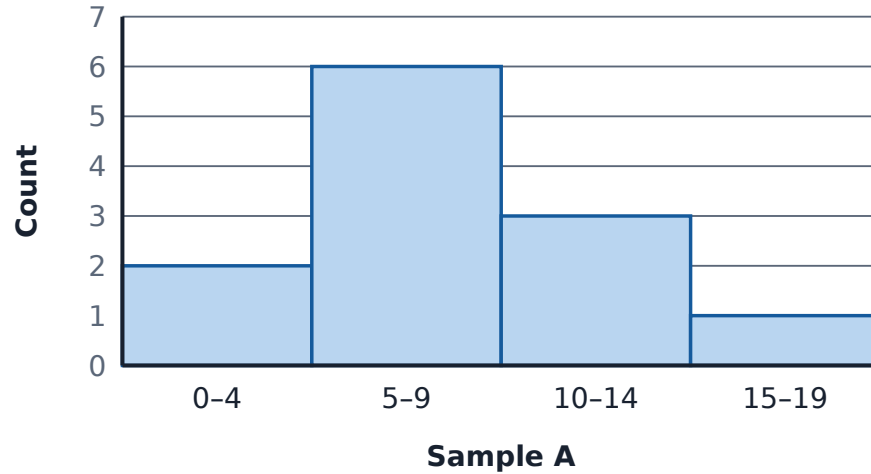
A. 0-4

B. 5-9

C. 10-14

D. 15-19

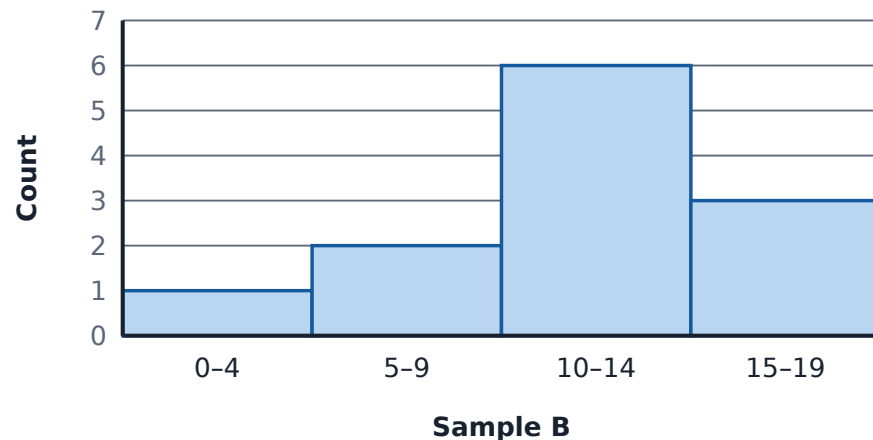
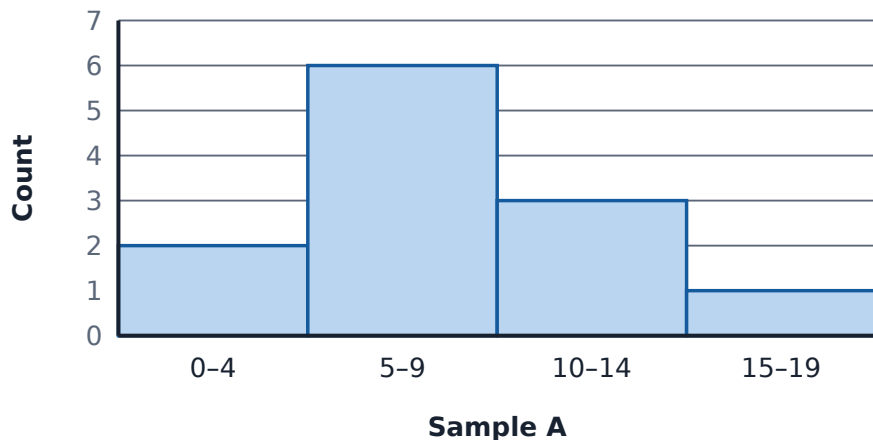
How many values are in each sample?



Values: _____

True or false?

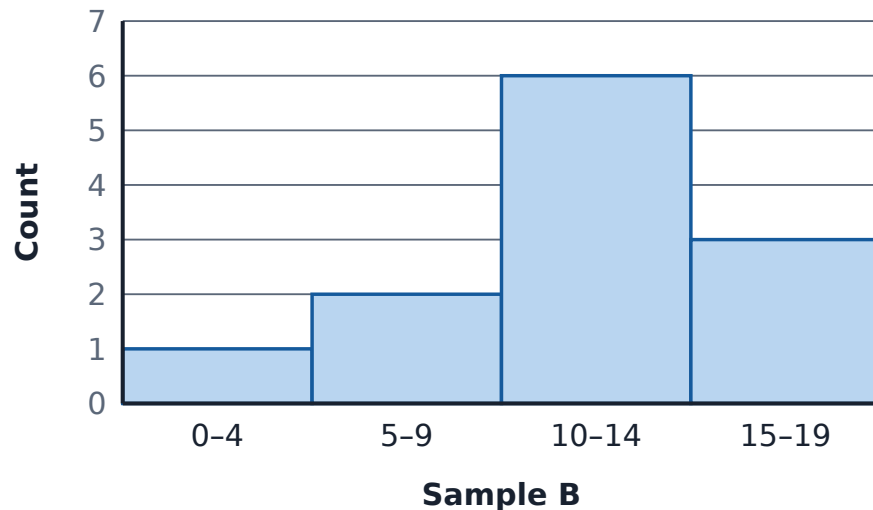
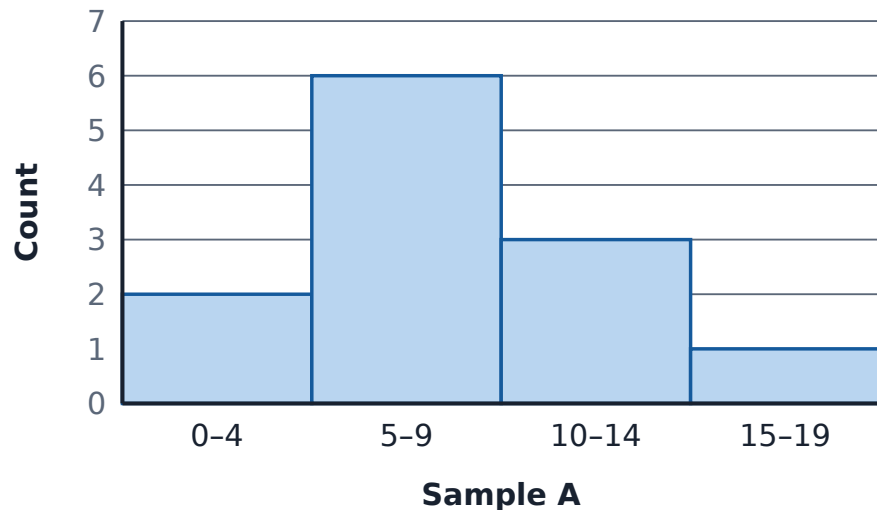
Sample B has more values from 15 to 19 than Sample A.



A. True

B. False

In which interval is the median of Sample B?



A. 0-4

B. 5-9

C. 10-14

D. 15-19

FORWARD 2

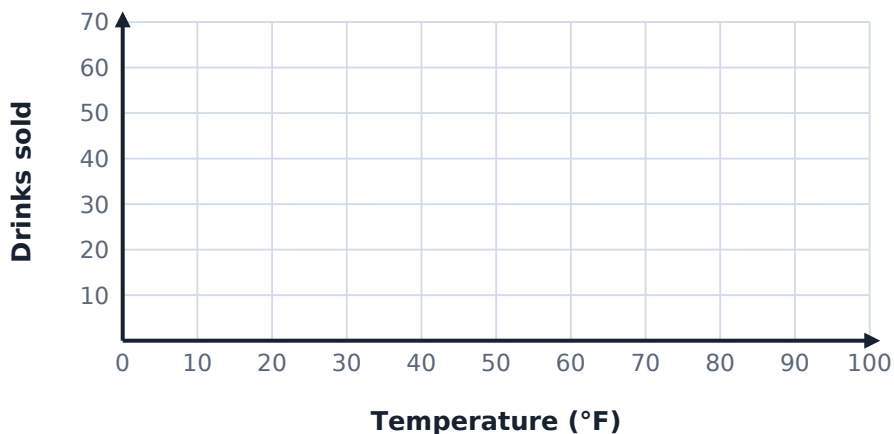
Display numerical data for two variables in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data.

Describe the association between temperature and drinks sold.

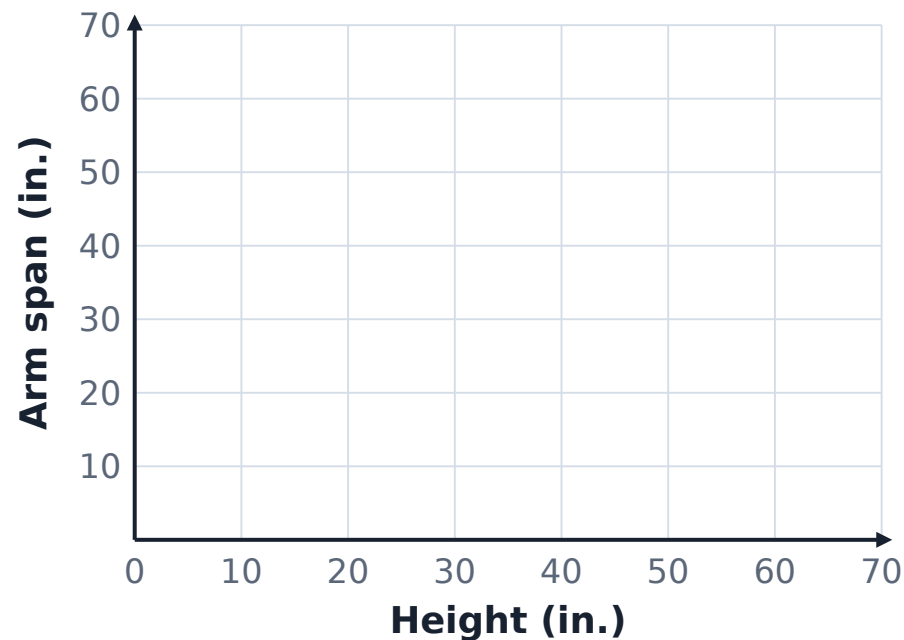
Temp (°F)	60	70	75	85	90
Drinks sold	20	35	40	55	65



Association: _____

Make a scatter plot of the data.

Height (in.)	50	54	58	62
Arm span (in.)	49	55	57	63



A data set gives the height and the shoe size of 20 students.

Which display shows how the two variables are related?

A. Dot plot

B. Scatter plot

C. Box plot

D. Histogram

True or false?

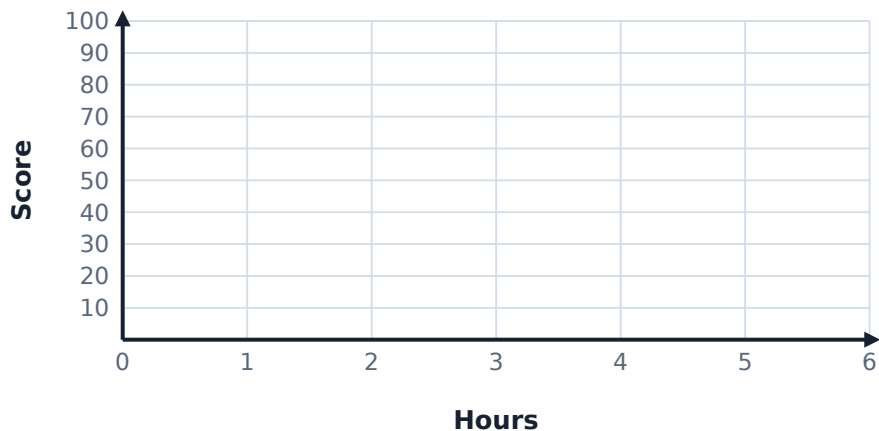
A histogram displays one numerical variable, while a scatter plot displays two numerical variables for each person or object.

- A.** True
- B.** False

Make a scatter plot of the data.

Which point does not fit the pattern?

Hours	1	2	3	4	5
Score	62	68	40	80	86



Point: _____

Display numerical data in a box plot

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	Halfway between two numbers	2.MD.B.6
BACKWARD 3	Reading a number line scale	4.MD.A.2
BACKWARD 4	Finding the median	6.SP.B.5.c
FORWARD 1	Compare populations with box plots	7.SP.B.4
FORWARD 2	Spread and outliers in scatter plots — nearest related (no direct G...	8.SP.A.1

SET 74 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Data: 3, 5, 7, 8, 10, 12, 15

Make a box plot of the data on the number line.



Data: 12, 14, 15, 18, 20, 21, 25, 28

Make a box plot of the data on the number line.



Data: 40, 45, 45, 50, 55, 60, 70

Make a box plot of the data on the number line.



Data: 2, 4, 4, 5, 6, 8, 9, 11, 12

Make a box plot of the data on the number line.



Data: 21, 23, 24, 26, 28, 30, 31, 35

Make a box plot of the data on the number line.



SET 74 • 6.SP.B.4

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

15, 8, 23, 4, 16

Order: _____

Which list is in order from least to greatest?

A. 27, 21, 12, 7

B. 7, 21, 12, 27

C. 12, 7, 21, 27

D. 7, 12, 21, 27

True or false?

These numbers are in order from least to greatest: 30, 33, 31, 35

- A. True
- B. False

Order from least to greatest.

62, 26, 60, 20, 66

Order: _____

Which number is least?

A. 108

B. 118

C. 180

D. 801

SET 74 • 6.SP.B.4

BACKWARD 2

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 12 and 18?

Number: _____

What number is halfway between points A and B?



Number: _____

True or false?

The number halfway between 40 and 50 is 45.

- A. True
- B. False

What number is halfway between 6 and 10?

- A.** 4
- B.** 7
- C.** 8
- D.** 16

What number is halfway between 0 and 30?

Number: _____

SET 74 • 6.SP.B.4

BACKWARD 3

Reading a number line scale • Grade 4 • Measurement & Data • 4.MD.A.2

What number is at point A?



A = _____

What number is at point B?



B = _____

True or false?

Point C is at 12.



A. True

B. False

What number is at point D?



A. 150

B. 170

C. 175

D. 180

What number is at point E?



E = _____

SET 74 • 6.SP.B.4

BACKWARD 4

Finding the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

3, 5, 7, 8, 10, 12, 15

Median: _____

Find the median.

12, 14, 15, 18, 20, 21, 25, 28

Median: _____

True or false?

The median of 40, 45, 45, 50, 55, 60, 70 is 50.

A. True

B. False

What is the median of 2, 4, 4, 5, 6, 8, 9, 11, 12?

A. 5

B. 6

C. 7

D. 8

Find the median of the lower half.

3, 5, 7

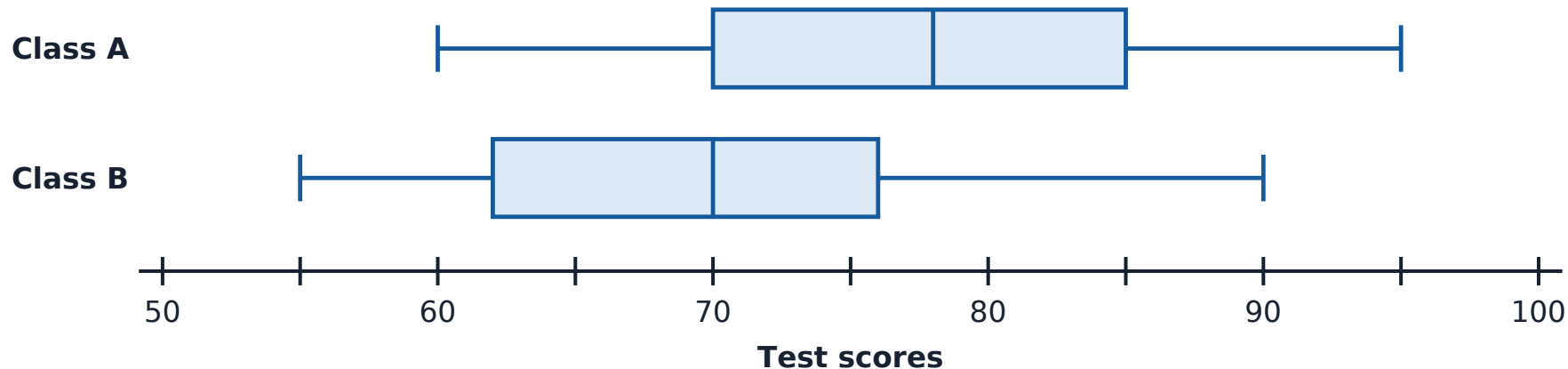
Median: _____

SET 74 • 6.SP.B.4

FORWARD 1

Compare populations with box plots • Grade 7 • Statistics & Probability •
7.SP.B.4

Which class has the greater median score?

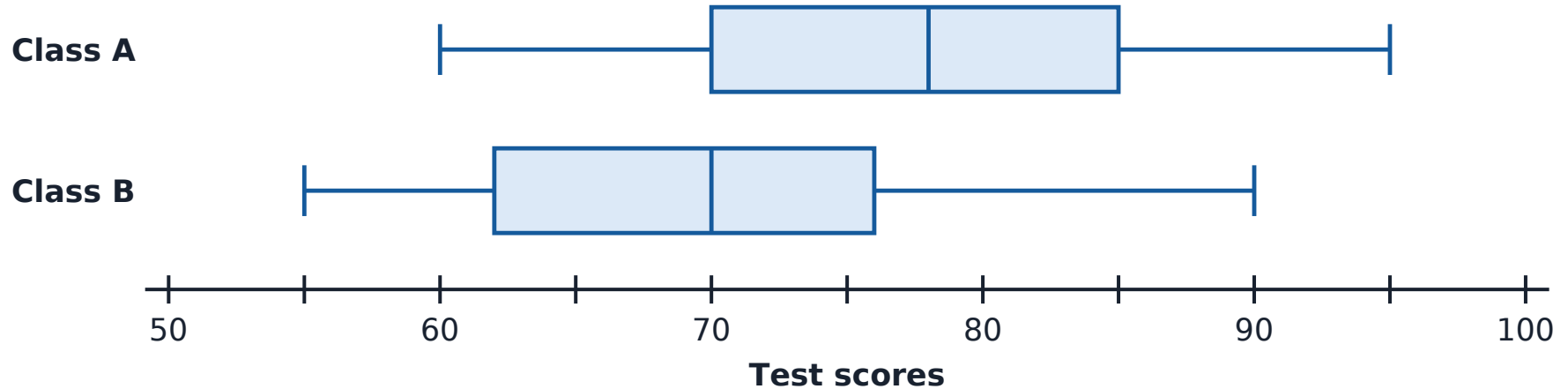


A. Class A

B. Class B

C. They are the same.

By how many points do the medians differ?



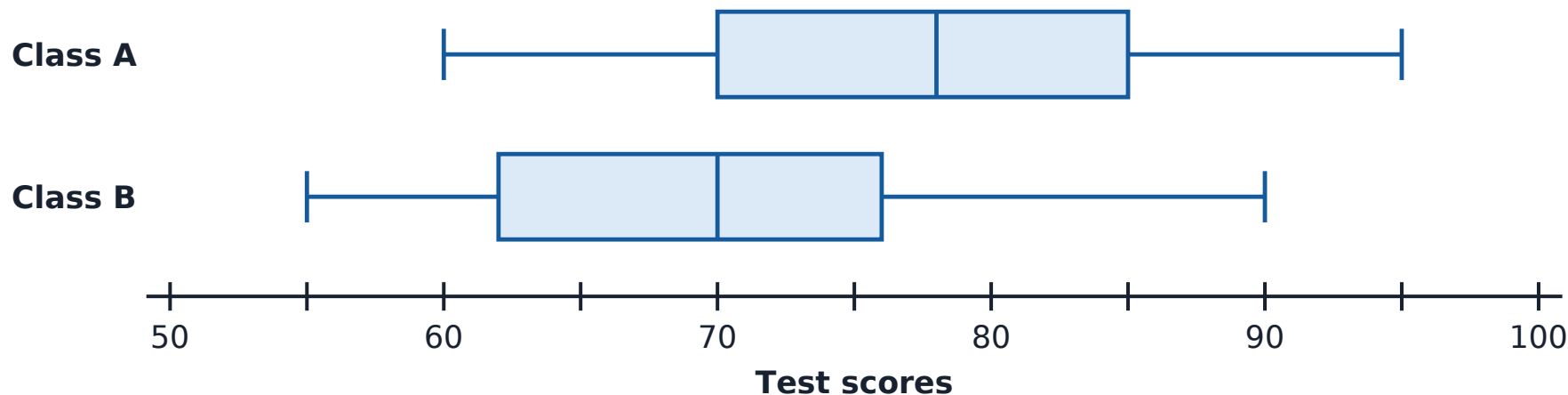
A. 3

B. 5

C. 8

D. 15

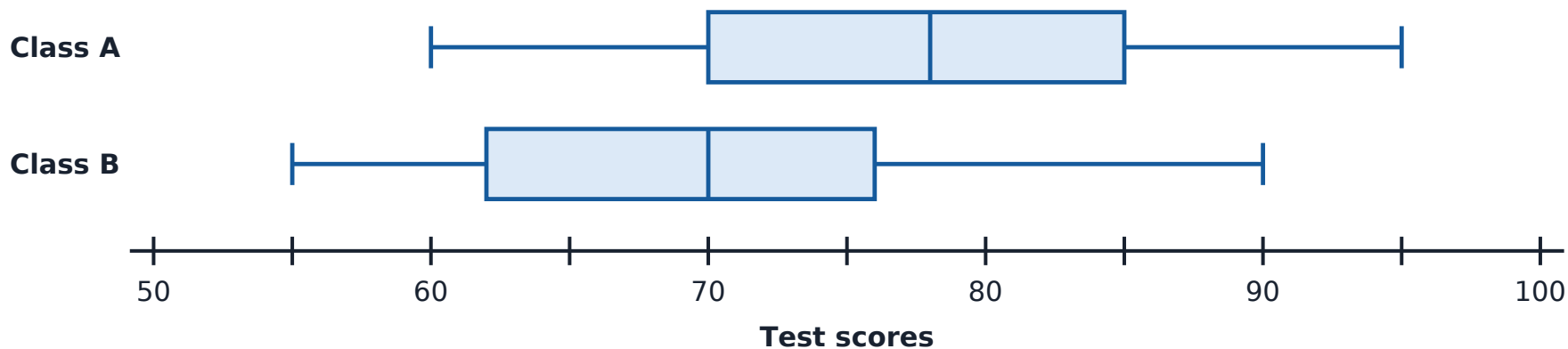
What is the interquartile range of Class A's scores?



IQR: _____

True or false?

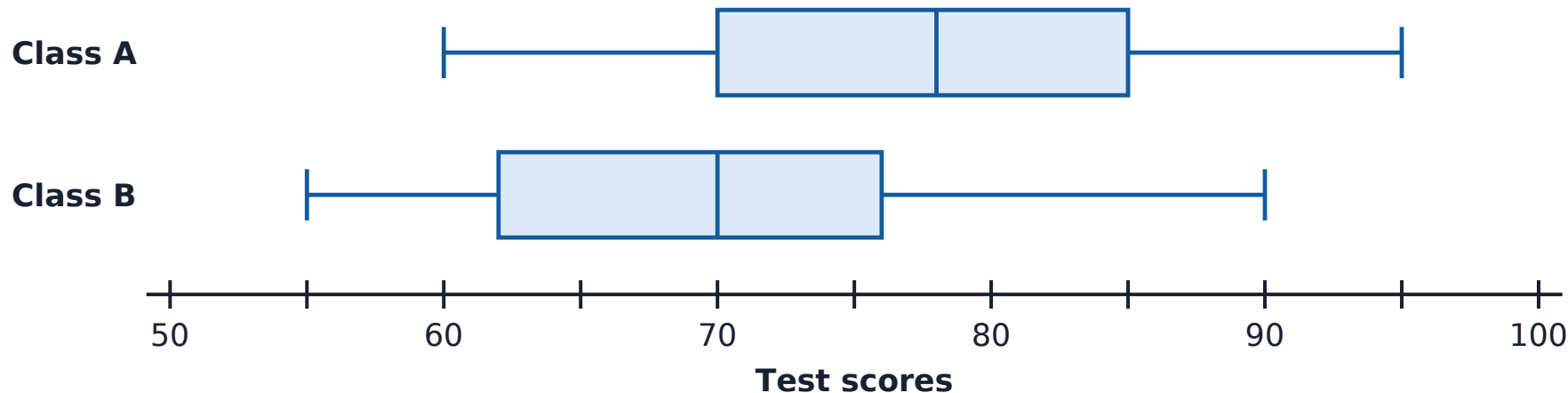
At least half of Class A scored 78 or higher.



A. True

B. False

Which class had the highest single score?



Class: _____

FORWARD 2

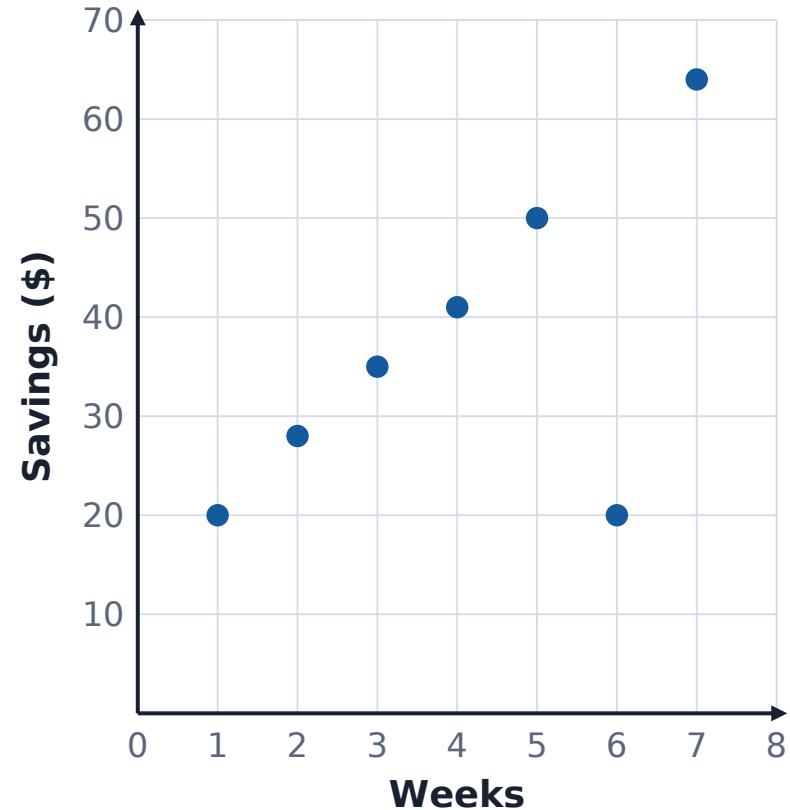
Spread and outliers in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point is an outlier?

Write its coordinates.

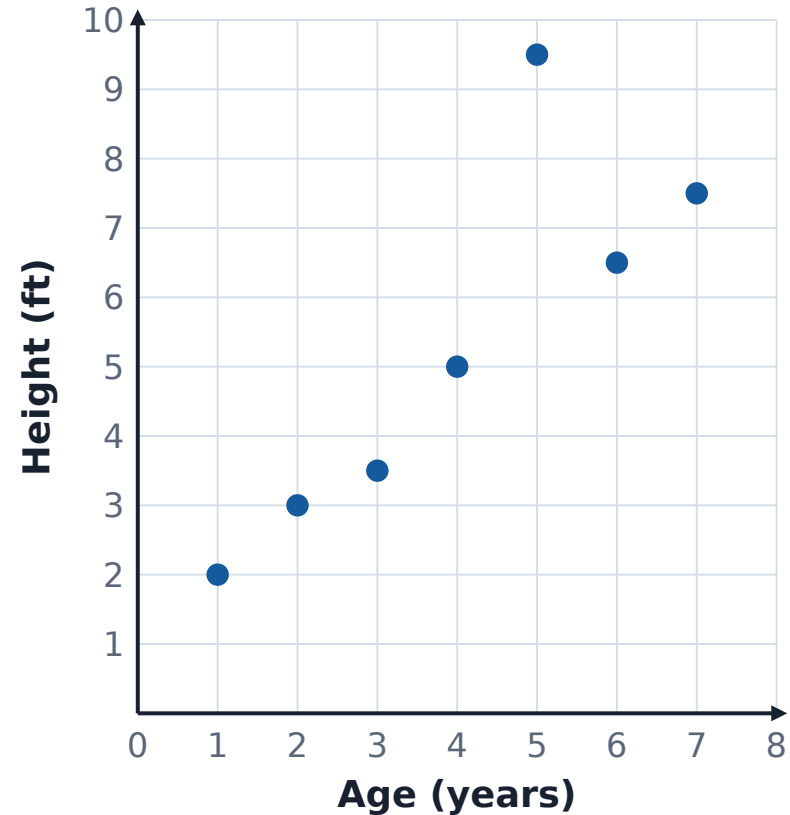
Outlier: _____



Which point is an outlier?

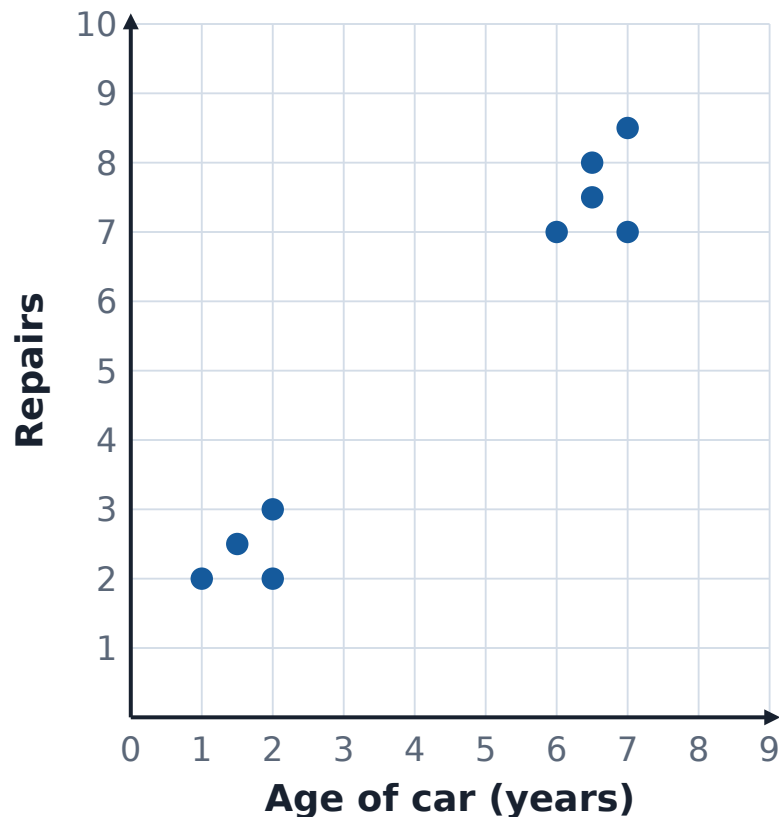
Write its coordinates.

Outlier: _____



Which describes the scatter plot?

- A.** A single outlier
- B.** A strong positive association
- C.** No pattern
- D.** Two clusters



True or false?

An outlier in a scatter plot is a point that is far from the overall pattern.

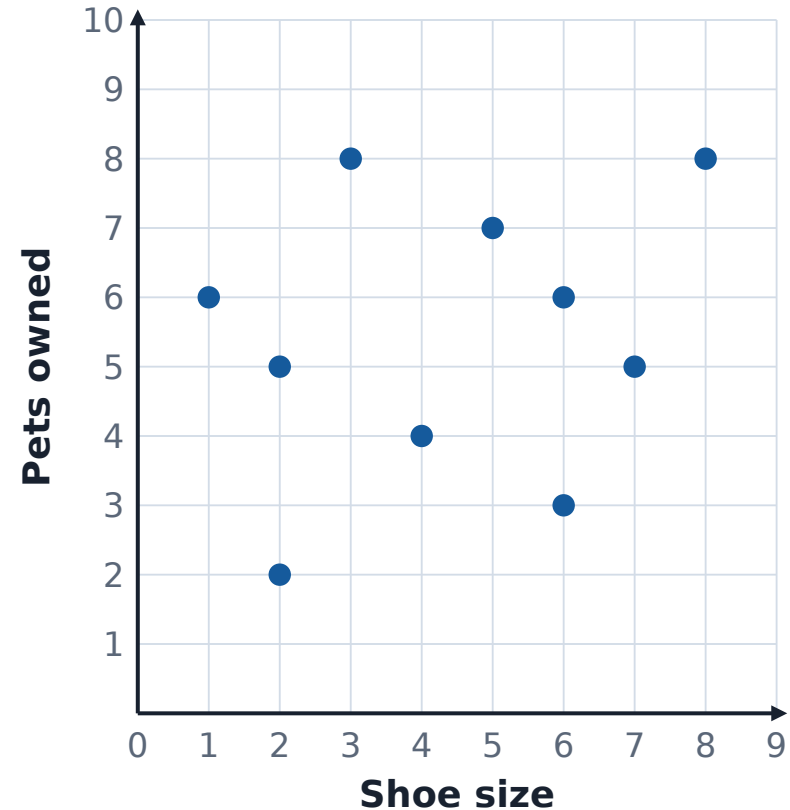
- A.** True
- B.** False

True or false?

The scatter plot shows no association.

A. True

B. False



Report the number of observations

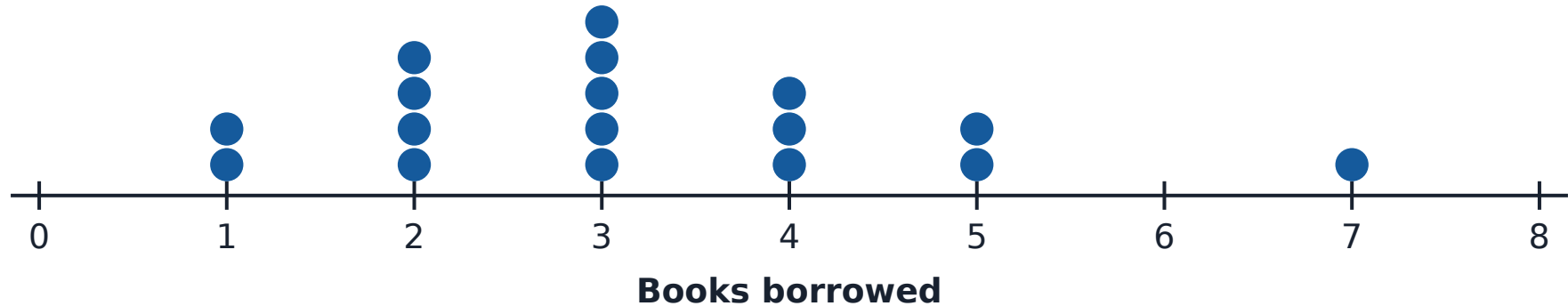
MAIN	Grade 6 main question	6.SP.B.5.a
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Totals from a bar graph	3.MD.B.3
BACKWARD 3	Adding several two-digit numbers	2.NBT.B.6
FORWARD 1	Use a random sample to make inferences	7.SP.A.2
FORWARD 2	Relative frequencies in two-way tables	8.SP.A.4

SET 75 • 6.SP.B.5.a

MAIN

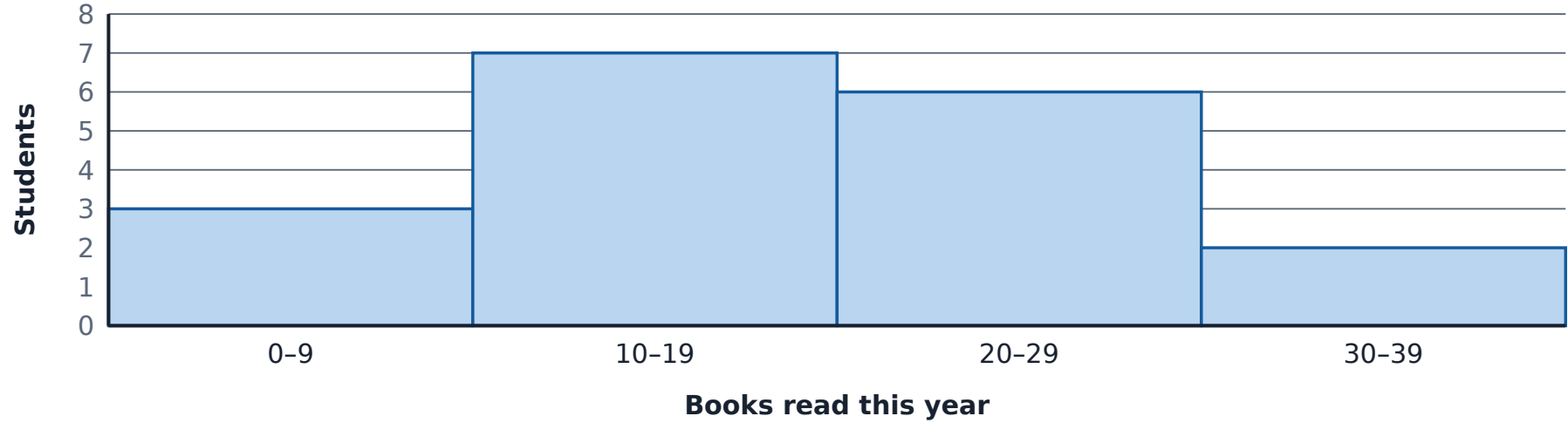
Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.a

The dot plot shows how many books students borrowed.
How many observations are in the data set?



Observations: _____

How many students are represented in the histogram?



Students: _____

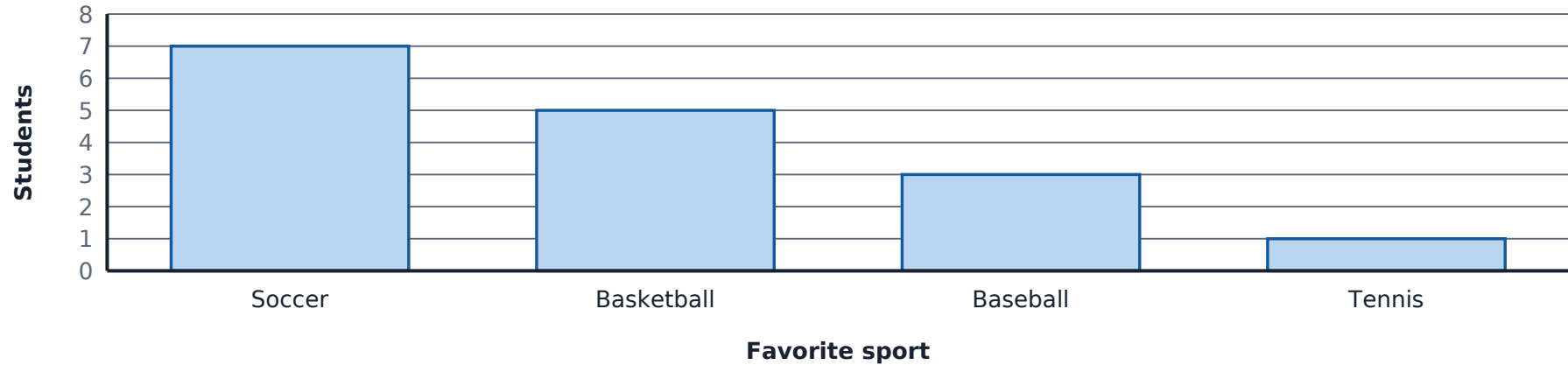
The table shows how many siblings students have.
How many observations are in the data set?

Siblings	0	1	2	3
Students	4	9	6	2

Observations: _____

The bar graph shows survey answers.

How many observations are in the data set?



Observations: _____

Data: 12, 15, 9, 15, 20, 11, 13, 18, 15, 10

How many observations are in the data set?

Observations: _____

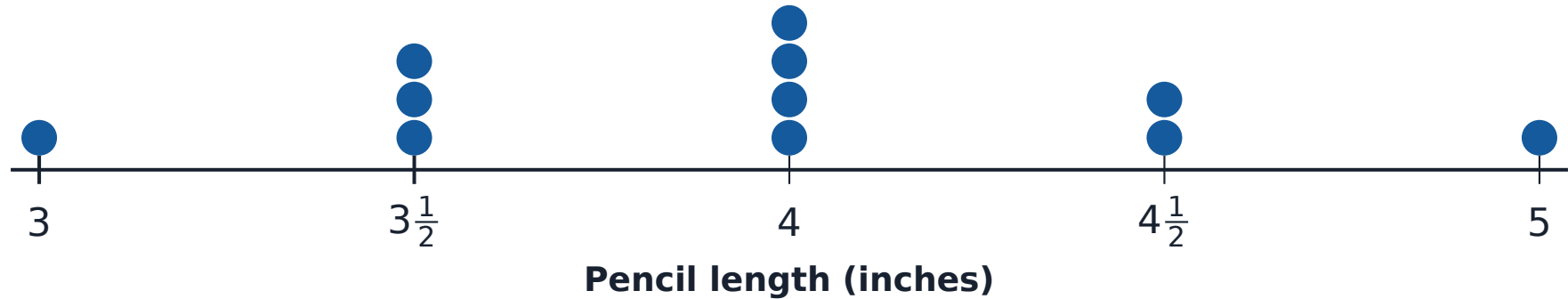
SET 75 • 6.SP.B.5.a

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

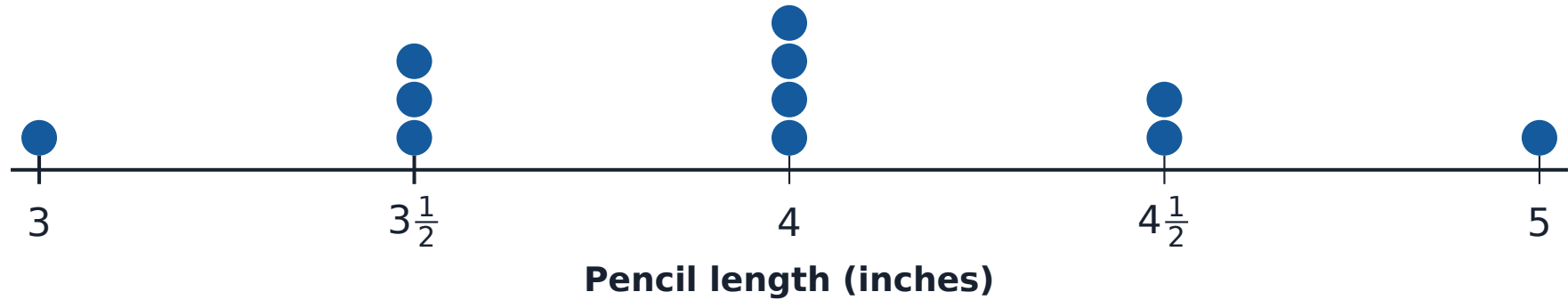
The line plot shows pencil lengths.

How many pencils are 4 inches long?



Pencils: _____

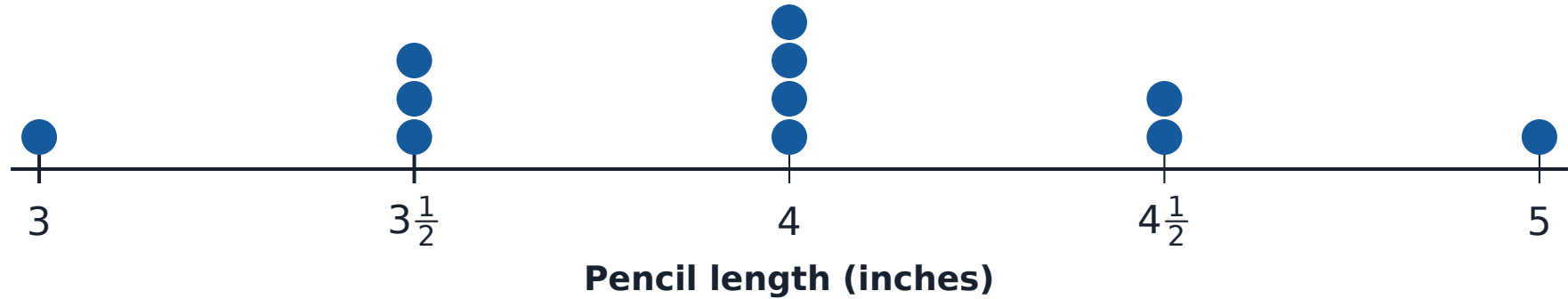
The line plot shows pencil lengths.
How many pencils were measured?



Pencils: _____

True or false?

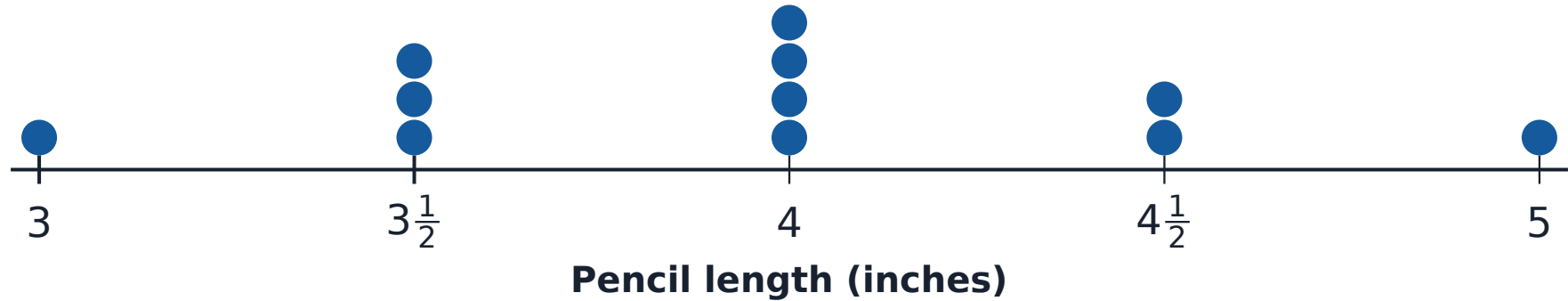
Two pencils are $4\frac{1}{2}$ inches long.



A. True

B. False

How many pencils are longer than 4 inches?



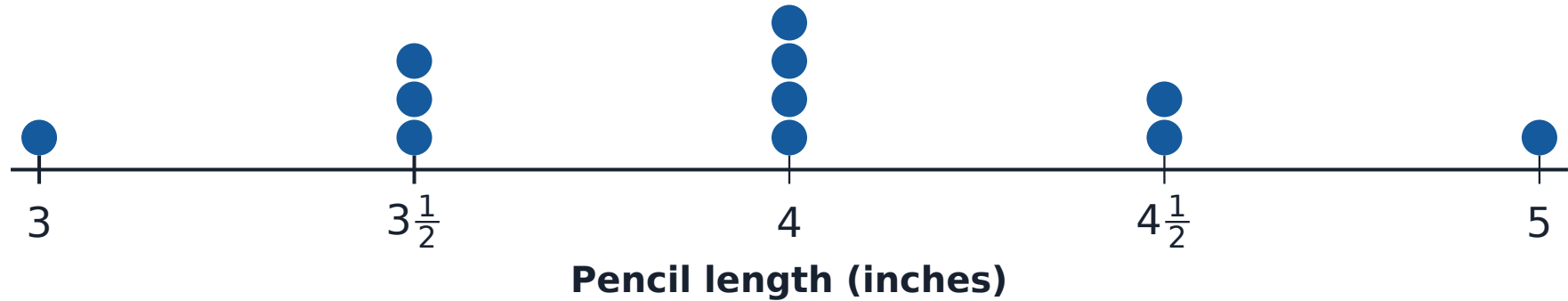
A. 2

B. 3

C. 4

D. 7

How many pencils are 5 inches long?



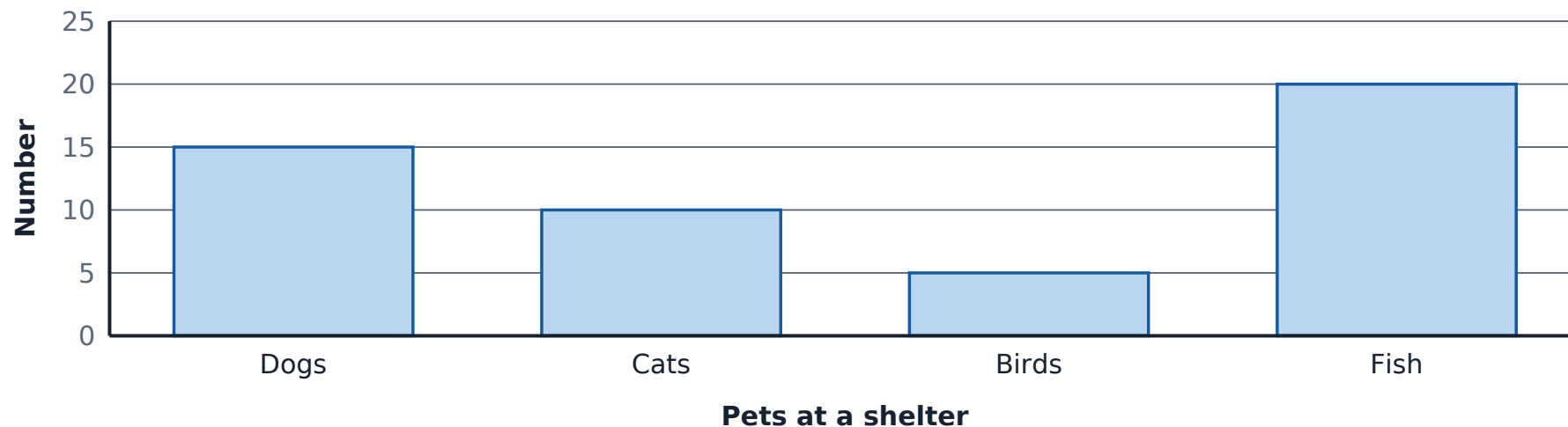
Pencils: _____

SET 75 • 6.SP.B.5.a

BACKWARD 2

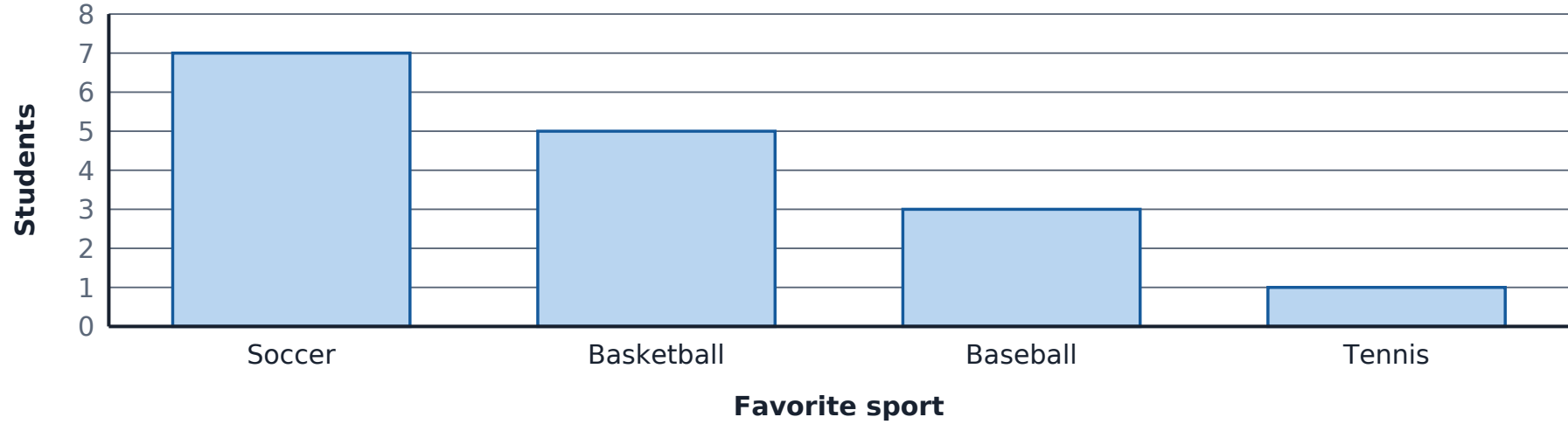
Totals from a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many pets are at the shelter in all?



Pets: _____

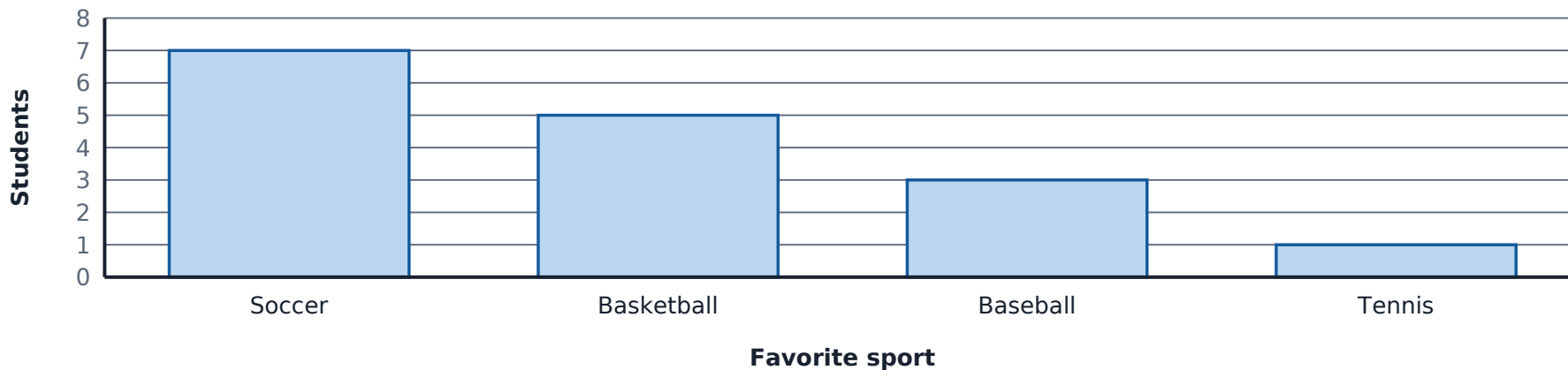
How many students chose soccer or tennis?



Students: _____

True or false?

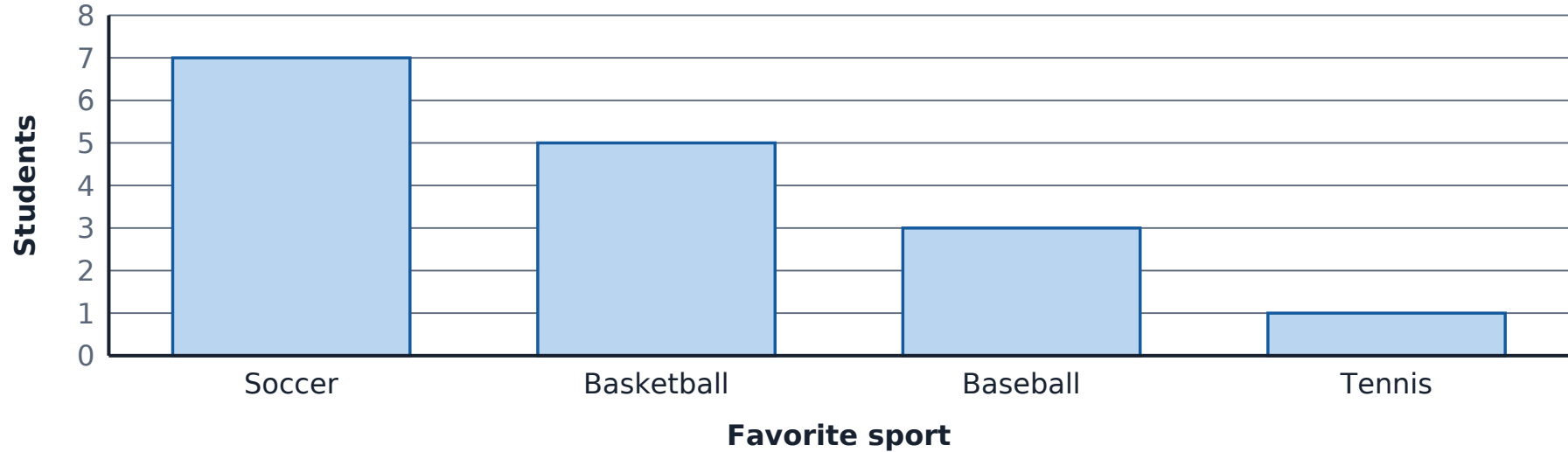
12 students chose soccer or basketball.



A. True

B. False

How many students did NOT choose soccer?



A. 5

B. 7

C. 9

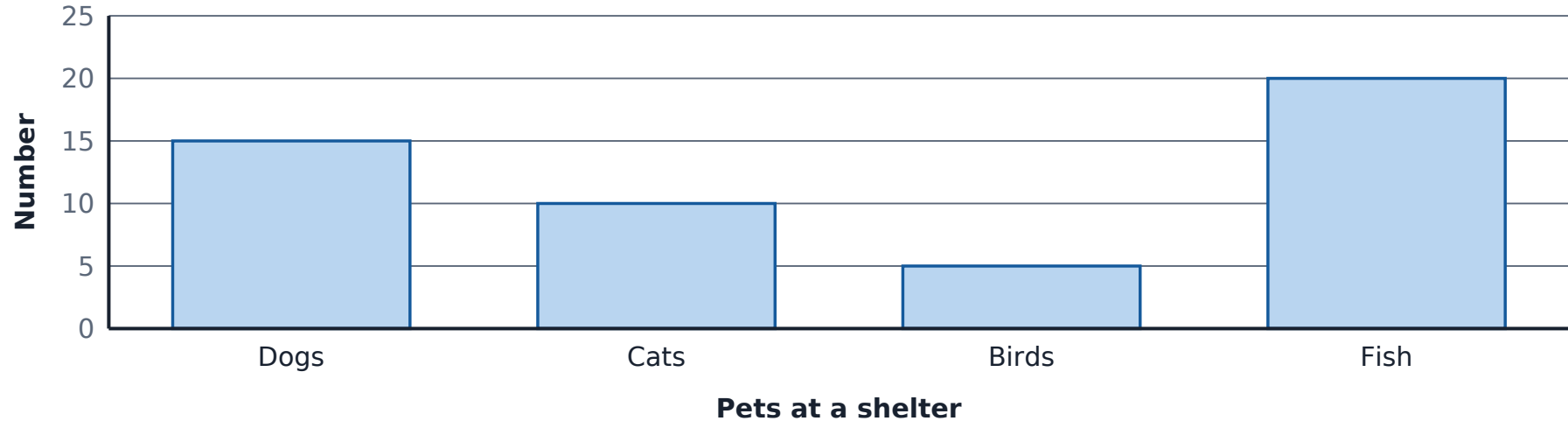
D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q4/5

How many dogs and cats are at the shelter together?



Number: _____

SET 75 • 6.SP.B.5.a

BACKWARD 3

Adding several two-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6

Add.

$$14 + 23 + 31 + 12$$

Sum: _____

Add.

$$25 + 18 + 30$$

Sum: _____

True or false?

$$16 + 16 + 20 + 8 = 60$$

A. True

B. False

$$41 + 27 + 13 + 9 = ?$$

- A.** 80
- B.** 90
- C.** 91
- D.** 100

Add.

$$35 + 22 + 18 + 15$$

Sum: _____

SET 75 • 6.SP.B.5.a

FORWARD 1

Use a random sample to make inferences • Grade 7 • Statistics & Probability •
7.SP.A.2

A researcher surveys 80 randomly chosen students out of 900.

What is the sample size?

A. 80

B. 820

C. 900

D. 980

True or false?

A larger random sample generally gives a better estimate about a population.

A. True

B. False

In a random sample of 25 students out of 500, 5 like jazz.

Estimate how many of the 500 students like jazz.

Estimate: _____

In a random sample of 50 voters, 30 support a new park. There are 2,000 voters.

About how many voters support the park?

A. 30

B. 600

C. 1,200

D. 1,500

Two random samples of 20 students each found 12 and 14 students who like pizza best.

What is the mean of the two sample results?

Mean: _____

SET 75 • 6.SP.B.5.a

FORWARD 2

Relative frequencies in two-way tables • Grade 8 • Statistics & Probability •
8.SP.A.4

What fraction of 7th graders have a pet? What fraction of 8th graders have a pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

Grade 7: _____

Grade 8: _____

Compare the relative frequencies. Which grade has a greater fraction of pet owners?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

A. Grade 7

B. Grade 8

C. They are the same.

What is the relative frequency of instrument players who also play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. 0.35

B. 0.47

C. 0.7

D. 14

Among students with no instrument, what fraction play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

Fraction: _____

True or false?

Students with no instrument are more likely to play sports than students who play an instrument.

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. True

B. False

Describe the attribute measured and its units

MAIN	Grade 6 main question	6.SP.B.5.b
BACKWARD 1	Choosing measurement units	4.MD.A.1
BACKWARD 2	What a bar graph shows	3.MD.B.3
BACKWARD 3	Choosing a measuring tool	2.MD.A.1
FORWARD 1	How data are collected: sampling methods	7.SP.A.1
FORWARD 2	Association between two categorical variables	8.SP.A.4

SET 76 • 6.SP.B.5.b

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.b

A data set lists the heights of 25 plants in centimeters.
What attribute is being measured, and in what unit?

Attribute: _____

Unit: _____

A data set records the time each student spends on homework.

Which unit is most likely?

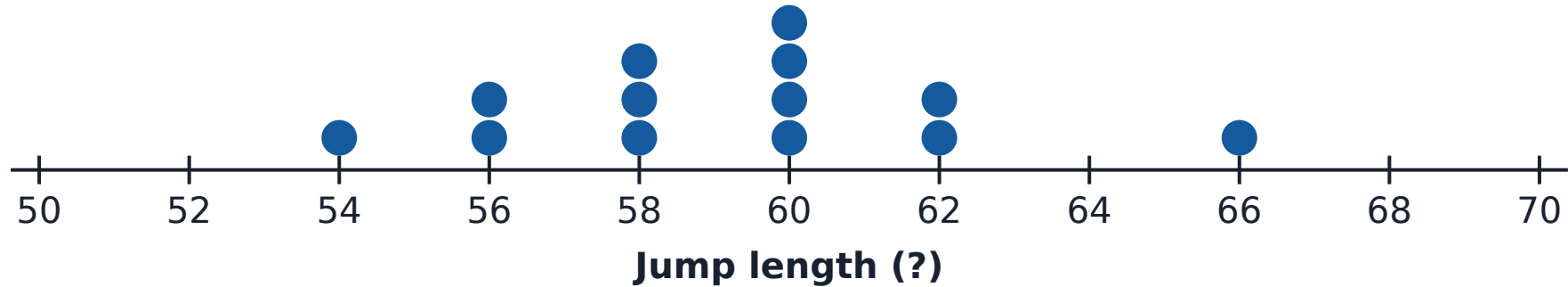
A. Degrees

B. Pounds

C. Inches

D. Minutes

The dot plot shows how far students jumped.
Which unit is most likely?



- A.** Inches **B.** Seconds **C.** Pounds **D.** Miles

A data set lists the mass of each of 30 apples.

What attribute is being measured, and what is a reasonable unit?

Attribute: _____

Unit: _____

A data set records how long each student's trip to school takes.

Which describes the attribute and its unit?

- A.** Distance; minutes
- B.** Time; minutes
- C.** Speed; pounds
- D.** Time; miles

SET 76 • 6.SP.B.5.b

BACKWARD 1

Choosing measurement units • Grade 4 • Measurement & Data • 4.MD.A.1

Which unit is best for measuring the length of a pencil?

- A.** Centimeters
- B.** Kilograms
- C.** Meters
- D.** Kilometers

True or false?

A kilogram is heavier than a gram.

A. True

B. False

Which unit measures liquid volume?

- A.** Meters
- B.** Liters
- C.** Grams
- D.** Seconds

Name one unit used to measure time.

Unit: _____

True or false?

An inch is longer than a foot.

A. True

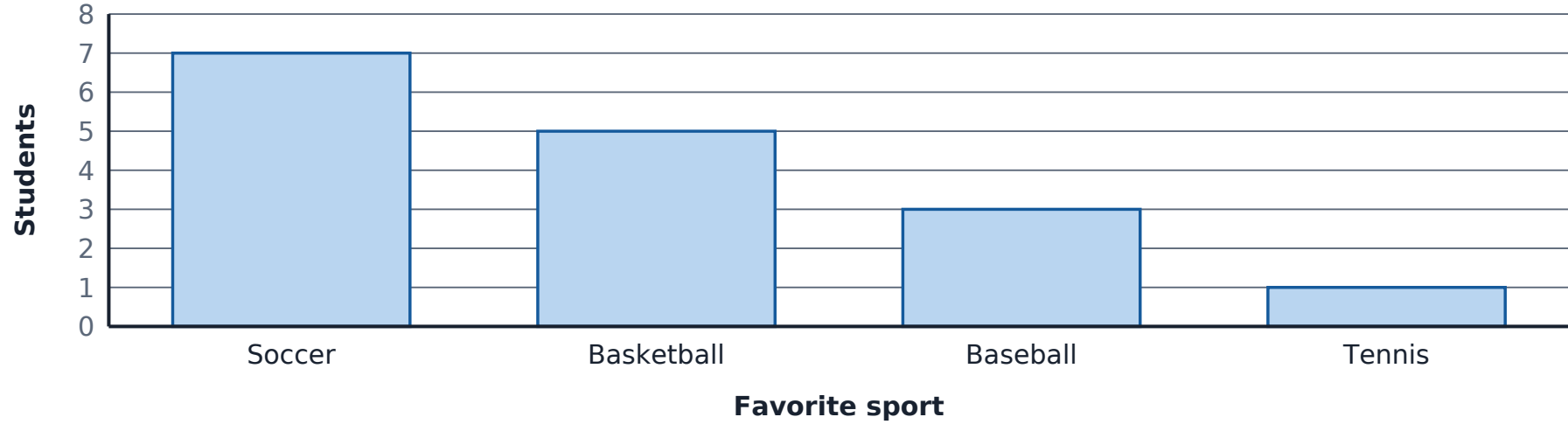
B. False

SET 76 • 6.SP.B.5.b

BACKWARD 2

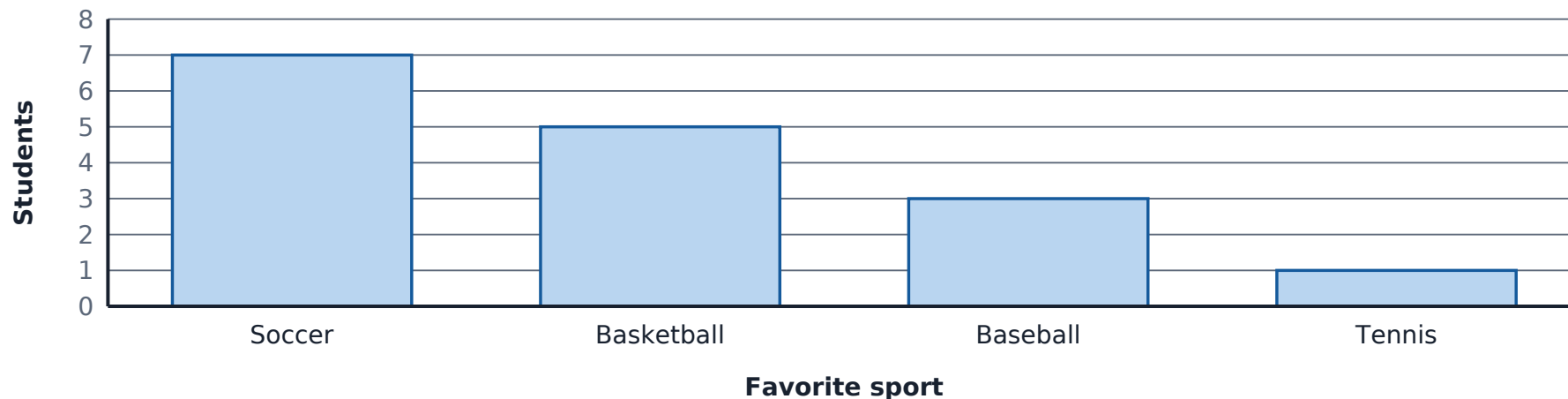
What a bar graph shows • Grade 3 • Measurement & Data • 3.MD.B.3

How many categories does the graph show?



Categories: _____

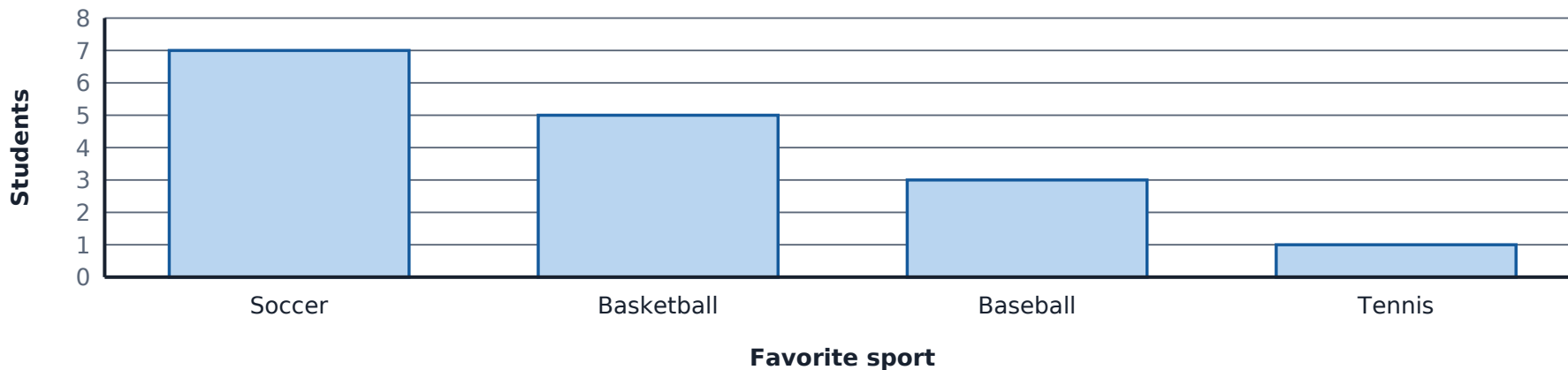
What does the height of each bar show?



- A.** The number of sports
- B.** The score of a game
- C.** The length of a game
- D.** The number of students

True or false?

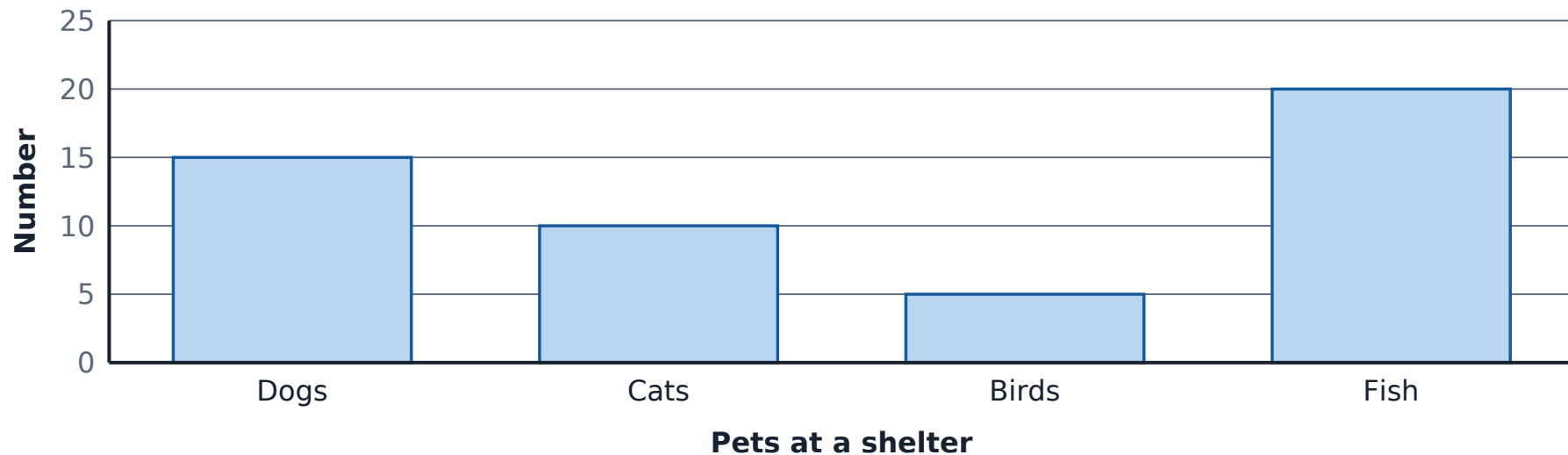
"Tennis" is one of the categories in the graph.



A. True

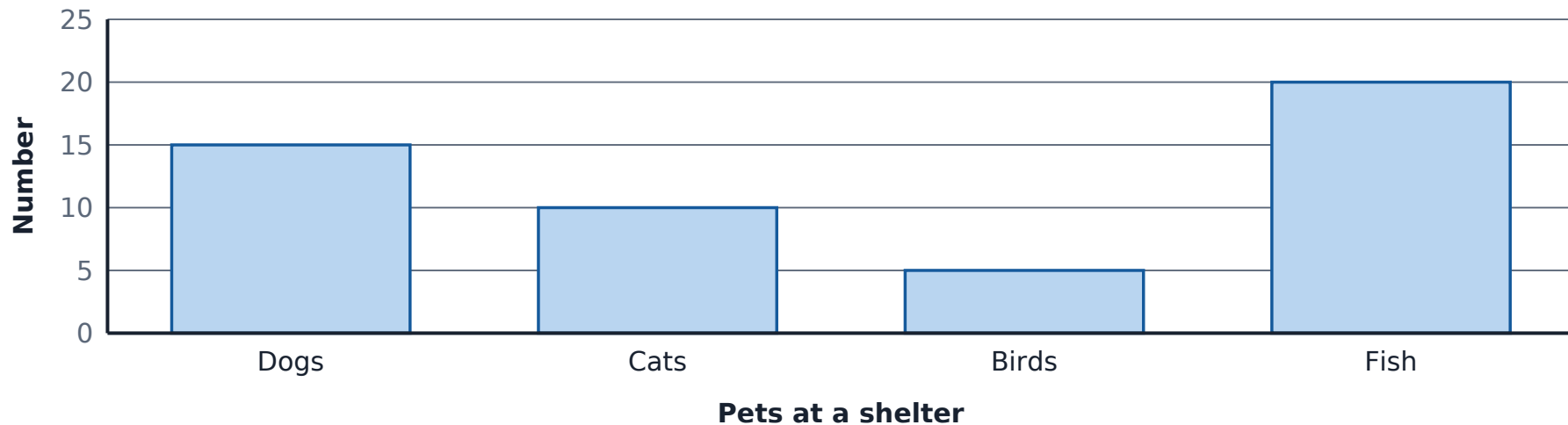
B. False

Which is a category in the graph?



- A.** 25 **B.** Birds **C.** Horses **D.** Soccer

What is being counted in the graph?



Answer: _____

SET 76 • 6.SP.B.5.b

BACKWARD 3

Choosing a measuring tool • Grade 2 • Measurement & Data • 2.MD.A.1

Which tool measures the length of a table?

- A.** Tape measure
- B.** Thermometer
- C.** Scale
- D.** Clock

Which tool measures how heavy an apple is?

- A.** Ruler
- B.** Clock
- C.** Measuring cup
- D.** Scale

True or false?

A clock measures time.

A. True

B. False

Which tool measures temperature?

- A.** Thermometer
- B.** Scale
- C.** Ruler
- D.** Yardstick

Name a tool that measures the length of a pencil.

Tool: _____

SET 76 • 6.SP.B.5.b

FORWARD 1

How data are collected: sampling methods • Grade 7 • Statistics & Probability •
7.SP.A.1

Which method gives a random sample of the students in a school?

- A.** Ask the students in the library
- B.** Ask the first 20 students to arrive
- C.** Ask your friends
- D.** Draw names from a hat that holds every student's name

True or false?

Surveying people who walk out of a gym is a good way to learn how often all people in a town exercise.

A. True

B. False

A town wants to know how residents feel about a new library.

Describe a way to choose a random sample.

Method: _____

Which sample is most likely to be biased?

- A.** Asking only people at a dog park whether they like dogs
- B.** Picking every 10th name from a school list
- C.** Choosing 50 residents at random from a town list
- D.** Drawing 30 names from a hat of all members

True or false?

In a random sample, every member of the population has an equal chance of being chosen.

- A. True
- B. False

SET 76 • 6.SP.B.5.b

FORWARD 2

Association between two categorical variables • Grade 8 • Statistics & Probability • 8.SP.A.4

Which pair of variables is categorical and could be shown in a two-way table?

- A.** Hours slept and test score
- B.** Height and weight
- C.** Age and arm span
- D.** Grade level and way of getting to school

What fraction of 6th graders walk to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

Fraction: _____

What fraction of 8th graders walk to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

Fraction: _____

What is the relative frequency of 8th graders who ride to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

A. $\frac{1}{10}$

B. $\frac{5}{23}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

True or false?

8th graders are more likely to walk to school than 6th graders.

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

A. True

B. False

Find the mean of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Adding multi-digit numbers	4.NBT.B.4
BACKWARD 2	Dividing a total into equal groups	4.NBT.B.6
BACKWARD 3	Sharing a total equally	5.MD.B.2
FORWARD 1	Compare populations using means	7.SP.B.4
FORWARD 2	Use a linear model to make predictions — nearest related (no dire...	8.SP.A.3

SET 77 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10, 15

Mean: _____

Five test scores are 82, 90, 75, 88, and 95.

What is the mean score?

Mean: _____

Find the mean.

12, 15, 18, 19

A. 15

B. 16

C. 16.5

D. 64

The dot plot shows goals scored in 8 games.
What is the mean number of goals?



Mean: _____

Find the mean.

3.5, 4.2, 5.0, 6.1, 6.2

Mean: _____

SET 77 • 6.SP.B.5.c

BACKWARD 1

Adding multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$82 + 90 + 75 + 88 + 95$$

Sum: _____

Add.

$$145 + 238 + 97$$

Sum: _____

True or false?

$$56 + 64 + 70 = 190$$

A. True

B. False

$$125 + 175 + 200 = ?$$

A. 400

B. 450

C. 500

D. 550

Add.

$$1,250 + 875$$

Sum: _____

SET 77 • 6.SP.B.5.c

BACKWARD 2

Dividing a total into equal groups • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide.

$$430 \div 5$$

Quotient: _____

Divide.

$$64 \div 4$$

Quotient: _____

True or false?

$$108 \div 4 = 27$$

A. True

B. False

$$252 \div 6 = ?$$

A. 36

B. 41

C. 42

D. 48

Divide.

$$315 \div 7$$

Quotient: _____

SET 77 • 6.SP.B.5.c

BACKWARD 3

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Two friends have 5 and 9 shells. They share the shells equally.

How many shells does each friend get?

Shells: _____

Three boxes hold 2, 7, and 9 books. The books are shared equally among the boxes.

How many books are in each box?

Books: _____

True or false?

Jars holding 4, 4, and 10 cups are shared equally.
Each jar then holds 6 cups.

- A.** True
- B.** False

Four children have 1, 2, 6, and 7 cards. They share the cards equally.

How many cards does each child get?

A. 3

B. 4

C. 5

D. 16

Two cups hold $\frac{1}{3}$ cup and $\frac{2}{3}$ cup of milk. The milk is shared equally.

How much is in each cup?

Cups: _____

SET 77 • 6.SP.B.5.c

FORWARD 1

Compare populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

A random sample of 6th graders has a mean height of 58 in. A random sample of 8th graders has a mean height of 64 in.

How much greater is the 8th graders' mean?

Difference: _____

Two random samples of plant heights have means of 24 cm and 30 cm. Each has a MAD of 3 cm.

How many MADs apart are the means?

A. 1

B. 2

C. 3

D. 6

True or false?

To compare two populations, you can compare the means of random samples from each population.

- A.** True
- B.** False

Sample A: mean 15.2 minutes. Sample B: mean 18.7 minutes.

Which population probably has greater values?

Population: _____

Ten random samples of 20 students gave these mean hours of sleep: 8.1, 8.3, 7.9, 8.0, 8.2, 8.1, 8.4, 7.8, 8.0, 8.2.

What is a good estimate of the population mean?

- A.** About 20 hours
- B.** About 7 hours
- C.** About 8.1 hours
- D.** About 9 hours

FORWARD 2

Use a linear model to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.3

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The equation $y = 2.5x + 40$ models test score y after x hours of study.

Predict the score for 8 hours of study.

Score: _____

The model $y = 15x + 100$ gives the cost y of x tickets.

Predict the cost of 20 tickets.

Cost: _____

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What height does the model predict after 5 weeks?

A. 15 cm

B. 27 cm

C. 36 cm

D. 60 cm

The model $y = -0.5x + 30$ gives the value y of a toy after x years.

Predict the value after 6 years.

Value: _____

True or false?

The model $y = 4x + 10$ predicts $y = 50$ when $x = 10$.

A. True

B. False

Find the median of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	The number halfway between two numbers	5.NBT.B.7
BACKWARD 3	Reading a line plot	4.MD.B.4
FORWARD 1	Compare populations using medians	7.SP.B.4
FORWARD 2	Draw and use a line of best fit — nearest related (no direct Grade...	8.SP.A.2

SET 78 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

12, 5, 9, 20, 7, 14

Median: _____

True or false?

The median of 3, 8, 8, 10, 21 is 8.

A. True

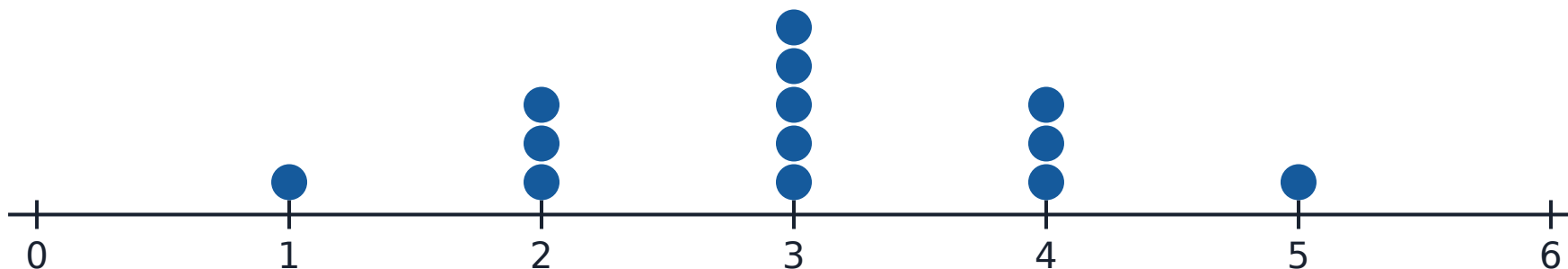
B. False

Heights (in inches): 58, 60, 61, 63, 63

What is the median height?

- A.** 60
- B.** 61
- C.** 61.5
- D.** 63

What is the median of the data in the dot plot?



Median: _____

Find the median.

4.5, 2.1, 3.8, 6.0

Median: _____

SET 78 • 6.SP.B.5.c

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

12, 5, 9, 20, 7, 14

Order: _____

Which list is in order from least to greatest?

A. 58, 60, 61, 63

B. 58, 61, 60, 63

C. 60, 58, 61, 63

D. 63, 61, 60, 58

True or false?

These numbers are in order from least to greatest: 75, 82, 88, 90, 95

- A. True
- B. False

Order from least to greatest.

47, 74, 44, 77

Order: _____

When 3, 9, and 6 are put in order, which number is in the middle?

- A.** 3
- B.** 6
- C.** 9
- D.** 18

SET 78 • 6.SP.B.5.c

BACKWARD 2

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(9 + 12) \div 2$$

Value: _____

Find the value.

$$(3.8 + 4.5) \div 2$$

Value: _____

True or false?

$$(6 + 9) \div 2 = 7.5$$

A. True

B. False

$$(20 + 25) \div 2 = ?$$

A. 22

B. 22.5

C. 23.5

D. 45

Find the value.

$$(3.5 + 4.5) \div 2$$

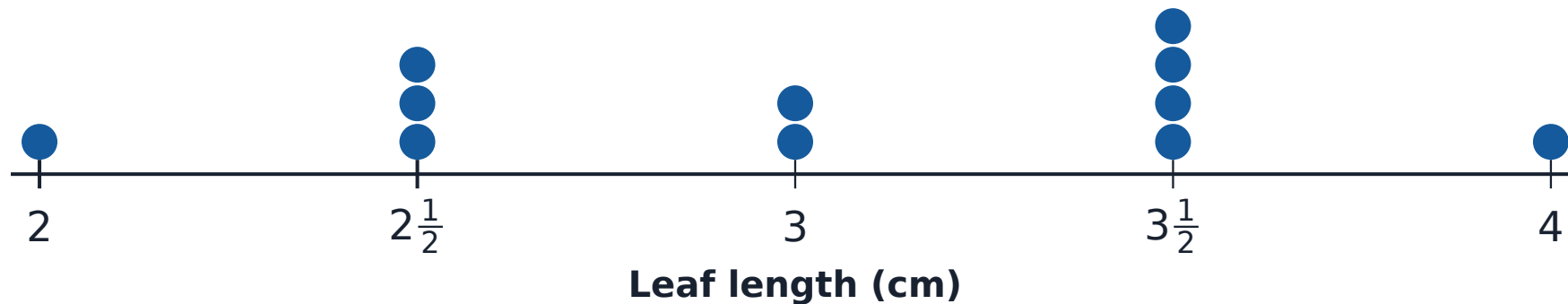
Value: _____

SET 78 • 6.SP.B.5.c

BACKWARD 3

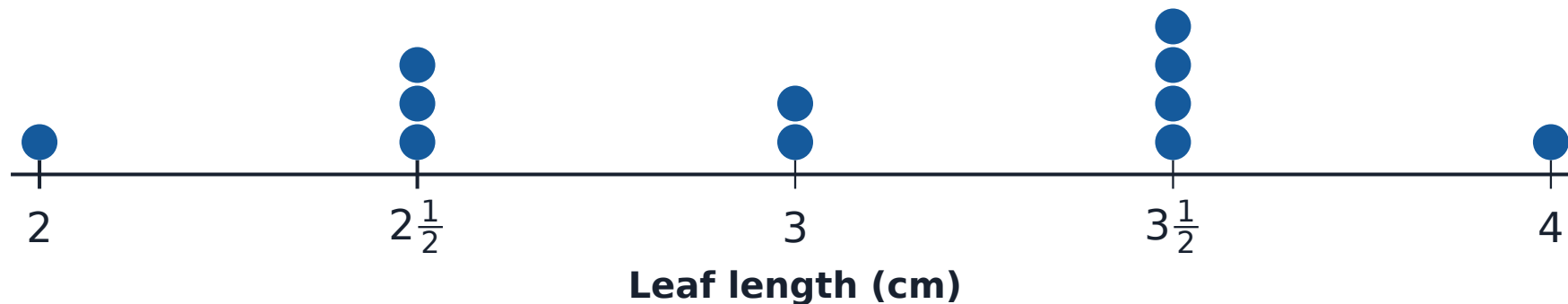
Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many leaves are $3\frac{1}{2}$ cm long?



Leaves: _____

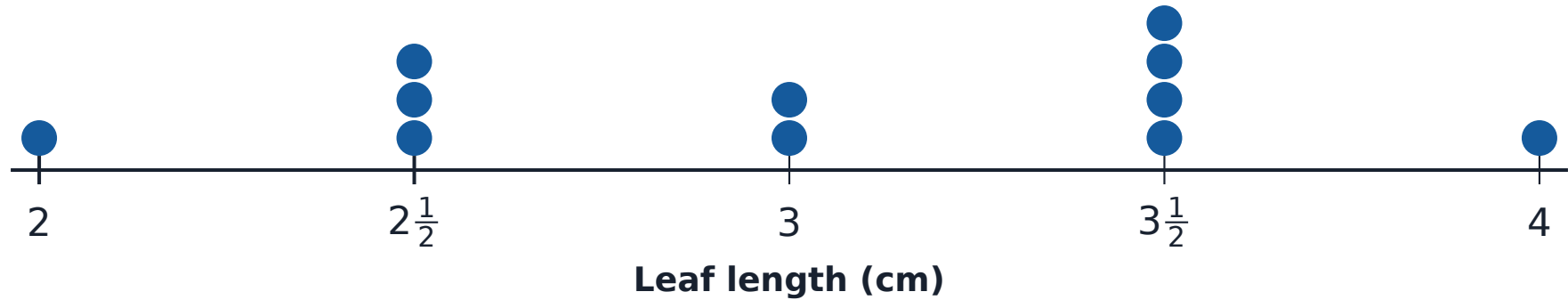
How many leaves were measured?



Leaves: _____

True or false?

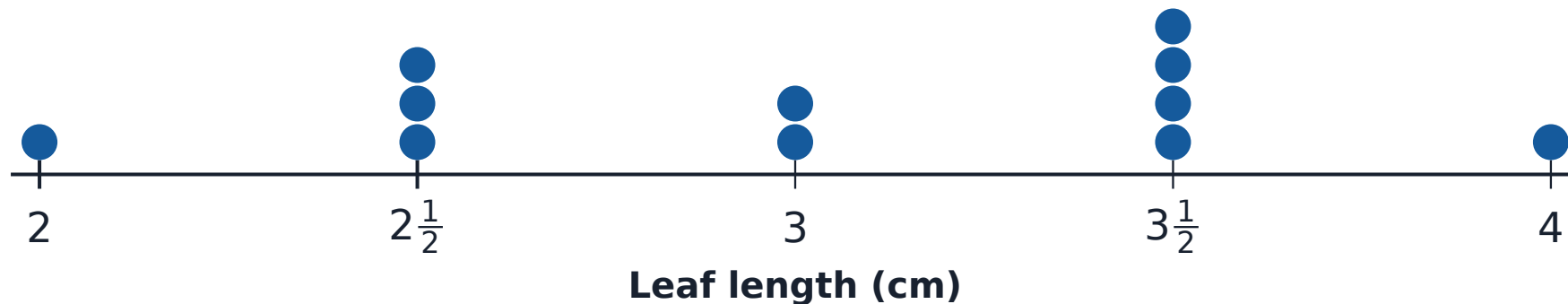
One leaf is 4 cm long.



A. True

B. False

Which leaf length is most common?



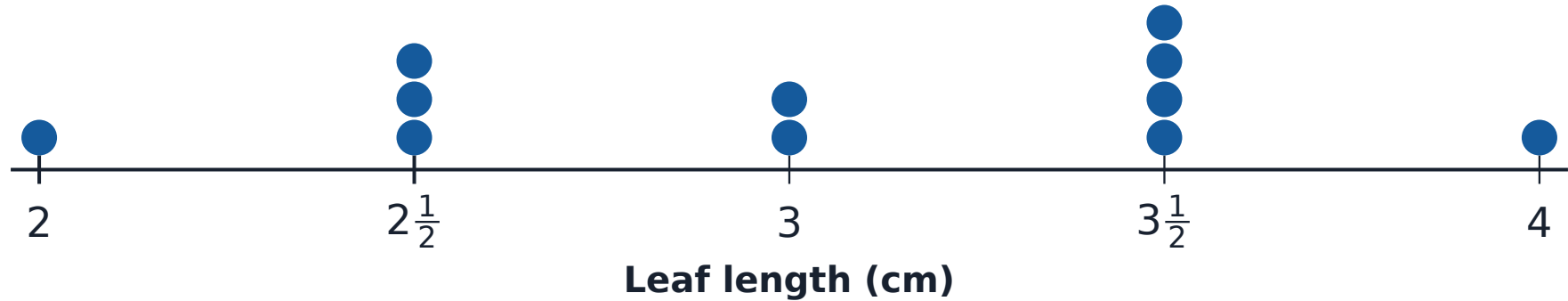
A. $2\frac{1}{2}$ cm

B. 3 cm

C. $3\frac{1}{2}$ cm

D. 4 cm

How many leaves are shorter than 3 cm?



Leaves: _____

SET 78 • 6.SP.B.5.c

FORWARD 1

Compare populations using medians • Grade 7 • Statistics & Probability •
7.SP.B.4

Sample A has a median of 12 and Sample B has a median of 15. The samples have similar variability.

Which conclusion is best?

- A.** Values in population A tend to be greater.
- B.** The populations are the same.
- C.** Values in population B tend to be greater.
- D.** No conclusion is possible.

In random samples, the median commute is 25 minutes in Town X and 18 minutes in Town Y. The IQRs are similar.

In which town are commutes typically longer?

Town: _____

Sample A: median 42, IQR 8. Sample B: median 50, IQR 8.

The difference in medians is how many times the IQR?

Times: _____

True or false?

When data are skewed, comparing medians is often better than comparing means.

- A.** True
- B.** False

Sample X has a median of 7.5 hours of sleep. Sample Y has a median of 8.5 hours. Both IQRs are 1 hour.

Which statement is best?

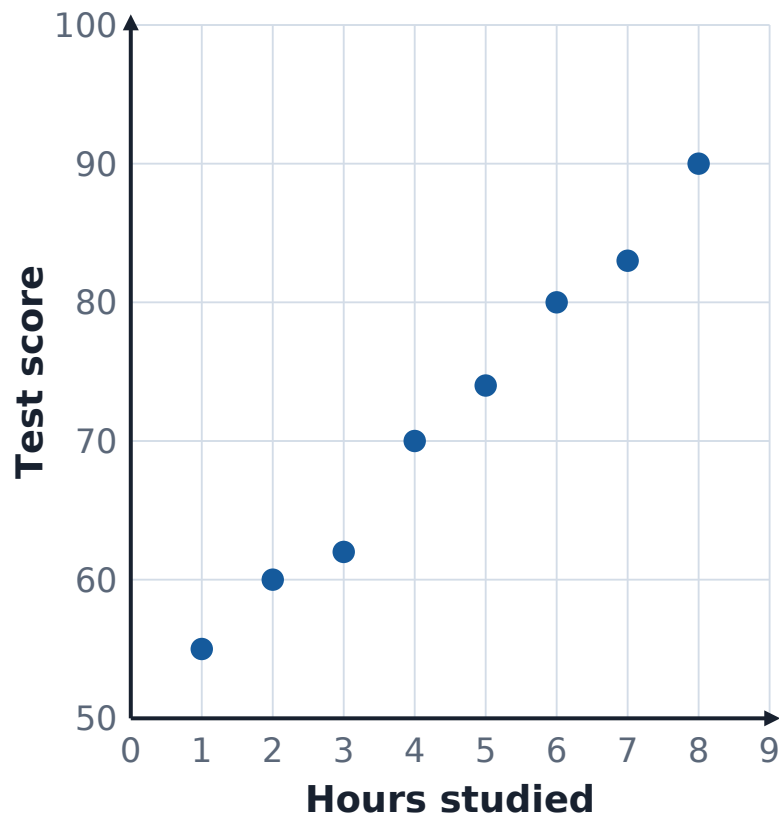
- A.** Population Y probably sleeps more.
- B.** Population X probably sleeps more.
- C.** Nothing can be said.
- D.** They sleep the same amount.

FORWARD 2

Draw and use a line of best fit • Grade 8 • Statistics & Probability • 8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Draw a line of best fit for the data.

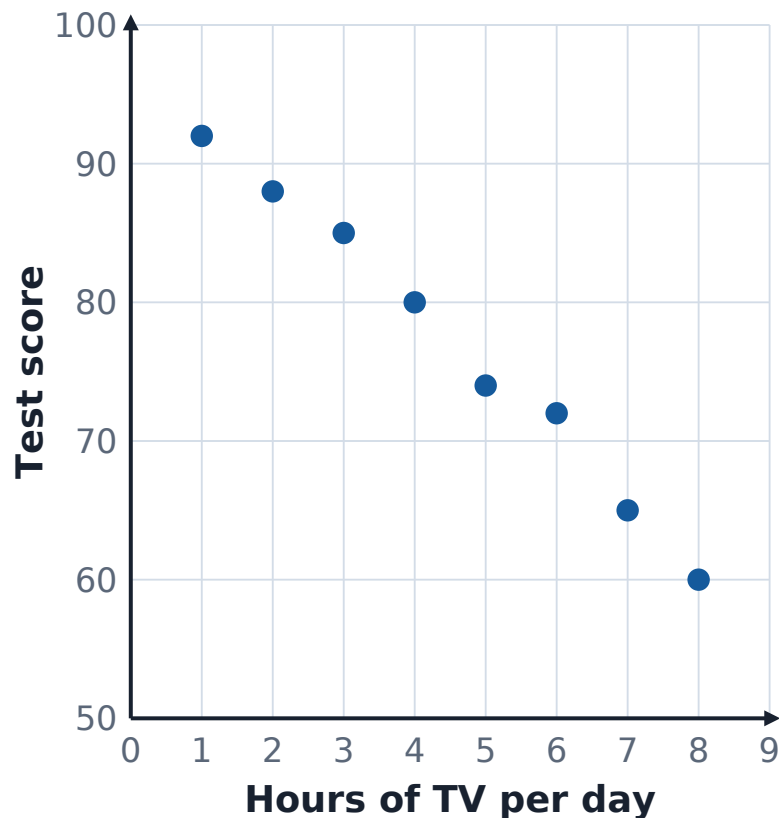


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q1/5 • NEAREST RELATED

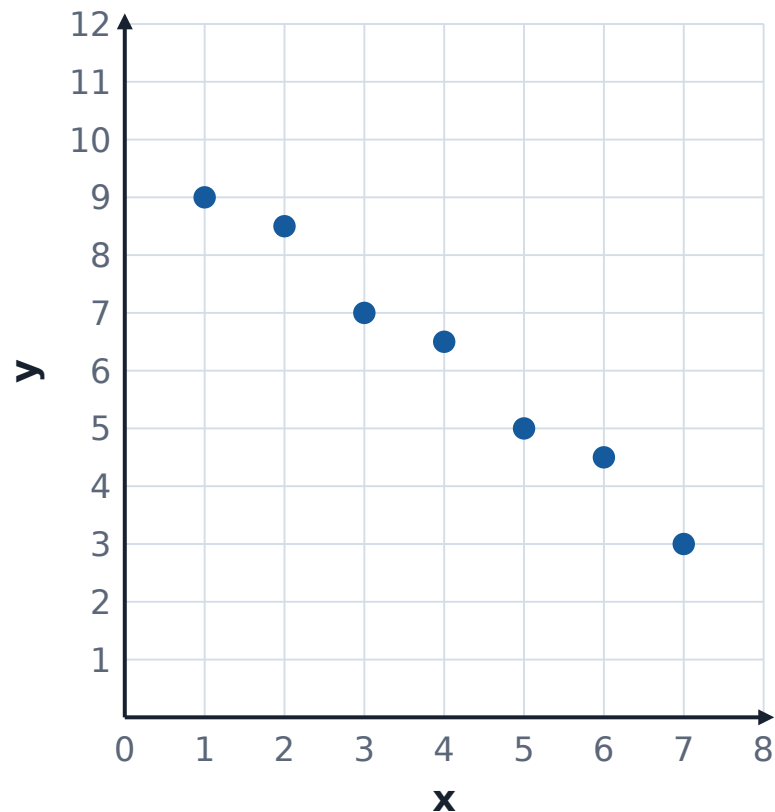
Draw a line of best fit for the data.



A line of best fit should have ____.

- A.** all points below it
- B.** no points near it
- C.** all points above it
- D.** about the same number of points above and below it

Draw a line of best fit for the data.



True or false?

A line of best fit must pass through the first and last points.

A. True

B. False

Find the interquartile range of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	The number halfway between two numbers	5.NBT.B.7
BACKWARD 3	Finding a difference	2.NBT.B.5
FORWARD 1	Difference in centers as a multiple of the IQR	7.SP.B.3
FORWARD 2	Outliers and clusters in scatter plots — nearest related (no direct...	8.SP.A.1

SET 79 • 6.SP.B.5.c

MAIN

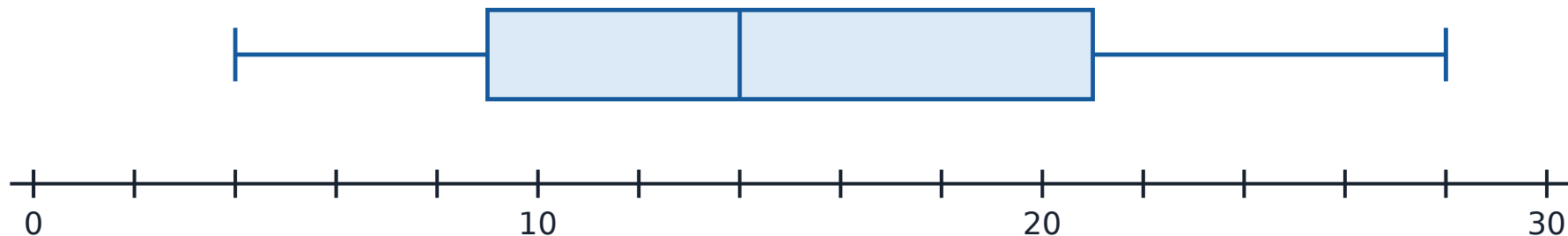
Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the interquartile range.

2, 4, 5, 7, 9, 11, 12, 15

IQR: _____

What is the interquartile range of the data in the box plot?



IQR: _____

What is the interquartile range?

10, 12, 15, 18, 20, 25, 30

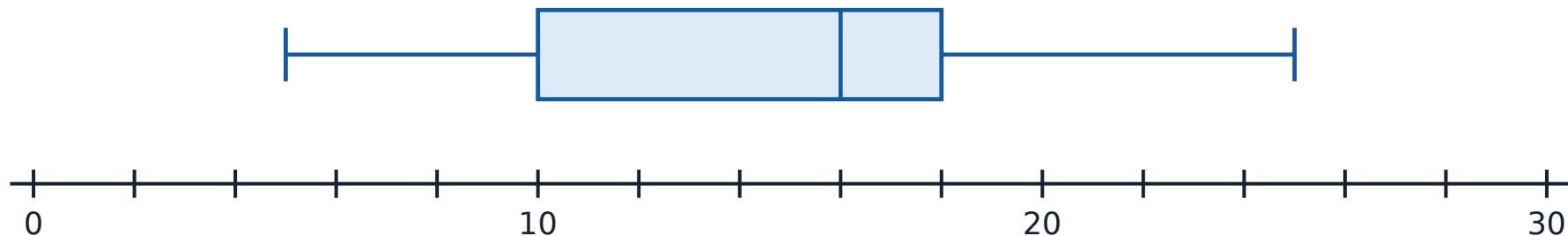
- A.** 5
- B.** 13
- C.** 18
- D.** 20

Find the interquartile range.

3, 6, 6, 8, 9, 12, 14, 15, 18

IQR: _____

What is the interquartile range of the data in the box plot?



IQR: _____

SET 79 • 6.SP.B.5.c

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

15, 9, 2, 12, 5

Order: _____

Which list is in order from least to greatest?

A. 18, 15, 12, 10

B. 10, 15, 12, 18

C. 12, 10, 15, 18

D. 10, 12, 15, 18

True or false?

These numbers are in order from least to greatest: 3, 6, 9, 8

A. True

B. False

Order from least to greatest.

25, 20, 30, 10

Order: _____

Which number is least?

A. 2

B. 5

C. 9

D. 12

SET 79 • 6.SP.B.5.c

BACKWARD 2

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(4 + 5) \div 2$$

Value: _____

Find the value.

$$(11 + 12) \div 2$$

Value: _____

True or false?

$$(7 + 10) \div 2 = 8.5$$

A. True

B. False

$$(15 + 18) \div 2 = ?$$

A. 16

B. 16.5

C. 17.5

D. 33

Find the value.

$$(2.5 + 3.5) \div 2$$

Value: _____

SET 79 • 6.SP.B.5.c

BACKWARD 3

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Subtract.

$$25 - 12$$

Difference: _____

Subtract.

$$21 - 9$$

Difference: _____

True or false?

$$30 - 14 = 16$$

A. True

B. False

$$18 - 10 = ?$$

- A.** 7
- B.** 8
- C.** 9
- D.** 28

Subtract.

$$40 - 23$$

Difference: _____

SET 79 • 6.SP.B.5.c

FORWARD 1

Difference in centers as a multiple of the IQR • Grade 7 • Statistics & Probability • 7.SP.B.3

Two data sets have medians of 40 and 55. Each has an IQR of 5.

The difference in medians is how many times the IQR?

Times: _____

Two data sets have medians of 20 and 26. Each has an IQR of 3.

How many IQRs apart are the medians?

A. 2

B. 3

C. 6

D. 9

True or false?

Two data sets with medians that differ by less than one IQR probably overlap a lot.

- A. True
- B. False

Class A's median is 72 and Class B's median is 84. Each class has an IQR of 12.

The difference in medians is how many times the IQR?

Times: _____

Which pair of data sets has the LEAST overlap?

- A.** Medians 10 and 14, each IQR 8
- B.** Medians 10 and 30, each IQR 4
- C.** Medians 10 and 11, each IQR 3
- D.** Medians 10 and 12, each IQR 4

FORWARD 2

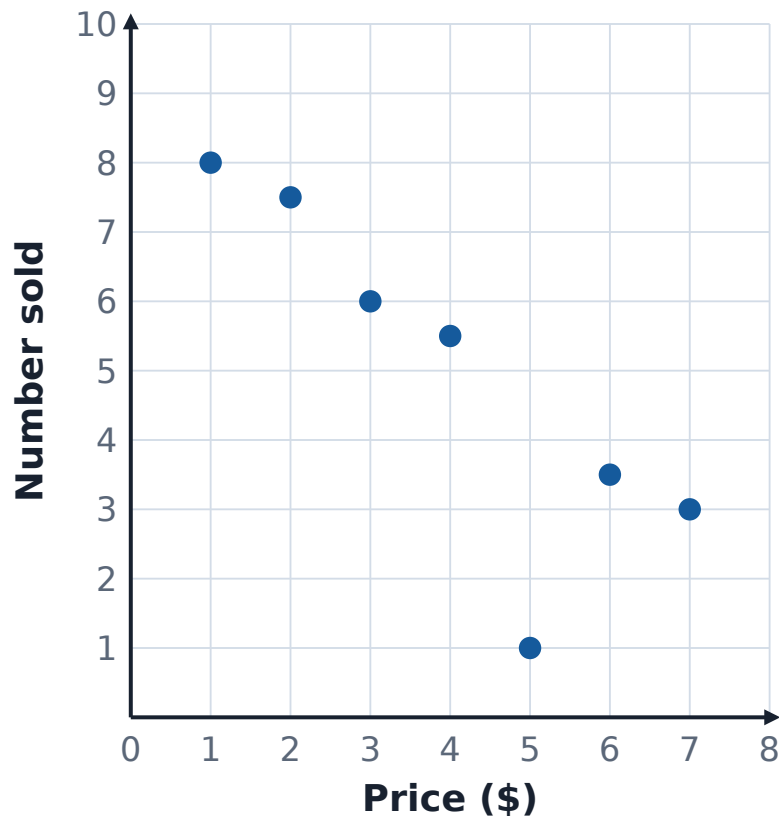
Outliers and clusters in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point is an outlier?

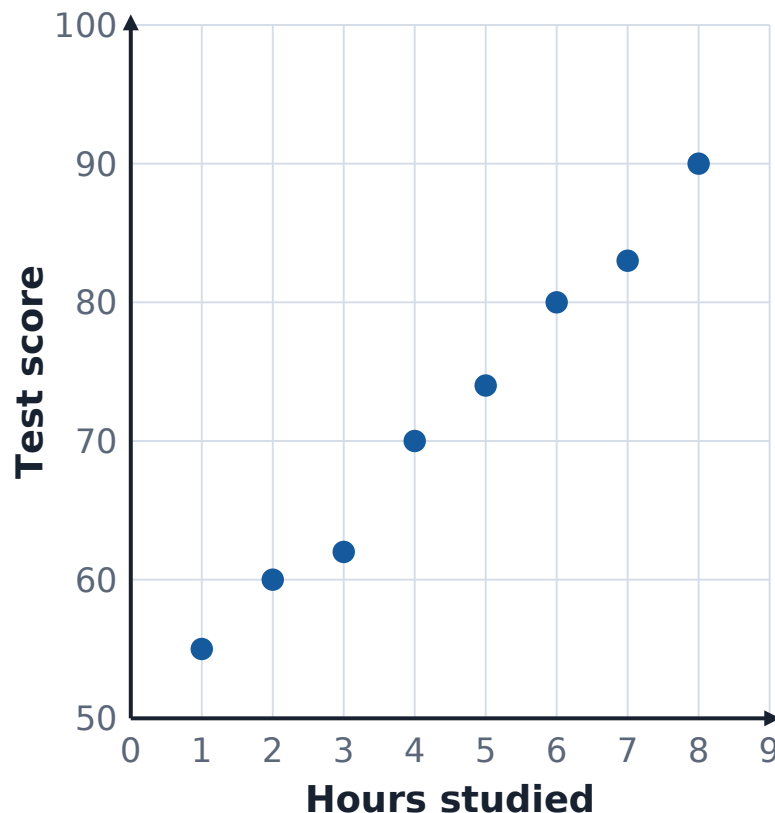
Write its coordinates.

Outlier: _____



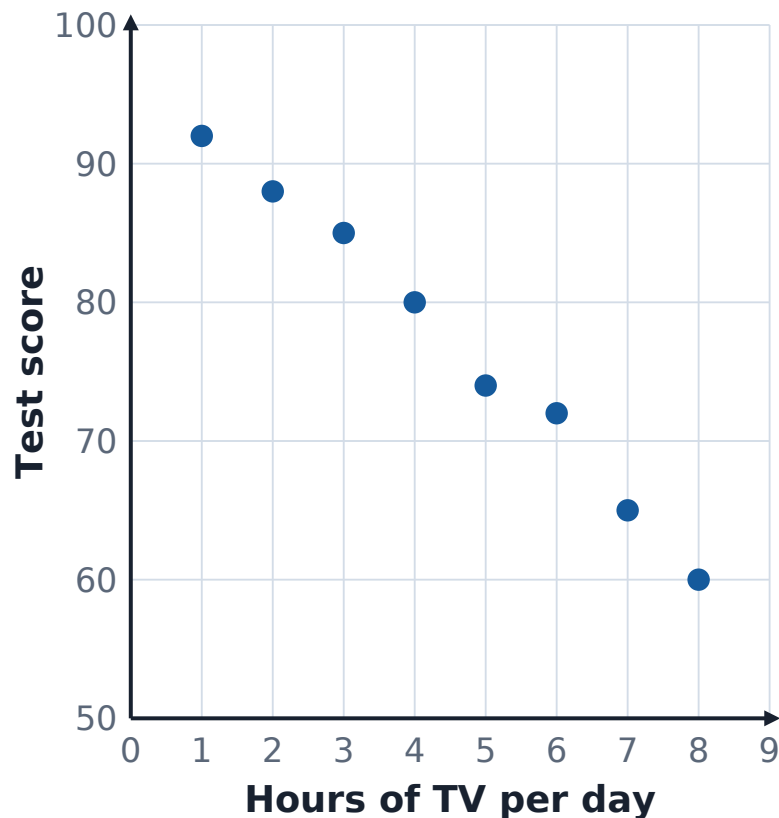
Which point would be an outlier if it were added to the scatter plot?

- A.** (5, 73)
- B.** (7, 50)
- C.** (7, 85)
- D.** (3, 63)



Which describes the association in the scatter plot?

- A.** No association
- B.** Negative linear
- C.** Nonlinear
- D.** Positive linear

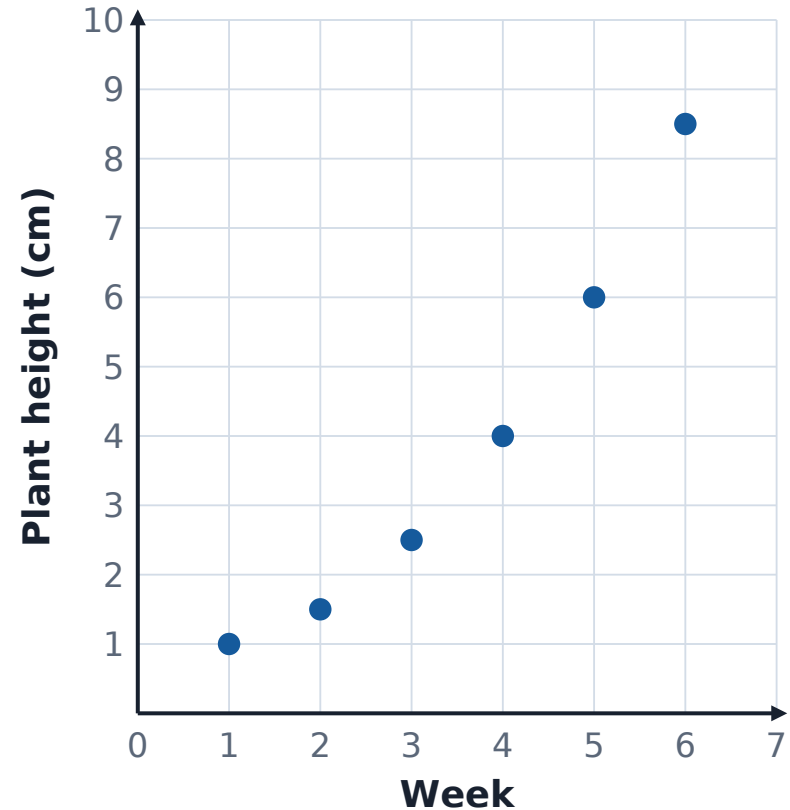


True or false?

The scatter plot has an outlier.

A. True

B. False



True or false?

Every scatter plot has at least one outlier.

- A. True
- B. False

Find the mean absolute deviation of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Sharing a total equally (the mean)	5.MD.B.2
BACKWARD 2	Distance as absolute value	6.NS.C.7.c
BACKWARD 3	Dividing to find an average distance	5.NBT.B.7
FORWARD 1	Difference in means as a multiple of the MAD	7.SP.B.3
FORWARD 2	Judge model fit by distances from the line	8.SP.A.2

SET 80 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean absolute deviation.

2, 4, 6, 8, 10

MAD: _____

What is the mean absolute deviation?

3, 3, 5, 7, 7

A. 0

B. 1.6

C. 2

D. 4

Find the mean absolute deviation.

5, 7, 8, 12

MAD: _____

Find the mean absolute deviation.

6, 8, 10, 12, 14, 16

MAD: _____

Find the mean absolute deviation.

20, 22, 25, 29

MAD: _____

SET 80 • 6.SP.B.5.c

BACKWARD 1

Sharing a total equally (the mean) • Grade 5 • Measurement & Data • 5.MD.B.2

Five jars hold 2, 4, 6, 8, and 10 cups of rice. The rice is shared equally among the jars.

How much is in each jar?

Cups: _____

Four friends have 3, 3, 7, and 7 marbles. They share the marbles equally.

How many marbles does each friend get?

Marbles: _____

True or false?

Bags with 1, 5, and 6 pounds of flour are shared equally. Each bag then has 4 pounds.

- A.** True
- B.** False

Six students have 2, 3, 3, 4, 5, and 7 pencils. They share equally.

How many pencils does each student get?

A. 3

B. 4

C. 5

D. 24

Three cups hold $\frac{1}{2}$, $\frac{1}{2}$, and 2 cups of water. The water is shared equally.

How much is in each cup?

Cups: _____

SET 80 • 6.SP.B.5.c

BACKWARD 2

Distance as absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|2 - 6|$$

Value: _____

How far is 9 from 6 on a number line?

Distance: _____

True or false?

$$|4 - 10| = 6$$

A. True

B. False

What is the distance between 3 and 8 on a number line?

A. -5

B. 5

C. 11

D. 24

Find the value.

$$|10 - 6|$$

Value: _____

SET 80 • 6.SP.B.5.c

BACKWARD 3

Dividing to find an average distance • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Divide.

$$12 \div 5$$

Quotient: _____

Divide.

$$8 \div 5$$

Quotient: _____

True or false?

$$18 \div 6 = 3$$

A. True

B. False

$$10 \div 4 = ?$$

A. 0.4

B. 2.4

C. 2.5

D. 40

Divide.

$$9 \div 4$$

Quotient: _____

SET 80 • 6.SP.B.5.c

FORWARD 1

Difference in means as a multiple of the MAD • Grade 7 • Statistics & Probability • 7.SP.B.3

Class A's mean is 72 and Class B's mean is 80. Each class has a MAD of 4.

The difference in means is how many times the MAD?

Times: _____

Two data sets have means of 15 and 21. Each has a MAD of 2.

How many MADs apart are the means?

A. 2

B. 3

C. 6

D. 12

True or false?

Two data sets have means that differ by 1 MAD.
The data sets probably overlap a lot.

- A. True
- B. False

Two teams have mean heights of 150 cm and 162 cm.
Each has a MAD of 4 cm.

The difference in means is how many times the MAD?

Times: _____

Which pair of data sets has the LEAST overlap?

- A.** Means 10 and 30, each MAD 2
- B.** Means 10 and 12, each MAD 2
- C.** Means 10 and 11, each MAD 3
- D.** Means 10 and 14, each MAD 4

SET 80 • 6.SP.B.5.c

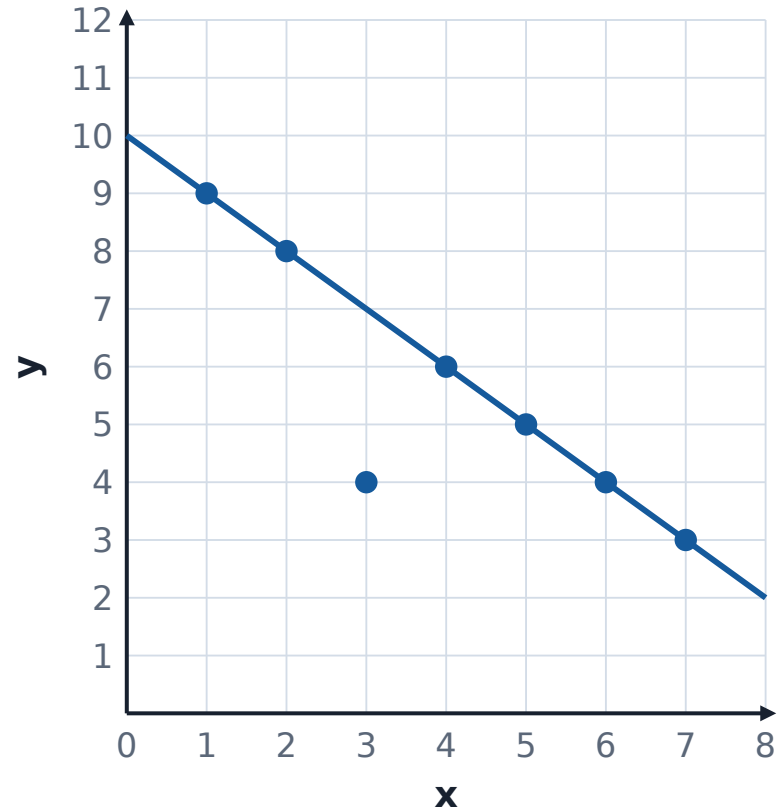
FORWARD 2

Judge model fit by distances from the line • Grade 8 • Statistics & Probability •
8.SP.A.2

Which point is farthest from the line?

Write its coordinates.

Point: _____

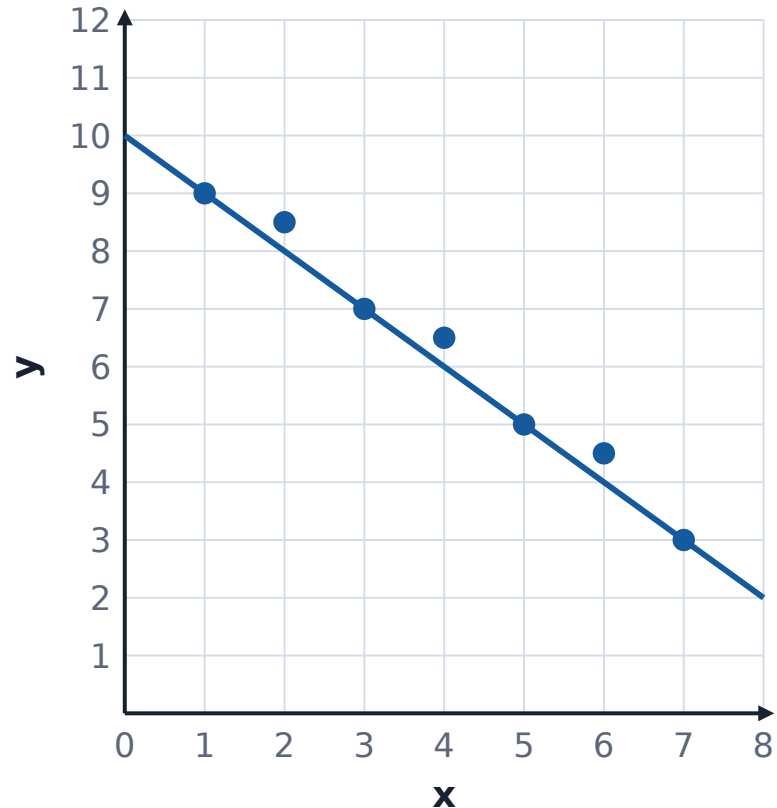


True or false?

The points are close to the line, so it is a good fit.

A. True

B. False

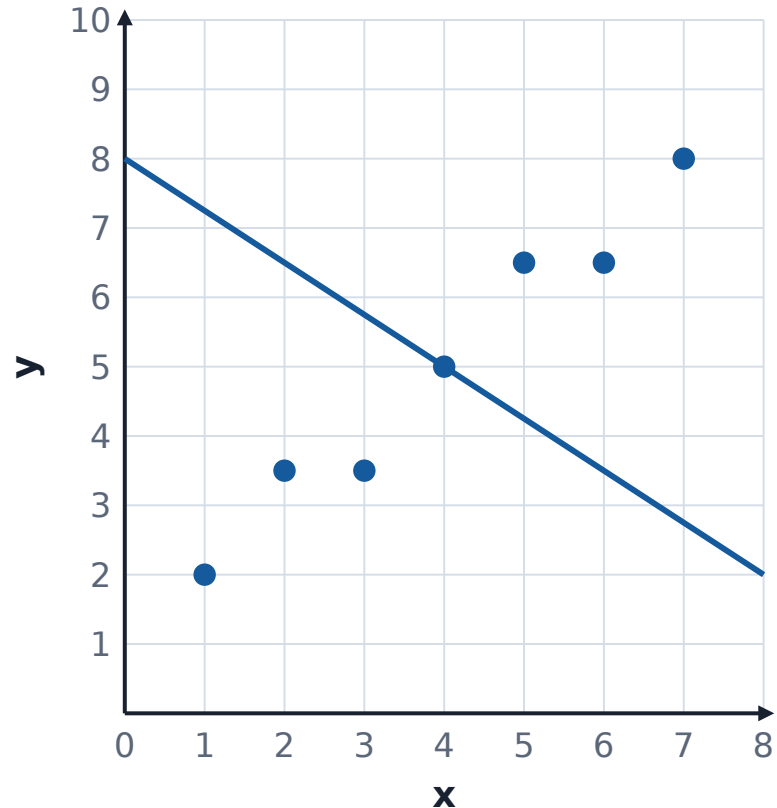


True or false?

The line is a good fit for the data.

A. True

B. False



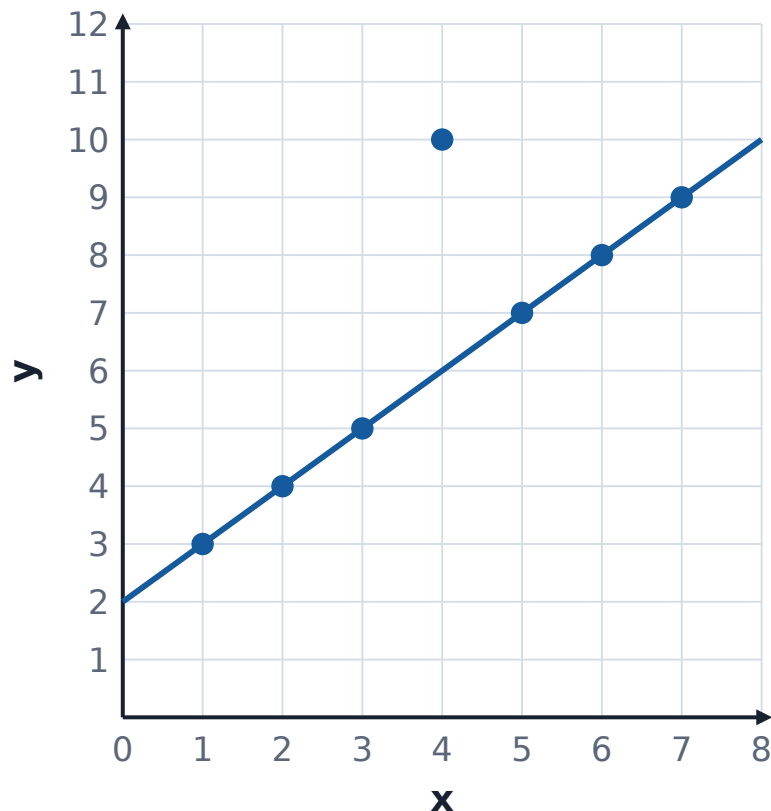
Which line fits data better?

- A.** A line that touches only one point
- B.** A line with all points far above it
- C.** A line with points close to it on both sides
- D.** A line with all points below it

For the point $(4, 10)$, the line gives $y = 6$.

How far is the point from the line, measured vertically?

Distance: _____



Describe the overall pattern and striking deviations in the context of the data

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Where data cluster and where there are gaps	6.SP.A.2
BACKWARD 2	Reading a dot plot in context	4.MD.B.4
BACKWARD 3	Greatest and least values	2.NBT.A.4
FORWARD 1	Compare the patterns of two distributions	7.SP.B.3
FORWARD 2	Describe patterns and deviations in scatter plots in context	8.SP.A.1

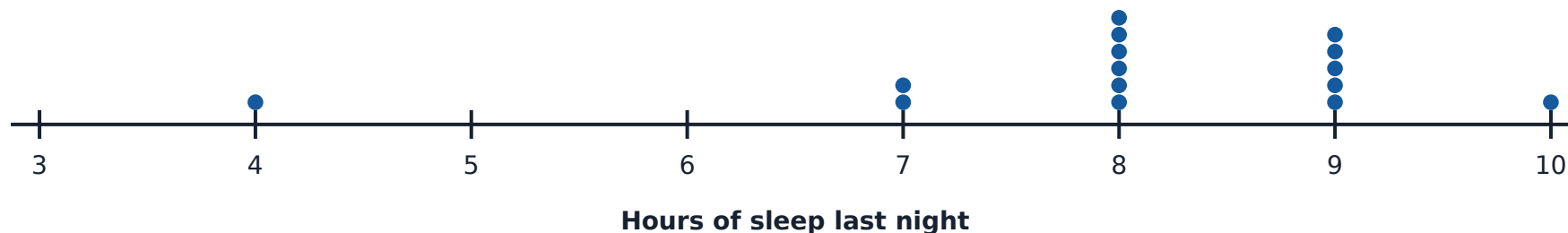
SET 81 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The dot plot shows how many hours students slept last night.

Describe the overall pattern. Which value stands apart, and what might explain it?

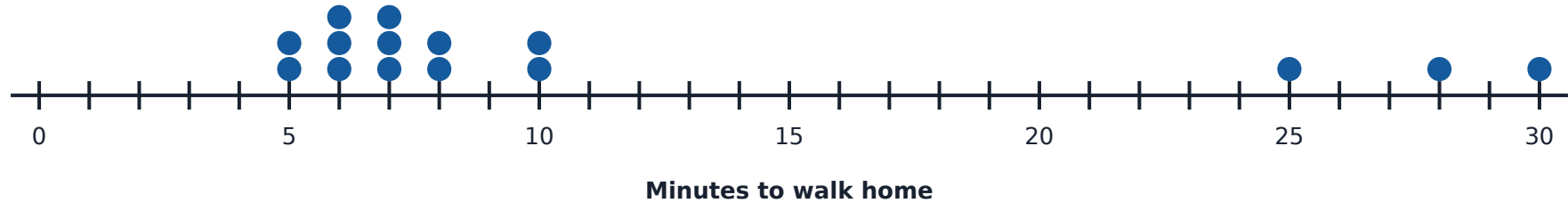


Overall pattern: _____

Value that stands apart: _____

The dot plot shows how long students take to walk home.

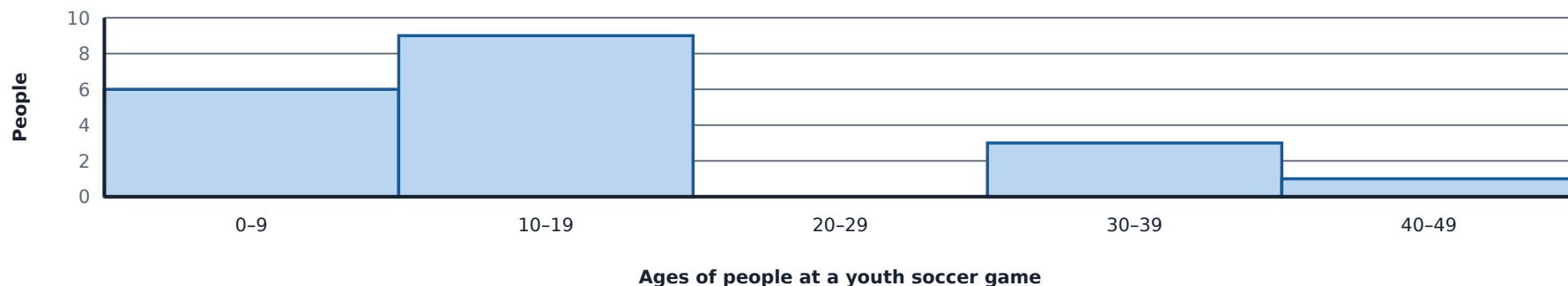
Which description best fits the data in context?



- A.** The times are spread evenly from 5 to 30 minutes.
- B.** All students take about the same amount of time.
- C.** Most students take 25 to 30 minutes.
- D.** Most students take 5 to 10 minutes; a few who live farther away take 25 to 30 minutes.

The histogram shows the ages of people at a youth soccer game.

Describe the overall pattern. Which ages stand apart, and who might they be?

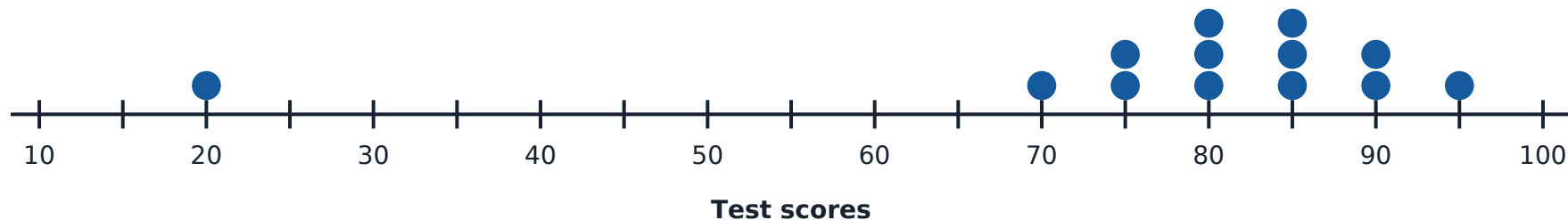


Overall pattern: _____

Ages that stand apart: _____

The dot plot shows the scores on a class test.

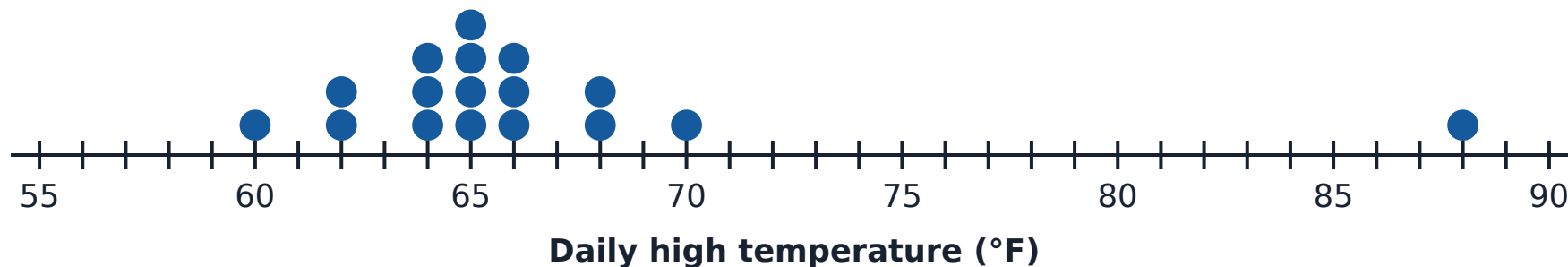
Which statement describes a striking deviation in context?



- A.** One student scored 20, far below the others, who scored from 70 to 95. **B.** Most students scored below 50.
- C.** The scores are spread evenly from 20 to 95. **D.** There are no unusual scores.

True or false?

The daily high temperatures cluster from 60°F to 70°F, with one unusually hot day at 88°F.



A. True

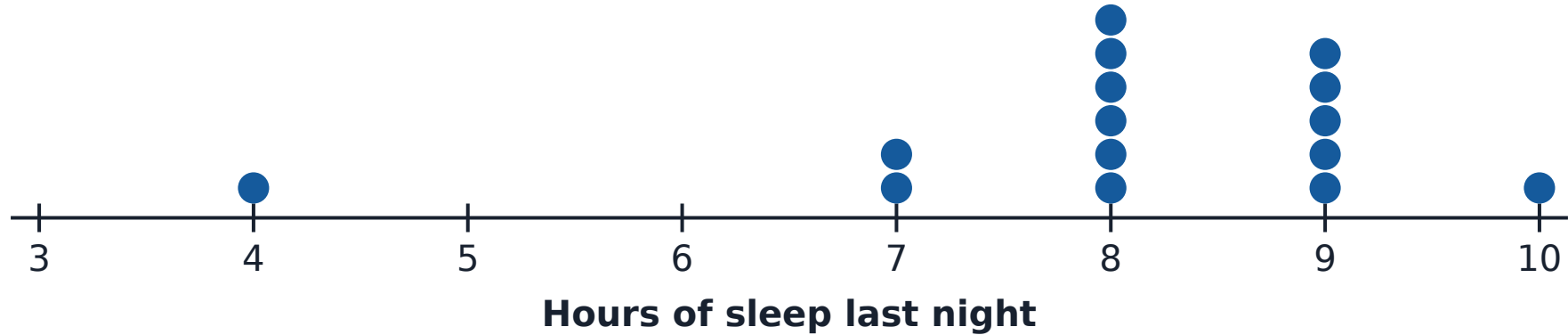
B. False

SET 81 • 6.SP.B.5.c

BACKWARD 1

Where data cluster and where there are gaps • Grade 6 • Statistics & Probability • 6.SP.A.2

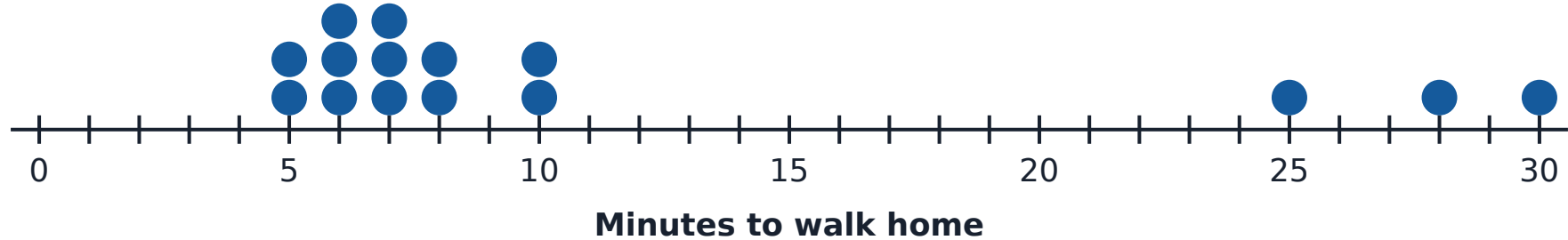
Between which two values do most of the data cluster?



Between: _____

True or false?

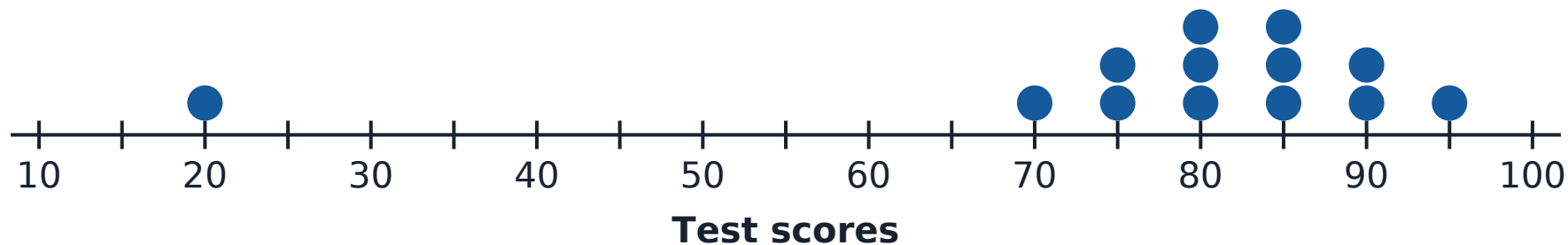
There is a gap in the data between 10 and 25 minutes.



A. True

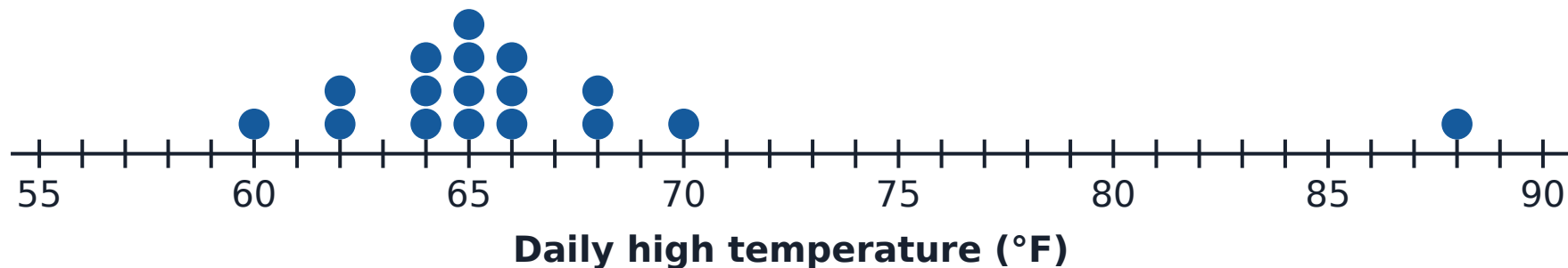
B. False

Where is there a gap in the data?



- A.** Between 70 and 80 **B.** Between 20 and 70
- C.** Between 85 and 95 **D.** There is no gap.

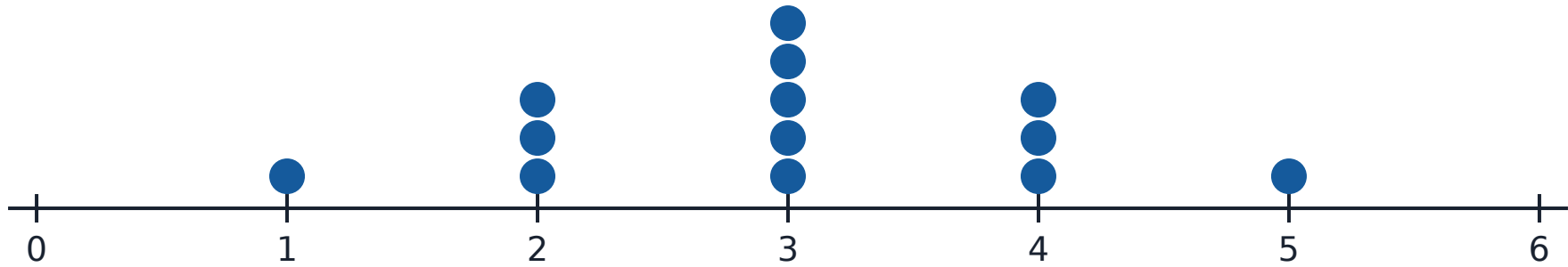
Between which two temperatures do most of the data cluster?



Between: _____

True or false?

The data have no gaps.



A. True

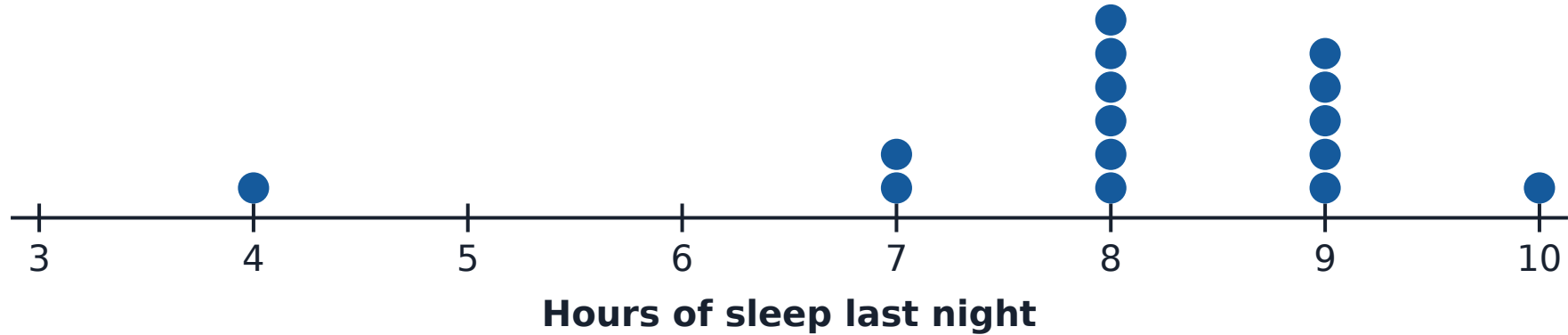
B. False

SET 81 • 6.SP.B.5.c

BACKWARD 2

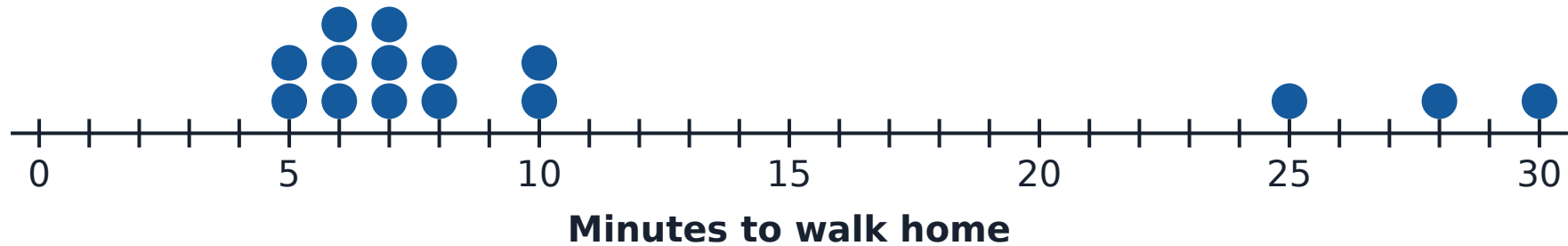
Reading a dot plot in context • Grade 4 • Measurement & Data • 4.MD.B.4

How many students slept 8 hours last night?



Students: _____

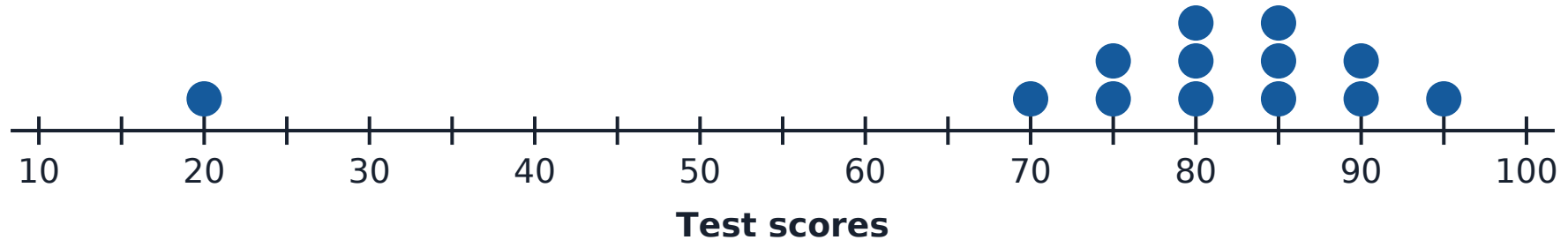
How many students take 7 minutes to walk home?



Students: _____

True or false?

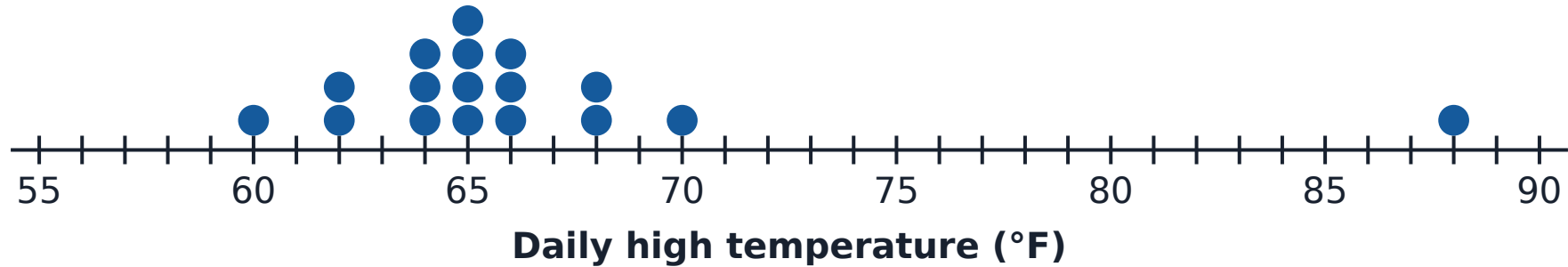
Three students scored 85 on the test.



A. True

B. False

How many days had a high temperature of 65°F?



A. 3

B. 4

C. 5

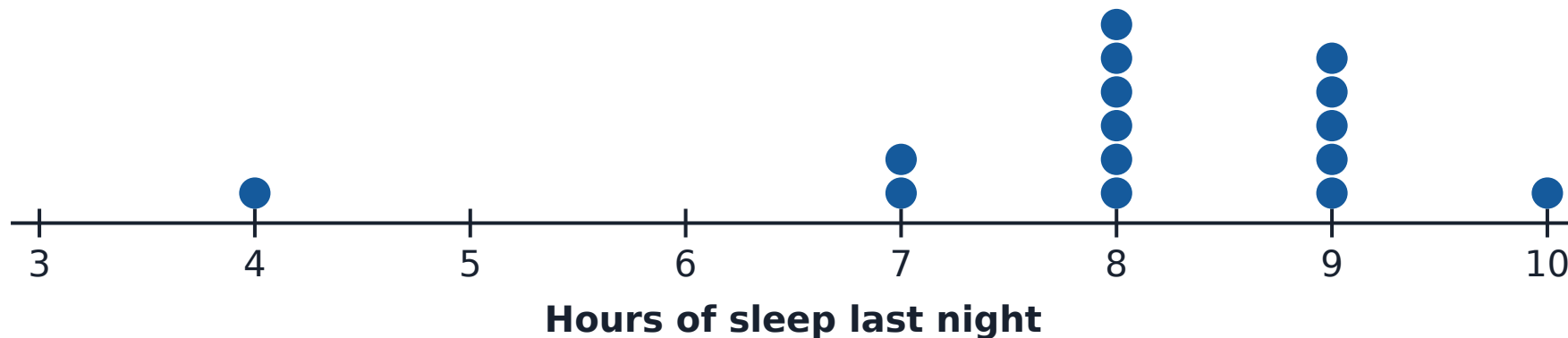
D. 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q4/5

What does each dot in the dot plot represent?



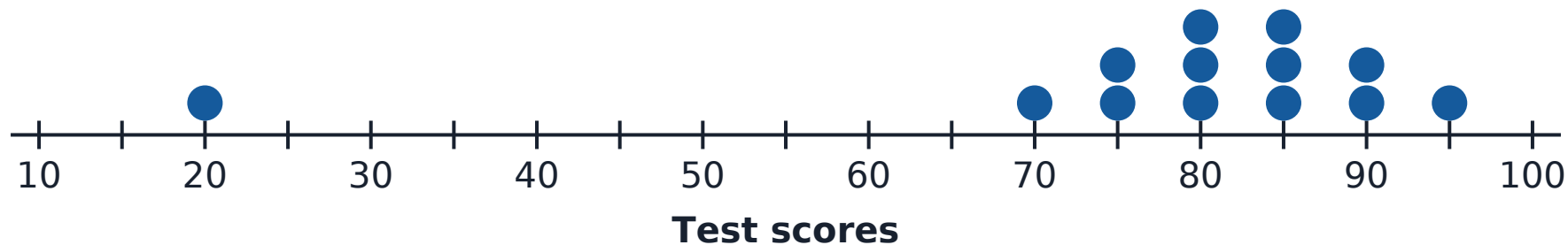
Each dot: _____

SET 81 • 6.SP.B.5.c

BACKWARD 3

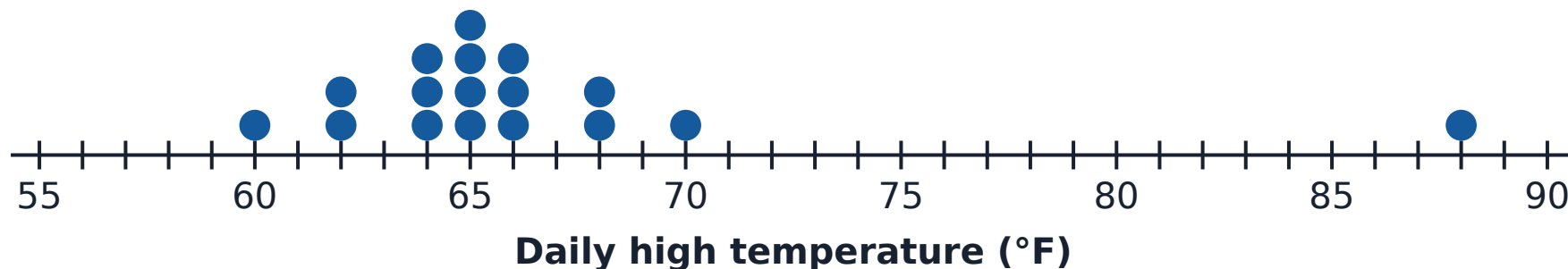
Greatest and least values • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

What is the least value in the data?



Least: _____

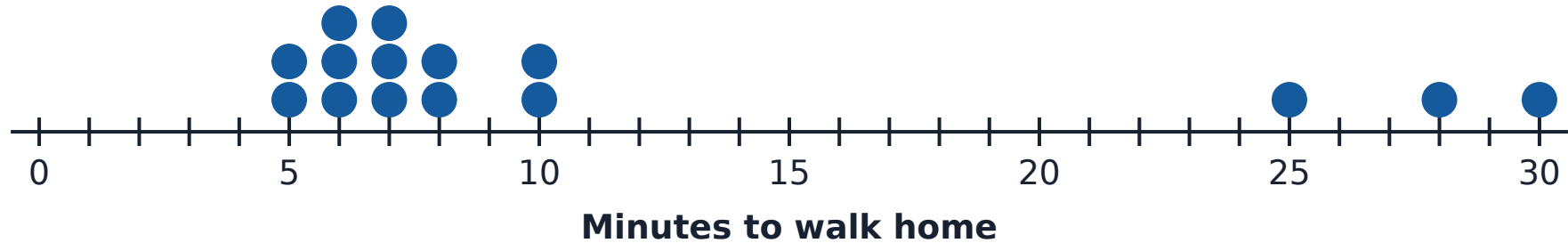
What is the greatest value in the data?



Greatest: _____

True or false?

The greatest walking time is 30 minutes.



A. True

B. False

Which number is least?

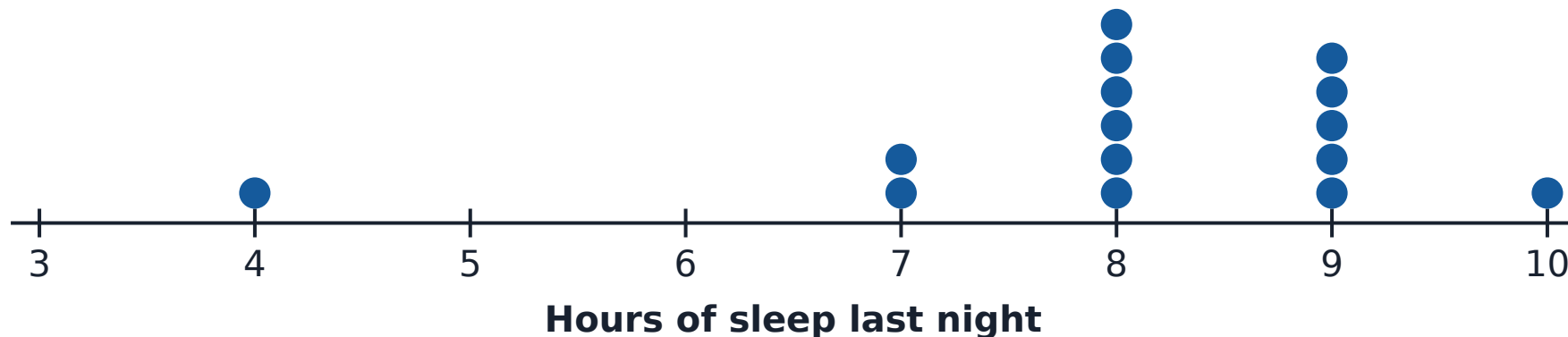
A. 20

B. 70

C. 75

D. 95

What is the least number of hours of sleep shown?



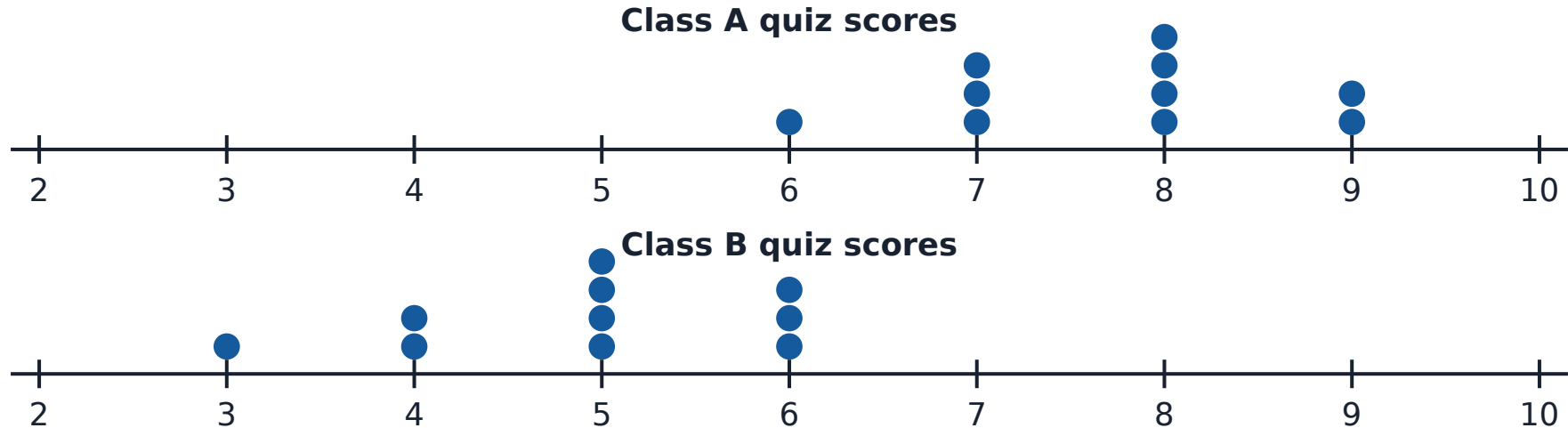
Least: _____

SET 81 • 6.SP.B.5.c

FORWARD 1

Compare the patterns of two distributions • Grade 7 • Statistics & Probability •
7.SP.B.3

Describe how the two distributions differ in context.



Comparison: _____

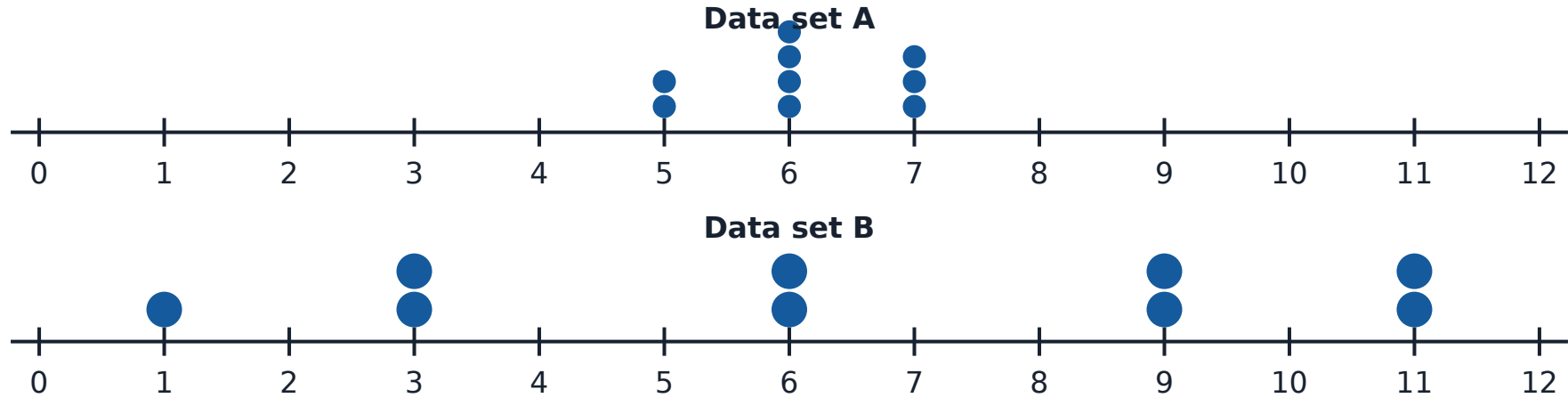
Which statement describes both gardens?



- A.** The gardens have the same center.
- B.** Garden A's plants are taller.
- C.** Garden B is much more spread out.
- D.** Both cluster tightly, and Garden B's plants are about 6 cm taller.

True or false?

Data set B is more spread out than data set A.



A. True

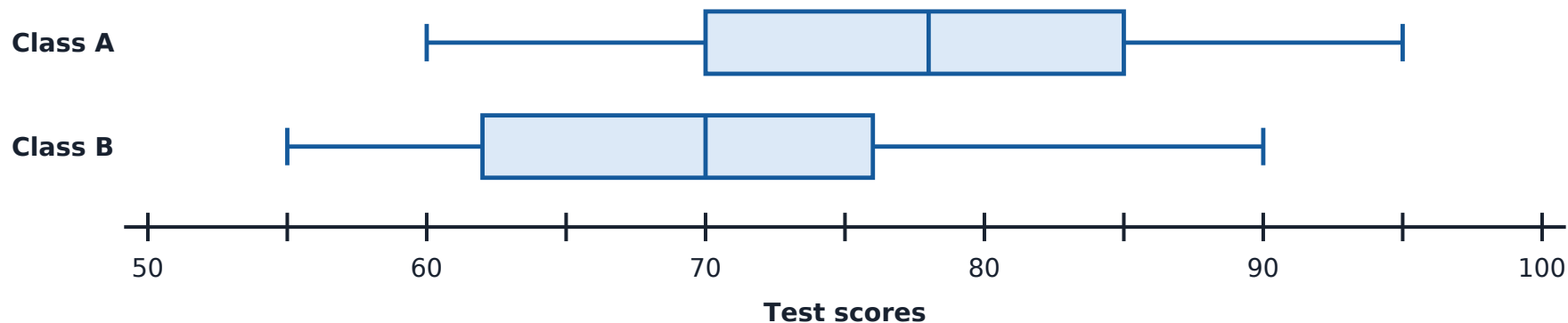
B. False

Class A's quiz scores cluster from 7 to 9, with one score of 2. Class B's scores cluster from 4 to 6 with no unusual values.

Which class has a striking deviation?

Class: _____

Which statement about the middle halves of the two box plots is true?



- A.** They overlap from 70 to 76.
- B.** They do not overlap.
- C.** They overlap from 55 to 95.
- D.** They are the same.

SET 81 • 6.SP.B.5.c

FORWARD 2

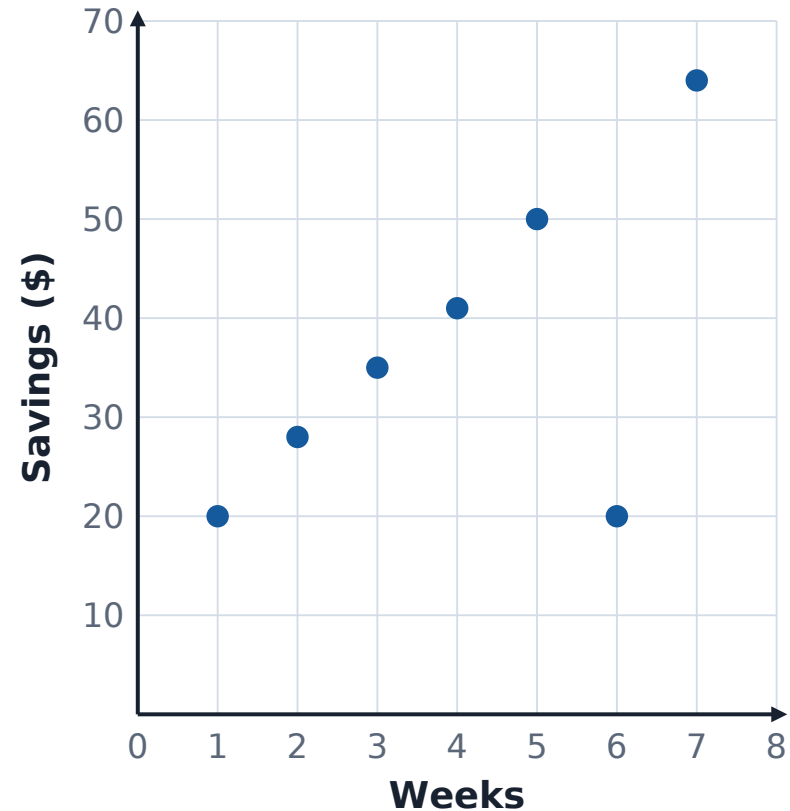
Describe patterns and deviations in scatter plots in context • Grade 8 •
Statistics & Probability • 8.SP.A.1

The scatter plot shows savings over several weeks.

Describe the overall pattern and the point that does not fit.

Pattern: _____

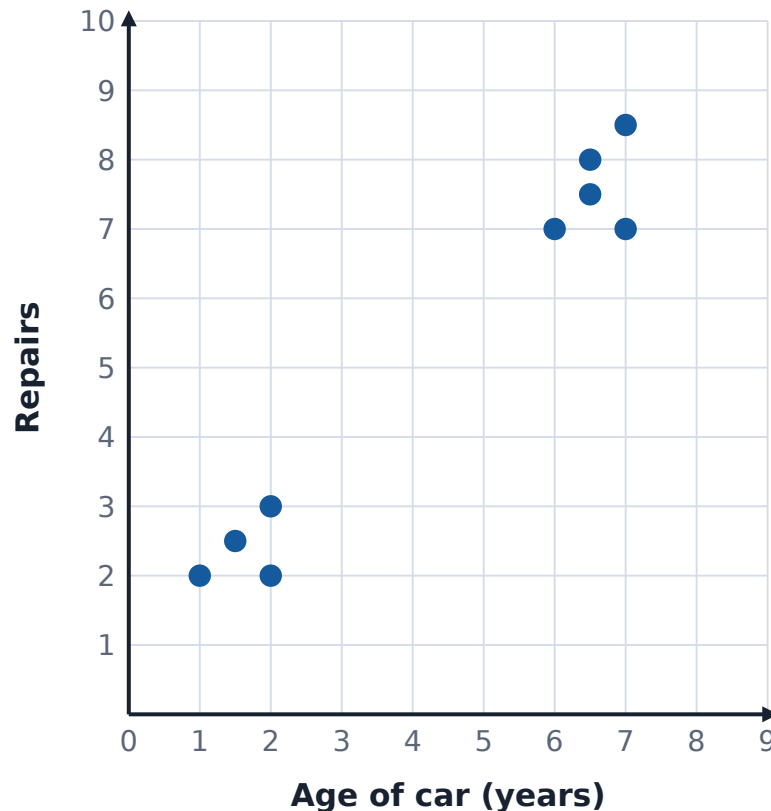
Point that does not fit: _____



The scatter plot shows the age of cars and the number of repairs.

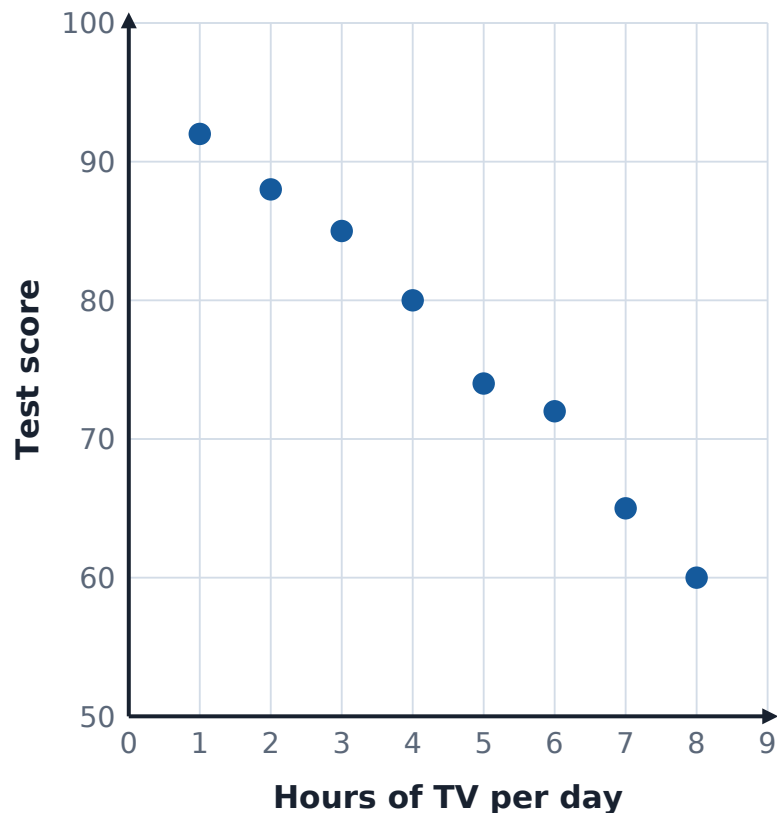
Which description fits the data in context?

- A.** Older cars have fewer repairs.
- B.** Newer cars have few repairs and older cars have many repairs, forming two groups.
- C.** All cars have the same number of repairs.
- D.** There is no pattern.



Which statement describes the association in context?

- A.** As hours of TV increase, test scores tend to increase.
- B.** Hours of TV and test scores are not related.
- C.** Every student scored the same.
- D.** As hours of TV increase, test scores tend to decrease.

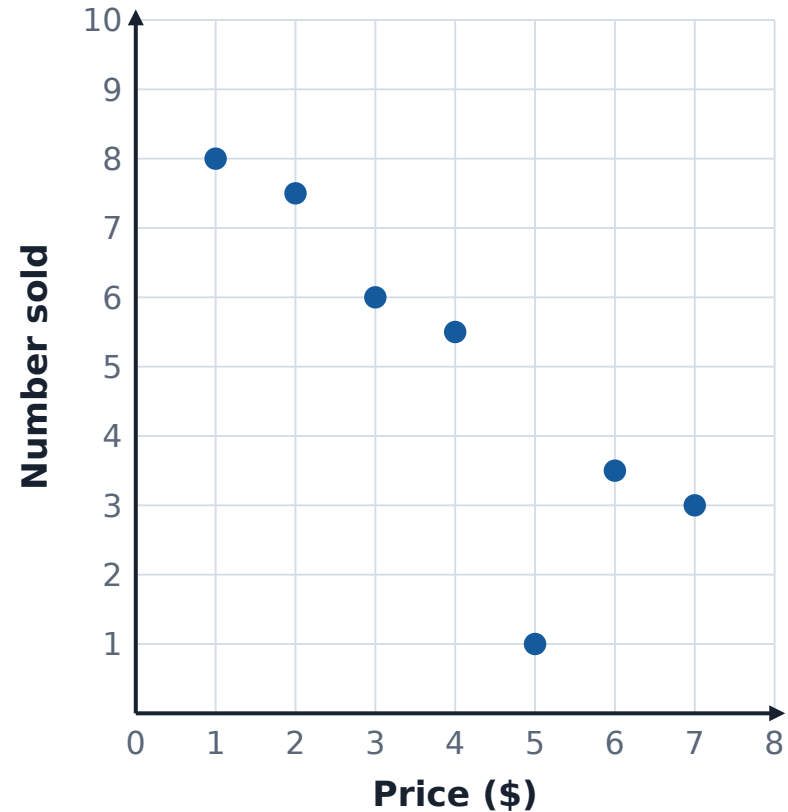


The scatter plot shows the price of an item and the number sold.

Which point does not fit the pattern? What might explain it?

Point: _____

Possible reason: _____

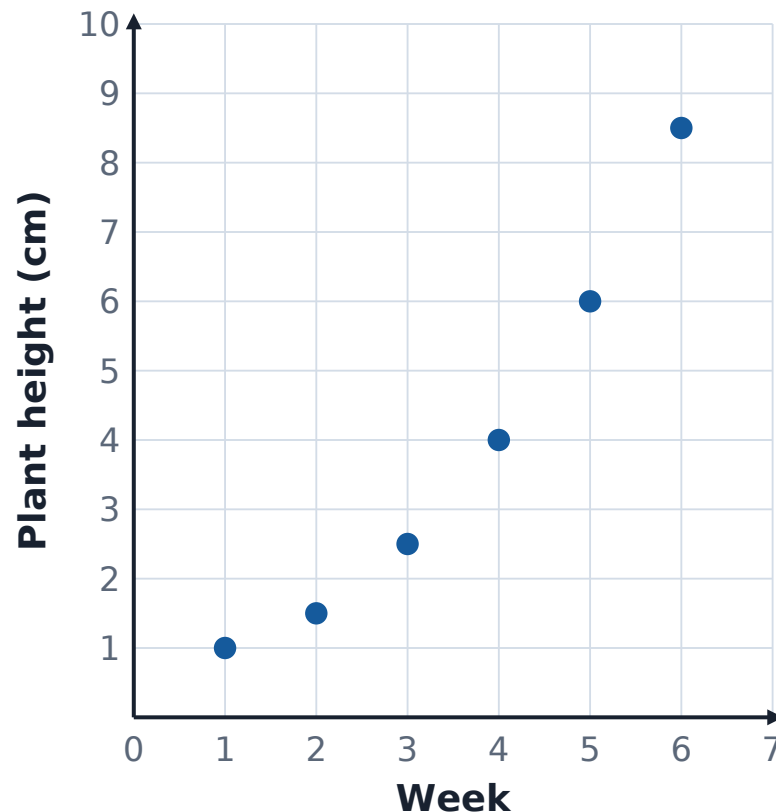


True or false?

The plant grows faster each week, so the pattern in the scatter plot is nonlinear.

A. True

B. False



Choose measures of center and variability based on the shape of the data

MAIN	Grade 6 main question	6.SP.B.5.d
BACKWARD 1	Shape of a distribution	6.SP.A.2
BACKWARD 2	Computing the mean and median	6.SP.B.5.c
BACKWARD 3	Reading a dot plot	4.MD.B.4
FORWARD 1	Compare populations using appropriate measures	7.SP.B.4
FORWARD 2	Judge how well a line fits the data — nearest related (no direct Gra...	8.SP.A.2

SET 82 • 6.SP.B.5.d

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.d

Data: 20, 22, 23, 25, 90

Which measure of center better describes a typical value?

A. Mean

B. Median

A data set is skewed right and has an outlier.

Which pair of measures best describes its center and spread?

A. Mode and range

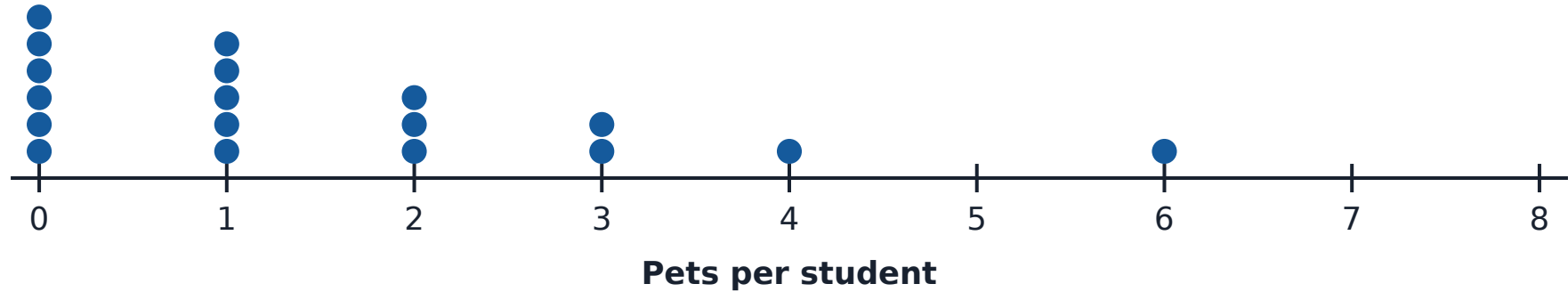
B. Median and IQR

C. Mean and range

D. Mean and MAD

The dot plot is skewed right.

Which measure of center better describes the typical number of pets?



A. Mean

B. Median

Salaries: \$30,000; \$32,000; \$35,000; \$38,000;
\$250,000

Which measure of center better describes a typical salary?

A. Mean

B. Median

True or false?

Heights of 20 sixth graders are symmetric with no outliers, so the mean is a good measure of center.

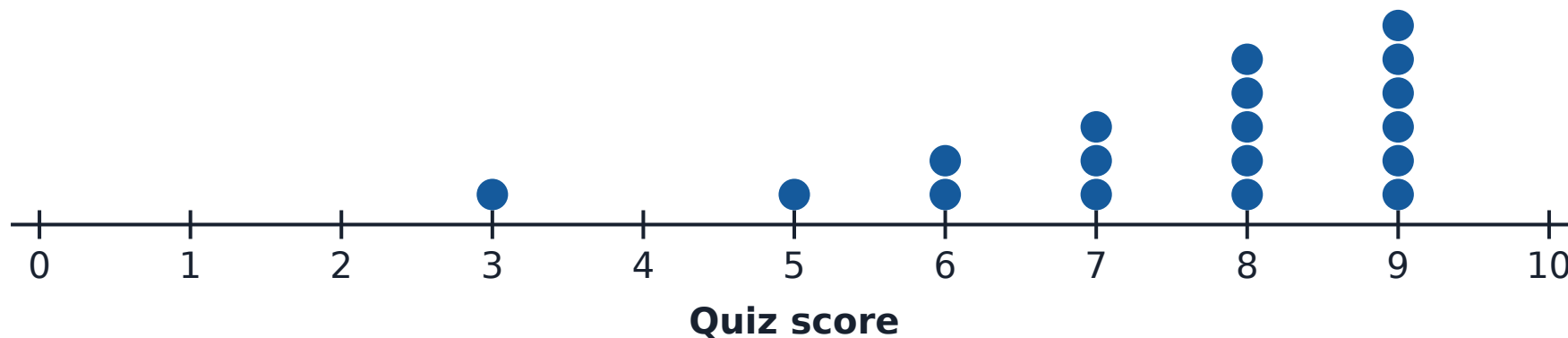
- A. True
- B. False

SET 82 • 6.SP.B.5.d

BACKWARD 1

Shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word describes the shape of the data?



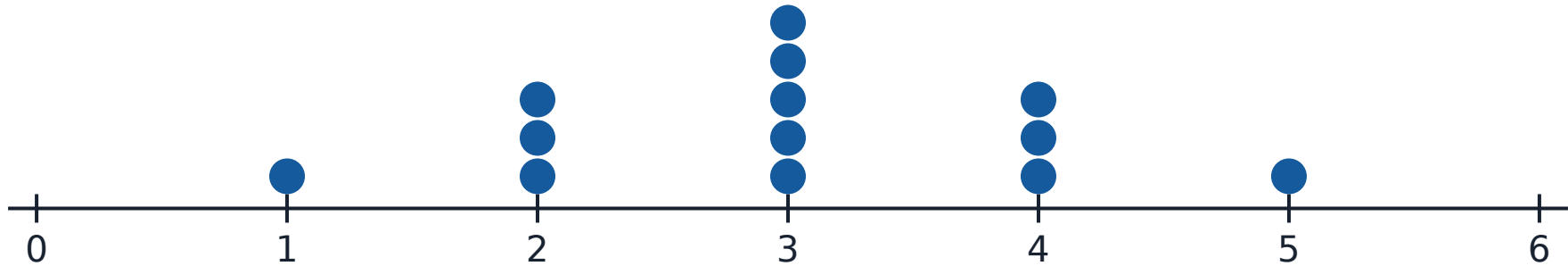
A. Skewed right

B. Symmetric

C. Skewed left

D. Uniform

Which word describes the shape of the data?



A. Skewed left

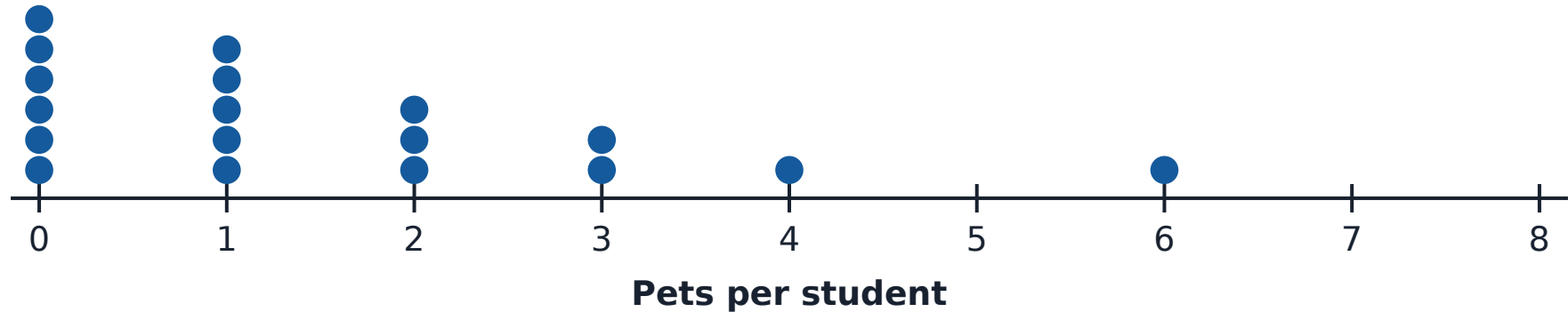
B. Symmetric

C. Uniform

D. Skewed right

True or false?

The data are skewed right.



A. True

B. False

True or false?

A data set with most values on the left and a long tail to the right is skewed right.

- A.** True
- B.** False

Which value in the data set is an outlier?

5, 6, 6, 7, 8, 30

- A.** 5
- B.** 6
- C.** 8
- D.** 30

SET 82 • 6.SP.B.5.d

BACKWARD 2

Computing the mean and median • Grade 6 • Statistics & Probability •
6.SP.B.5.c

Find the mean.

2, 3, 3, 4, 8

Mean: _____

Find the median.

2, 3, 3, 4, 8

Median: _____

True or false?

The mean of 10, 20, 30 is 20.

A. True

B. False

What is the median of 1, 4, 6, 9?

- A.** 4
- B.** 5
- C.** 6
- D.** 20

Find the mean.

5, 5, 6, 8, 11

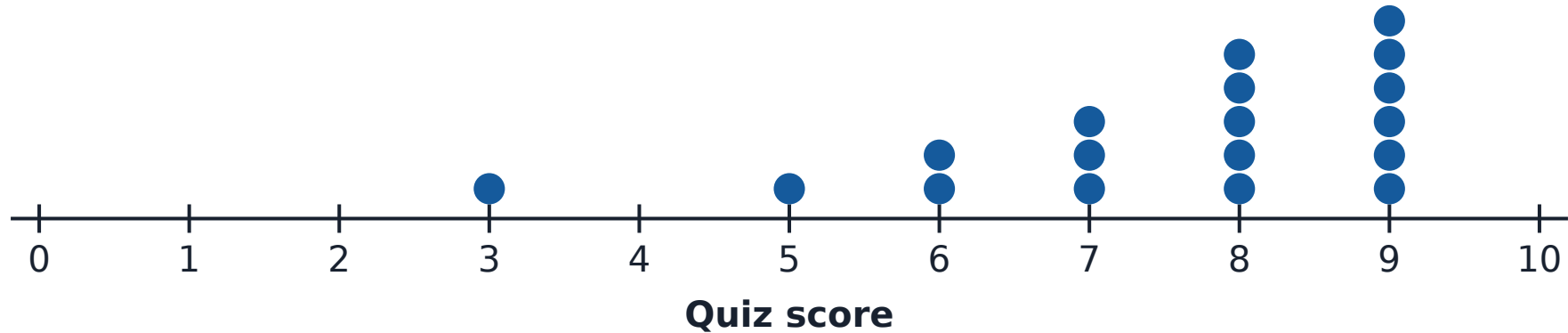
Mean: _____

SET 82 • 6.SP.B.5.d

BACKWARD 3

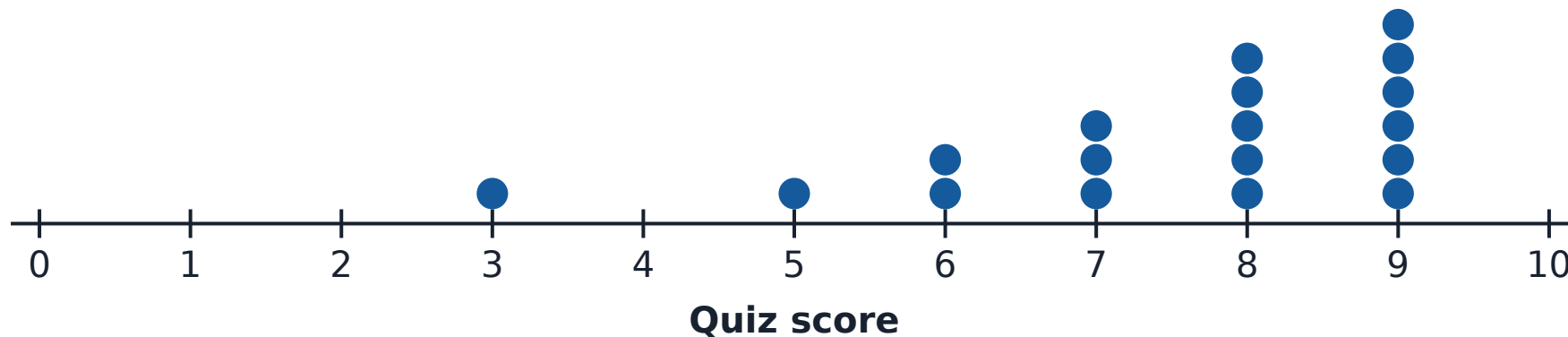
Reading a dot plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many students scored 9 on the quiz?



Students: _____

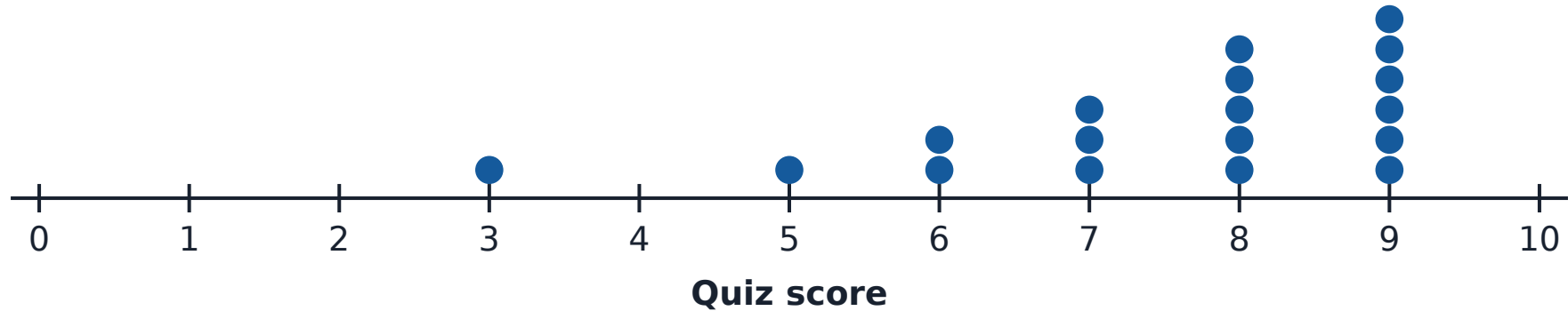
What is the lowest quiz score shown?



Score: _____

True or false?

Five students scored 8.



A. True

B. False

How many students took the quiz?



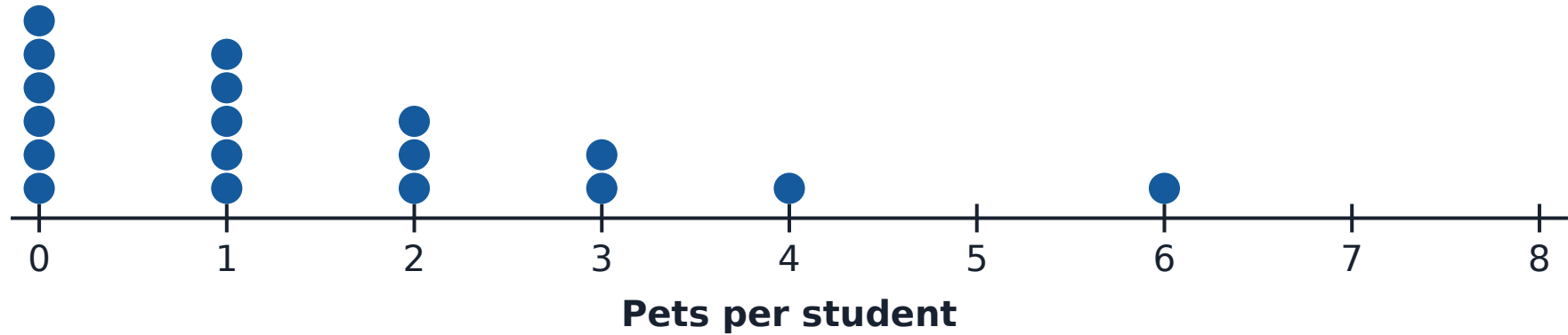
A. 6

B. 9

C. 15

D. 18

How many students had 0 pets?



Students: _____

SET 82 • 6.SP.B.5.d

FORWARD 1

Compare populations using appropriate measures • Grade 7 • Statistics & Probability • 7.SP.B.4

Two random samples of house prices are both skewed right with outliers.

Which measures are best for comparing them?

- A.** Medians and IQRs
- B.** Minimums
- C.** Means and MADs
- D.** Maximums

True or false?

When two samples are symmetric with no outliers, comparing their means is appropriate.

- A. True
- B. False

Sample A: median 30, IQR 6. Sample B: median 45, IQR 6.

The difference in medians is how many times the IQR?

Times: _____

Class A test scores have mean 78 and MAD 5. Class B test scores have mean 78 and MAD 12.

Which statement is true?

- A.** Class A scored higher.
- B.** Class B scored higher.
- C.** Class B's scores vary more.
- D.** Class A's scores vary more.

Random samples of puzzle times are both skewed right. Group X has a median of 14 minutes, and Group Y has a median of 20 minutes. Their IQRs are similar.

Which group typically takes longer?

Group: _____

FORWARD 2

Judge how well a line fits the data • Grade 8 • Statistics & Probability • 8.SP.A.2

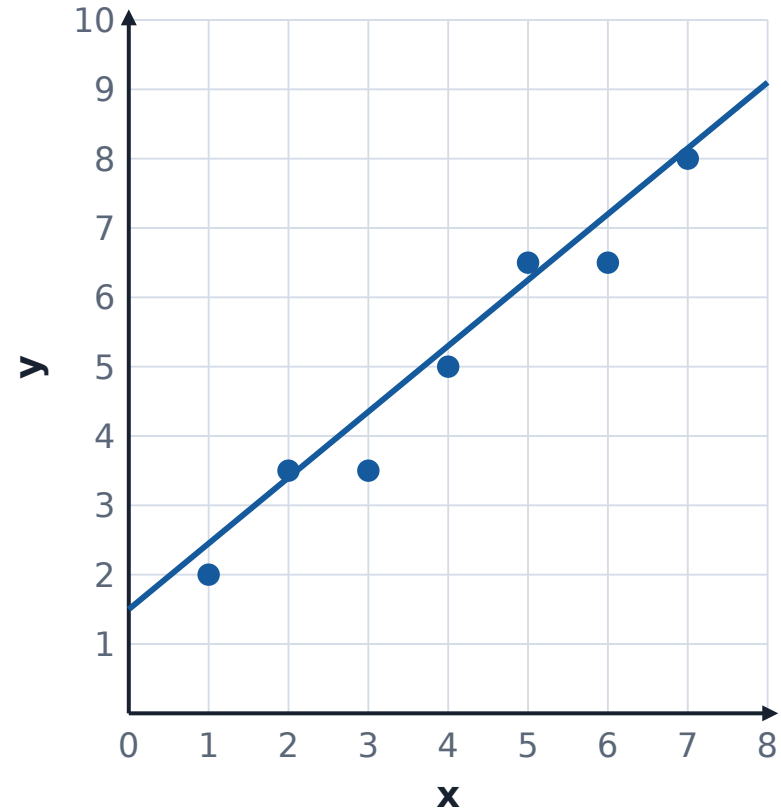
► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

The line is a good fit for the data.

A. True

B. False

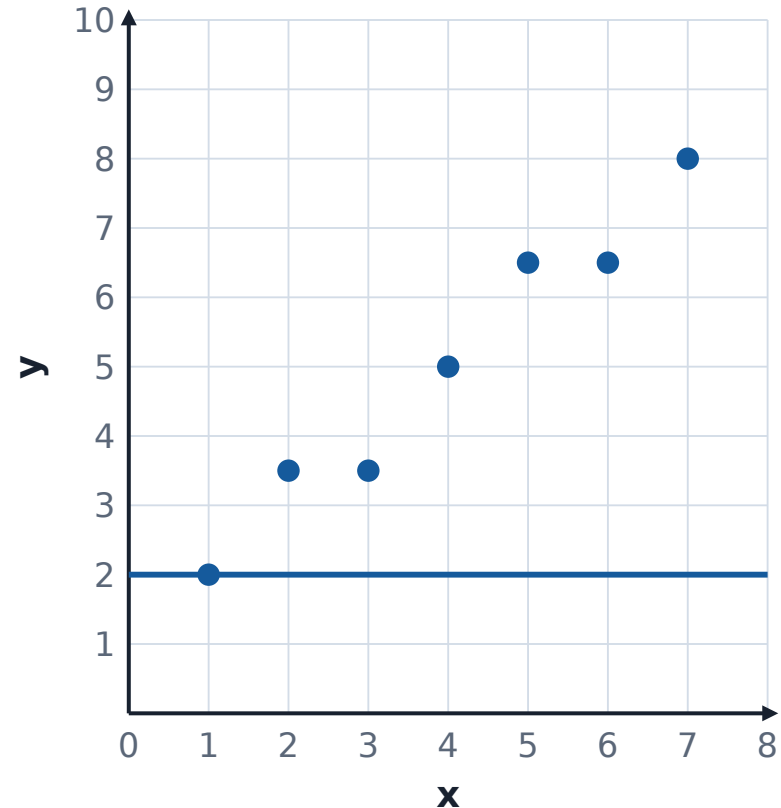


True or false?

The line is a good fit for the data.

A. True

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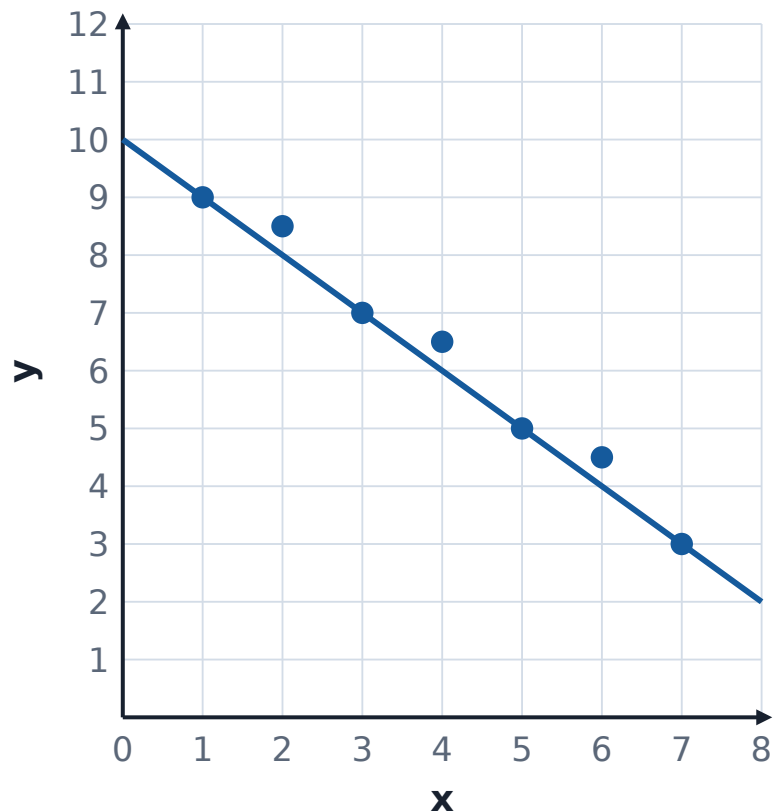


A line of best fit should show ____.

- A.** only the largest value
- B.** only the smallest value
- C.** only the first point
- D.** the overall trend of the data

Use the line of best fit to
predict y when $x = 2$.

$y =$ _____



Why should a scatter plot with no association NOT be modeled with a straight line?

- A.** There are too many points.
- B.** Lines must pass through the origin.
- C.** The axes are labeled.
- D.** There is no linear trend to model.

